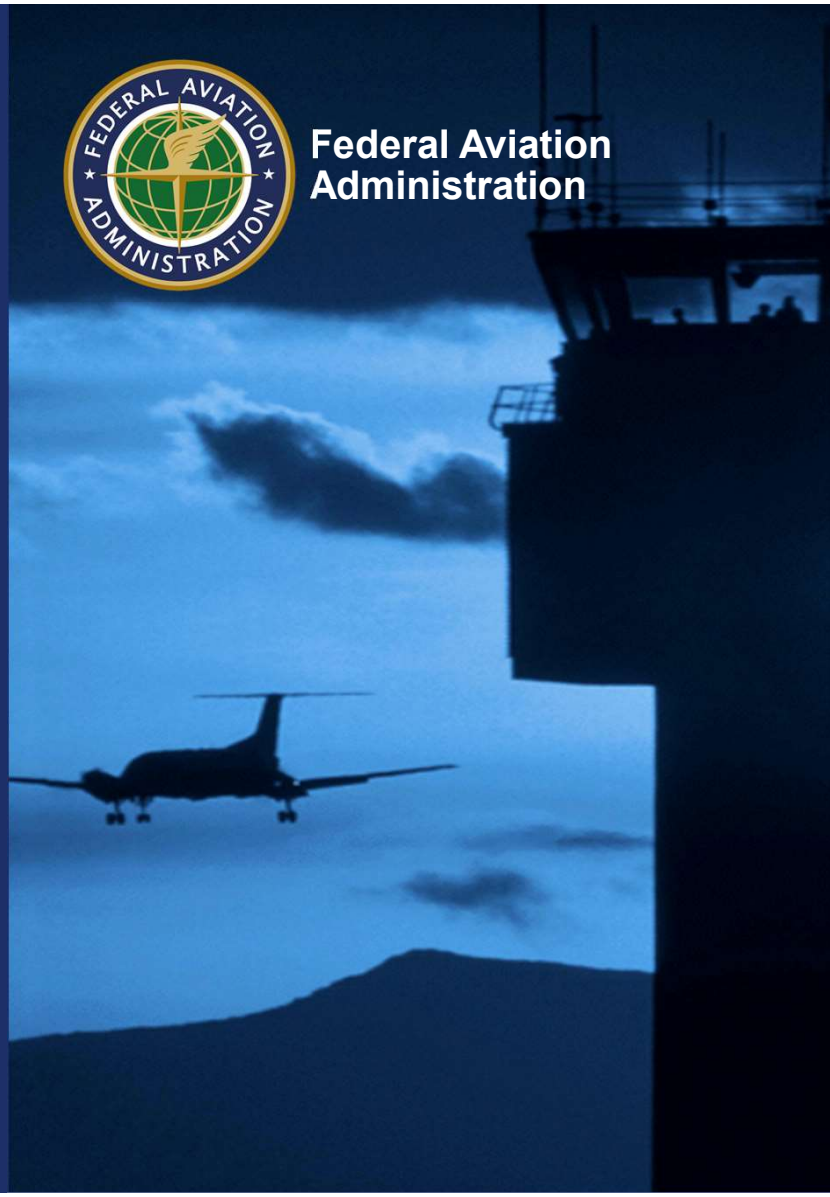
BAKFAA 3.4

Prepared by Applied Research Associates, Inc.
Task Order 17 Structural Airport Pavement NDT and Evaluation
Deliverable 5.2.1.1 Update Upgraded BAKFAA User Manual, Help Files and Presentation
March 3, 2023

Federal Aviation Administration

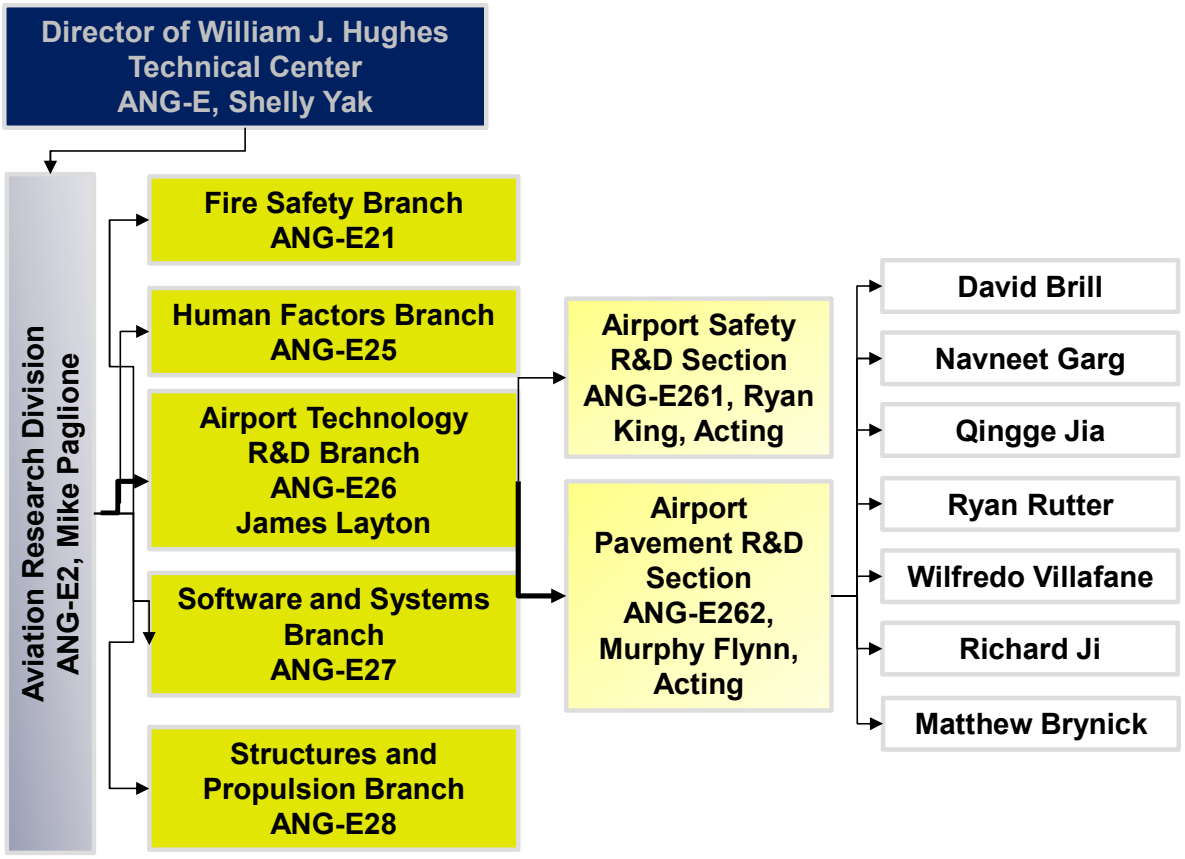


Presentation Outline

- FAA Airport Pavement R&D Overview
- BAKFAA Introduction
- BAKFAA Main Features
- BAKFAA Modular Interface
  - FWD Data
  - BAKFAA Inputs
  - Backcalculation Analysis
  - Load Transfer Efficiency (LTE) Calculation
  - Layered Elastic Analysis Program (LEAF)
- Software Demo
  - Flexible Pavement Example
  - Rigid Pavement Example
  - Composite Pavement Example
  - LTE Example
- Contact Us

FAA Airport Pavement R&D Overview

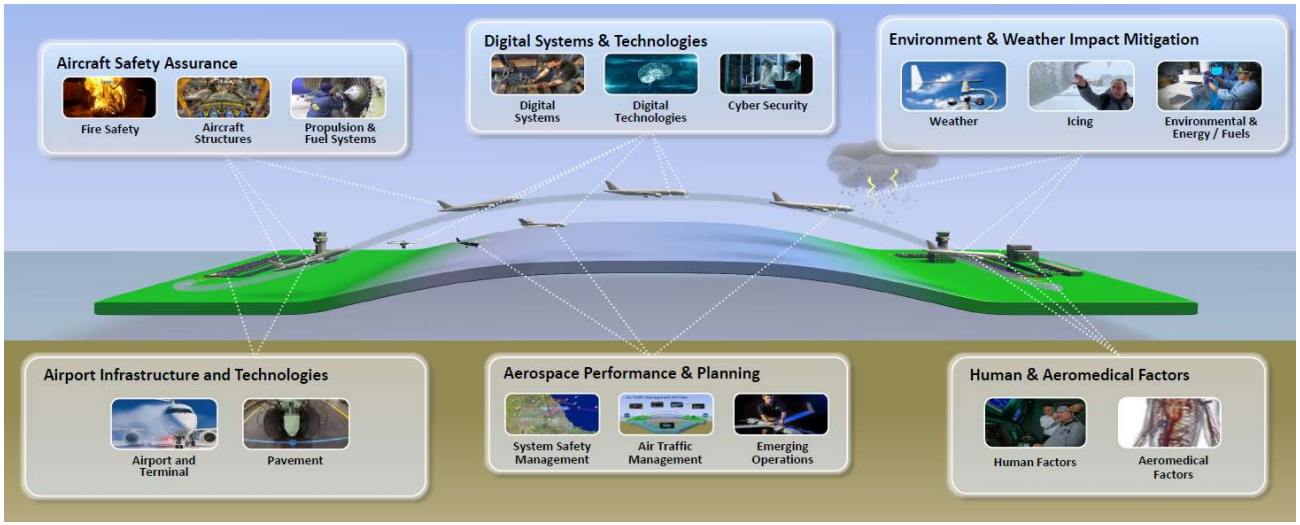
Who We Are



Pavement Consultant Support – ARA
Materials Lab O&M – ARA
Facilities O&M - GDIT

What We Do

- **Aviation Research Division Mission**
  - Develop scientific solutions to current and future air transportation challenges by conducting applied research and development in collaboration with industry, academia, and government.

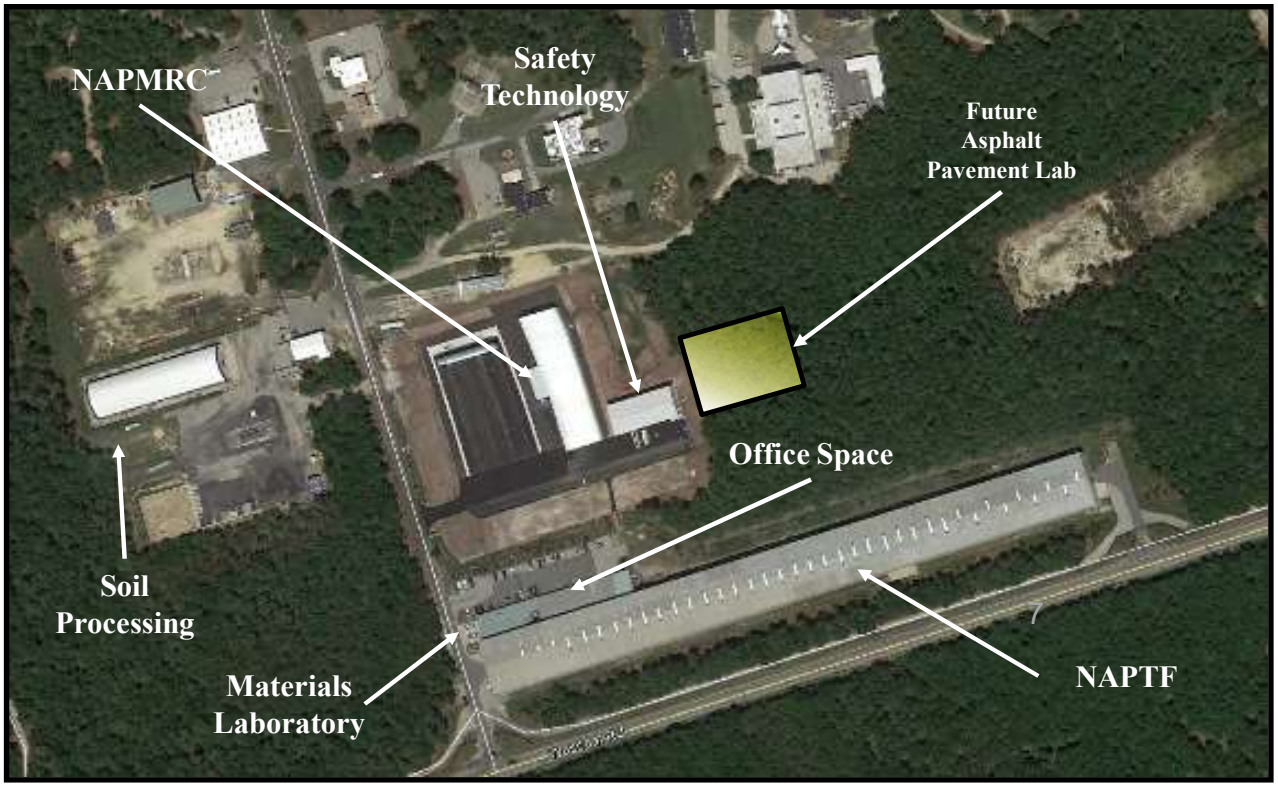


ANG-E262 Airport Pavement R&D

- **Research Conducted at the FAA William J. Hughes Technical Center, Atlantic City International Airport, NJ, USA.**
  - ANG-E262 Airport Pavement R&D Section
- **Sponsor: FAA Office of Airport Safety and Standards, Washington, DC.**
  - AAS-100 Airport Engineering Division (Michael Meyers)
  - AAS-110 Airport Engineering Division (Alberto Cruz, Harold Honey, Jeffrey Crislip)
  - AAS-120 Airport Data and Airspace
  - AAS-200 Airports Emerging Entrants Division
- **Provide Support for Development of FAA Pavement Standards**
  - Advisory Circulars
  - Engineering Briefs



Airport Pavement R&D Section Facility Layout



National Airport Pavement Test Facility (NAPTF)

NAPTF Facts:

- FAA / Boeing (CRDA) Partnership at $21M
- Opened April 1999
- Fully Enclosed Facility
- Accelerated Traffic Testing
- 900 ft. x 65 ft. of Test Pavement Surface
- Full-scale Pavement Structures and Landing Gear Loads





NAPTV Facts:

- Fully Automated & Programmed Wander Patterns
- Up to 5-dual wheel configuration
- Roughly 1.3 million lbs.
- Up to 75,000 lbs. per wheel

National Airport Pavement Materials Research Center (NAPMRC)

NAPMRC Facts:

- Dedication Ceremony August 2015
- Indoor and Outdoor Testing Capability
- Accelerated Traffic Testing
- Outdoor: 150ft. × 300ft. & Indoor: 72ft. × 300ft.
- Accelerated resurfacing

Heavy Vehicle Simulator – (HVS-A) Facts:

- Temperature Control Capability up to 150°F
- Wheel Load Capacity 10,000 to 100,000 lbs.
- Single Wheel – radial aircraft tire size 52×21.0R22
- Dual Wheel (B737-800)
- Fully Automated & Programmed Wander Patterns up to 6 ft.
- Test Speeds 0.17 to 5 mph



FAA NextGen Pavement Materials Lab

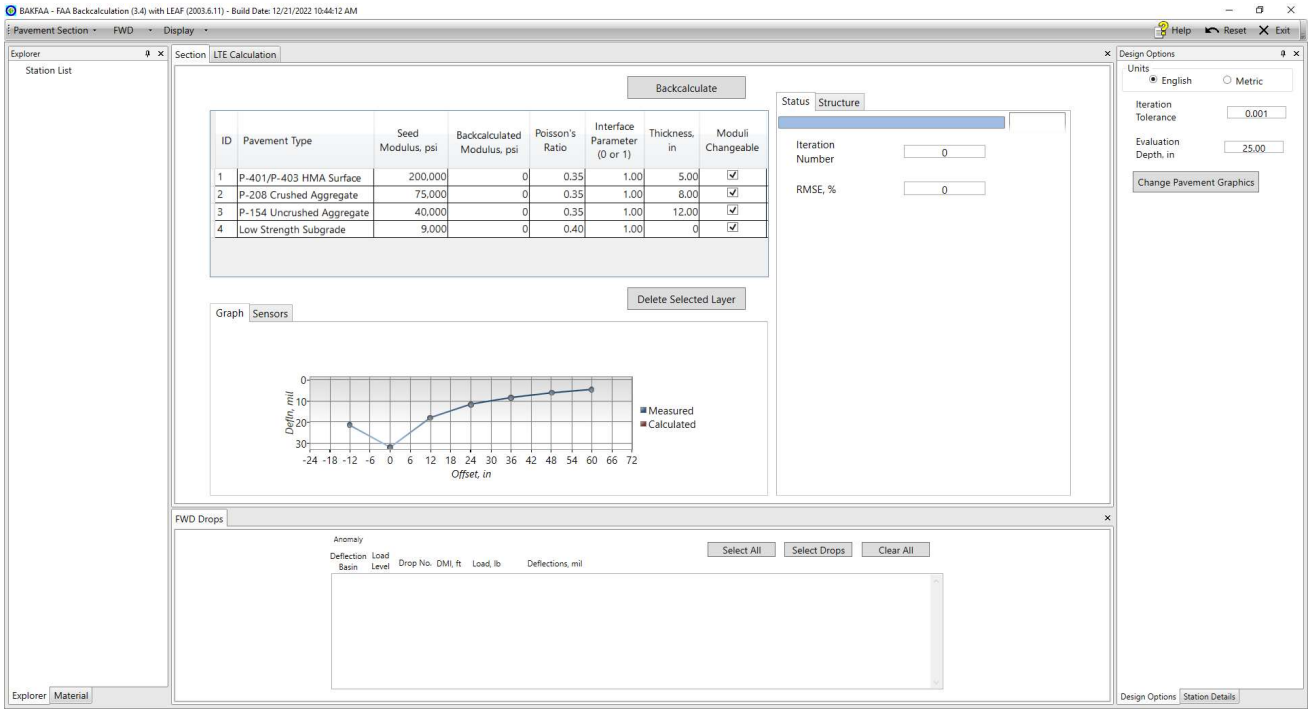
- **2010: Laboratory Opened**
- **2013: AASHTO Material Reference Laboratory (AMRL)**
- **2013: Cement and Concrete Reference Laboratory (CCRL)**
- **Full Test Capabilities: Asphalt, Concrete, Aggregate, Soils**
- **Advanced Test Capabilities:**
  - Asphalt Pavement Analyzer (APA)
  - Asphalt and Concrete Flexural Beam Fatigue
  - Semi-Circular Beam (SCB)
  - Disk-Shaped Compact Tension (DCT)
  - High Temperature Indirect Tension (IDT)
  - Overlay Tester
- **Benefits to the NAPTF:**
  - Quality Control of Testing
  - Expedient Testing of Materials During Construction
  - Perform Advanced Materials Characterization On-site
  - Development of Performance Based Specification



BAKFAA Introduction

About BAKFAA

- Computer program developed by the FAA for backcalculation of pavement layer modulus values using HWD/FWD data.
- Publicly available and free to use (source code available).
- Descriptions of use included in FAA AC 150/5370-11B ‘Use of Nondestructive Testing in the Evaluation of Airport Pavements’.



BAKFAA Main Features

BAKFAA Main Features

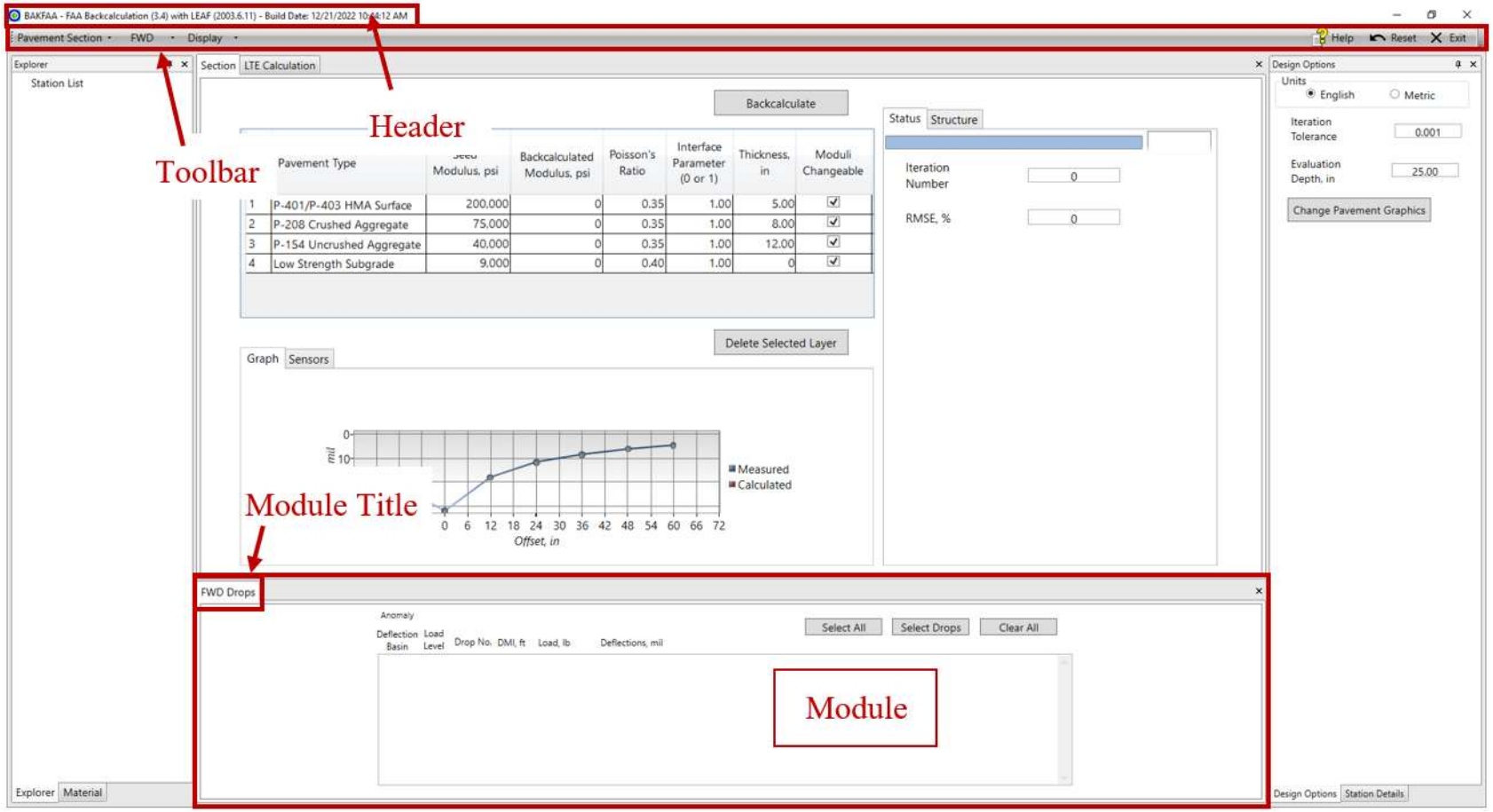
- Designed as a Visual Basic .Net application
- Ability to switch between English and Metric units
- Includes a library of FAA standard pavement material
- Ability to import different FWD file formats from leading FWD manufacturers
- Converts a proprietary manufacturer's FWD file into PDDX format file
- FWD File Parser: file processing class to read proprietary manufacturer's file
- Ability to identify and flag FWD drops with anomalies
- Allows for editing FWD data and deleting FWD test points
- Ability to perform backcalculation analysis in single analysis or batch analysis modes
- Includes graphs of backcalculated modulus values and measured and calculated deflections
- Calculates and displays Load Transfer Efficiency (LTE)
- LEAF analysis to calculate pavement responses under aircraft gear or FWD load

BAKFAA Modular Interface

BAKFAA Modular Interface

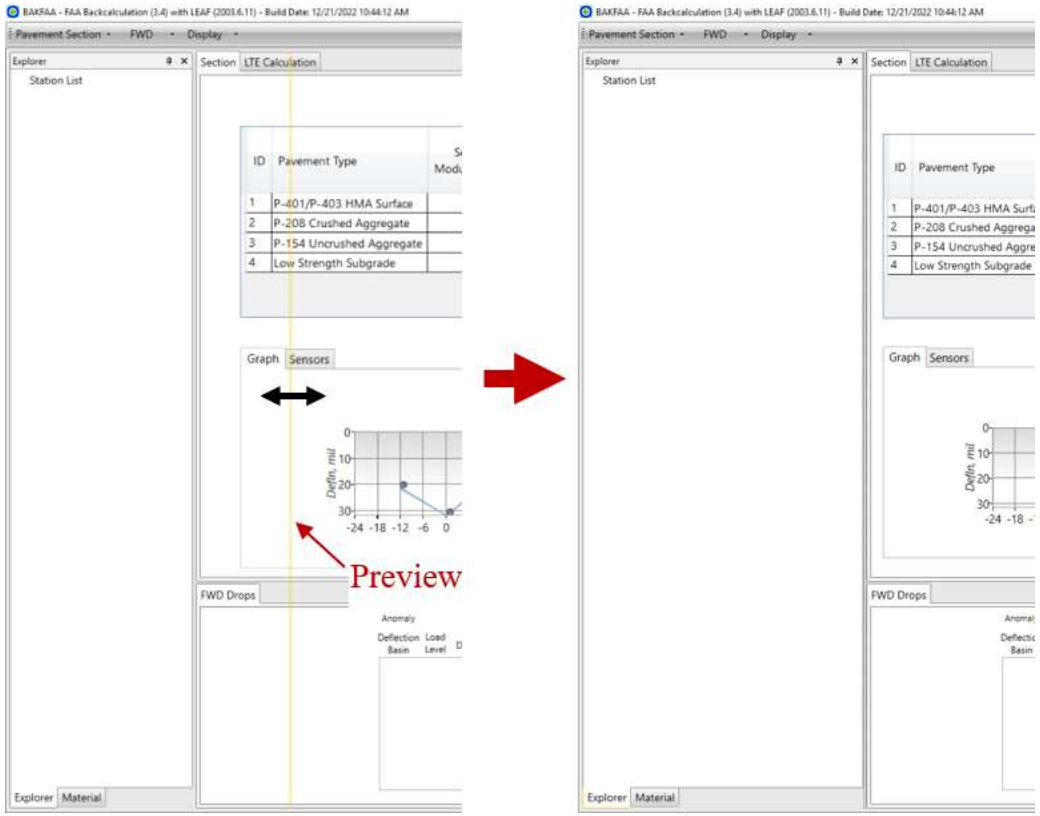
- Graphical user interface was overhauled
- Program functions are divided into ‘modules’ that serve a specific purpose
- Modules are user-configurable: resize, reposition, or remove as desired
- Main program controls relocated and organized in a toolbar
- BAKFAA Modules:
  - Section
  - FWD Drops/Select Load
  - Design Options
  - LTE Calculation
  - (FWD) Explorer
  - Material
  - Station Details

BAKFAA Modular Interface



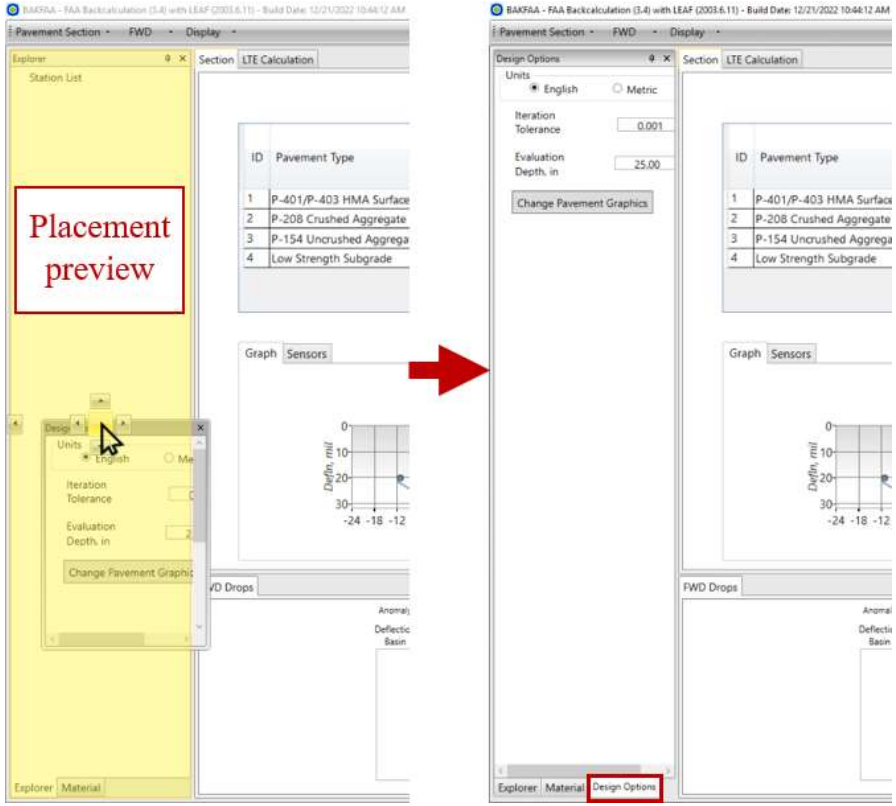
BAKFAA Modular Interface

Module Adjustment



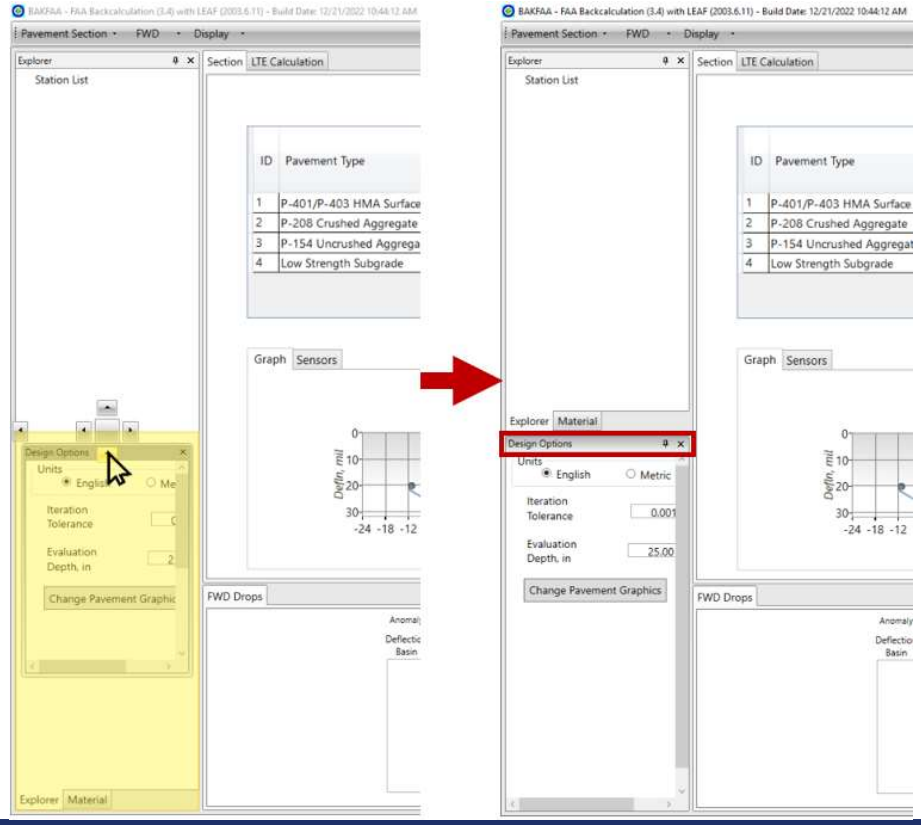
BAKFAA Modular Interface

Module Adjustment



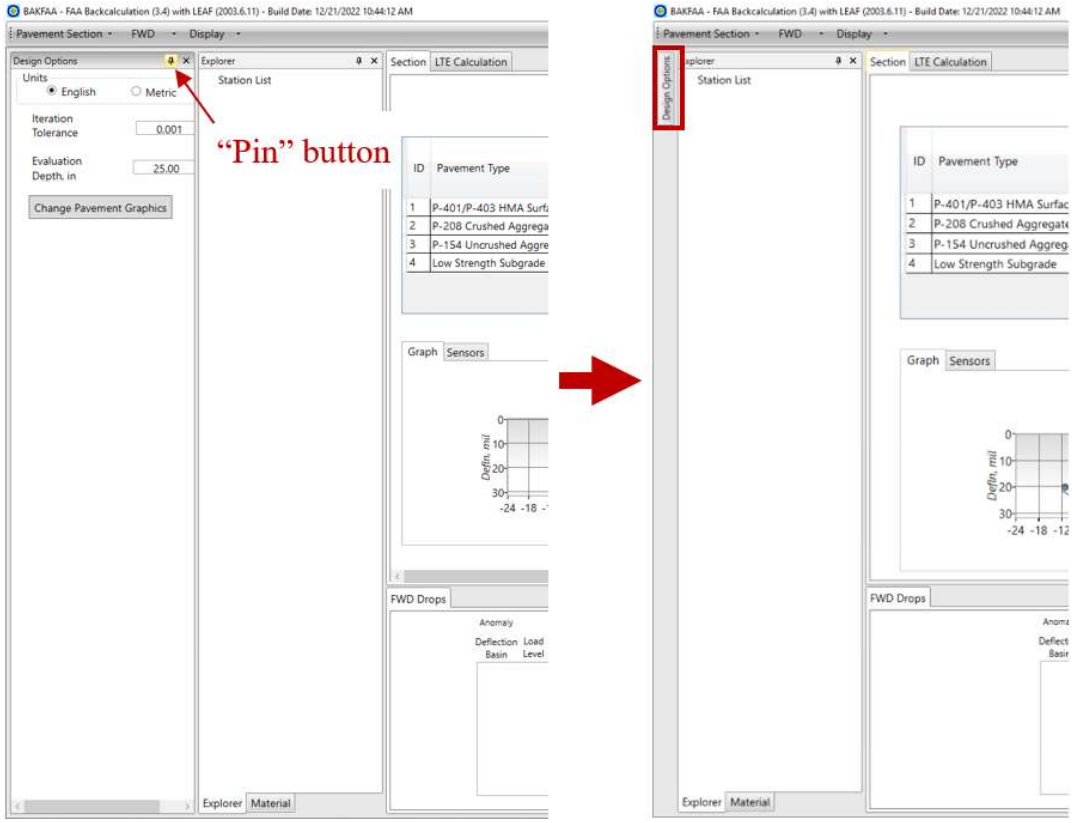
BAKFAA Modular Interface

Module Adjustment



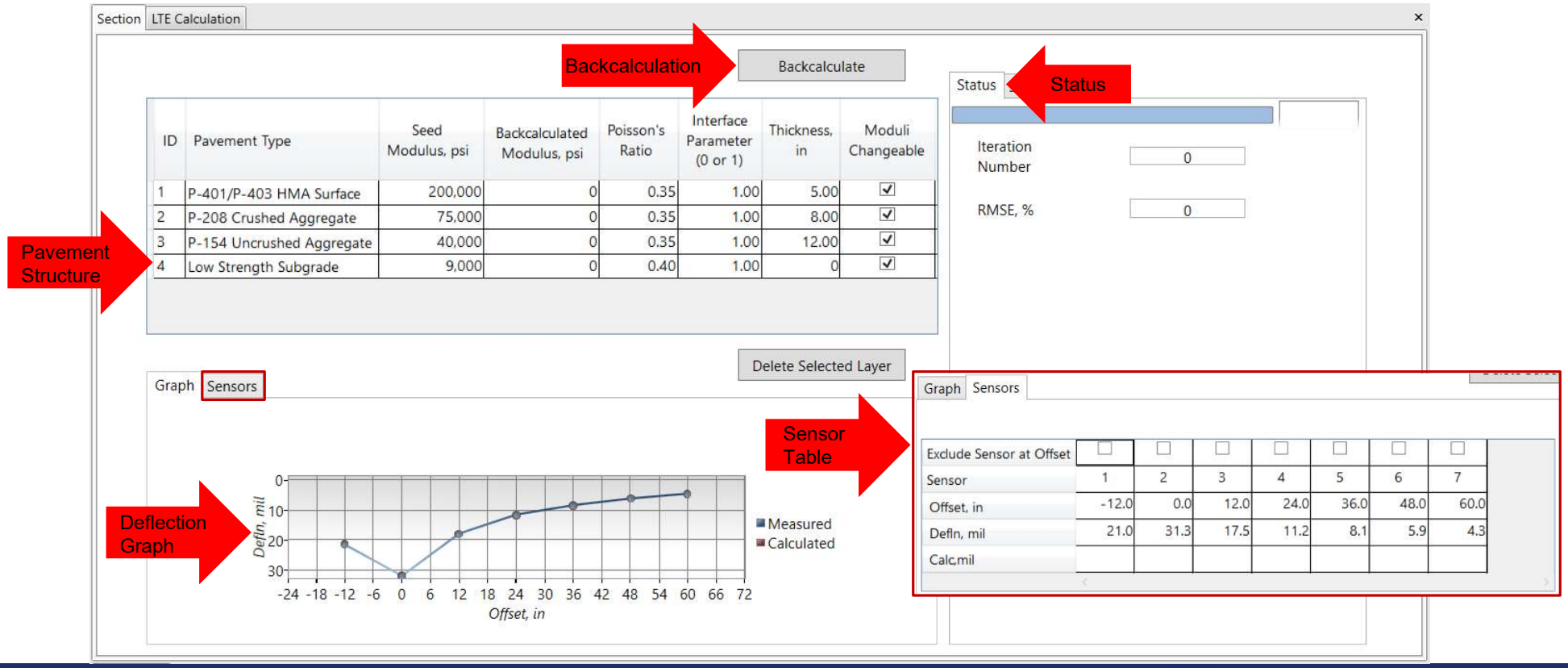
BAKFAA Modular Interface

Module Adjustment



BAKFAA Modular Interface

Section Module

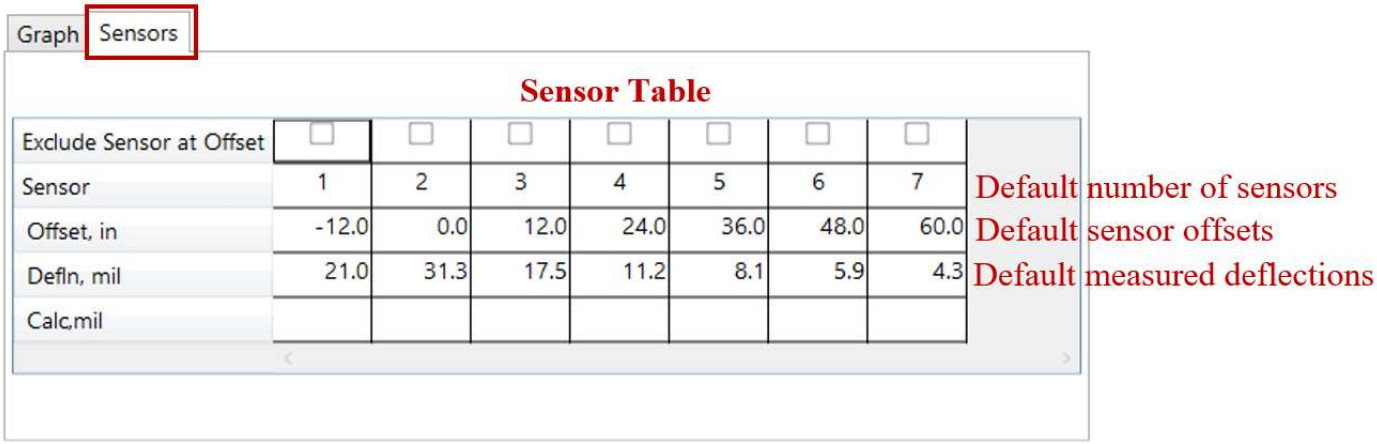


FWD Data

FWD Data

FWD Test Information

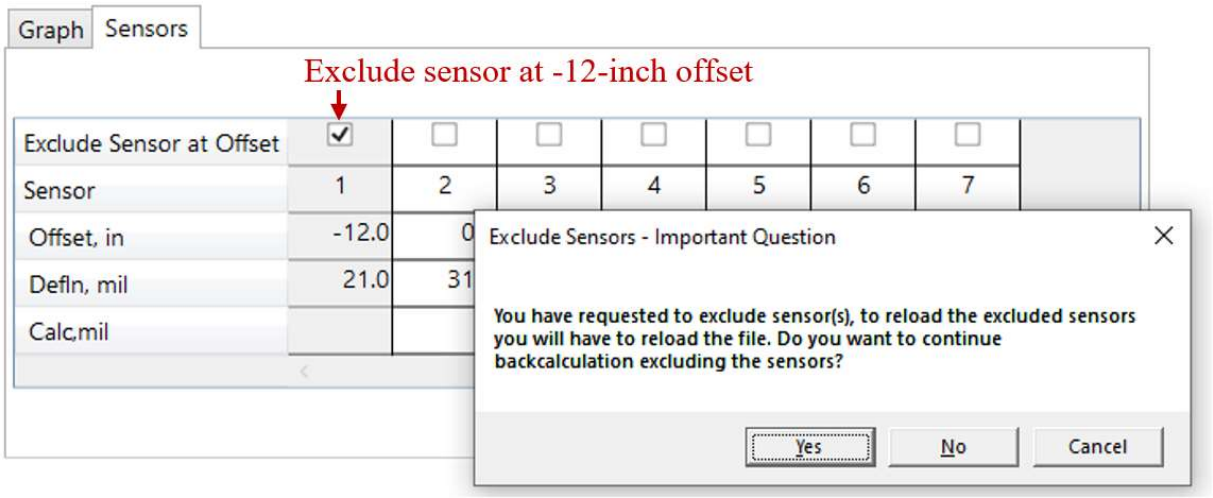
- Sensor: sensor number
- Offset: sensor offset from FWD load plate
- Defln: measured FWD sensor deflection



FWD Data

FWD Information updates after Backcalculation Analysis

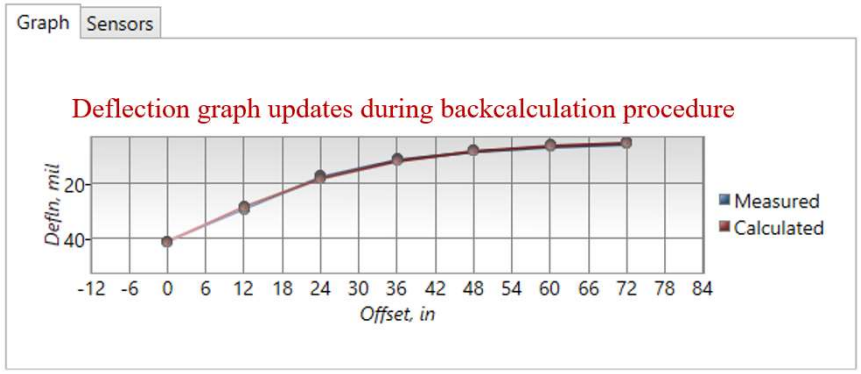
- Exclude Sensor at Offset: allows user to exclude any of the sensors in backcalculation analysis.

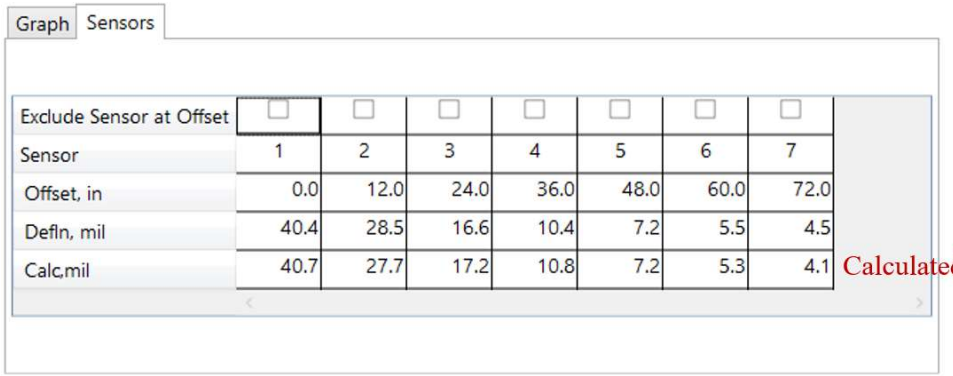


FWD Data

FWD Information updates after Backcalculation Analysis

- Deflection Graph: plot of measured and calculated deflections.
- Calc: calculated deflections. Will update after the backcalculation is done.

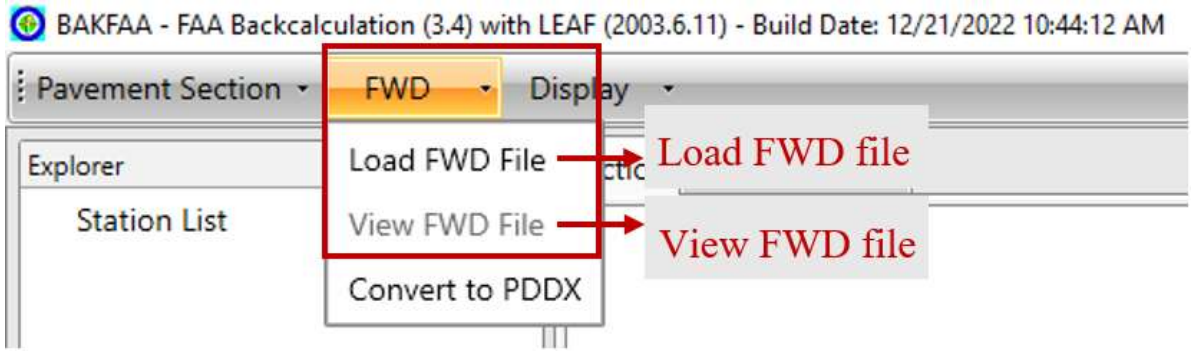




FWD Data

Load/View FWD File

- Load FWD File: import FWD file
- View FWD File: open FWD file viewer



FWD Data

FWD File Formats

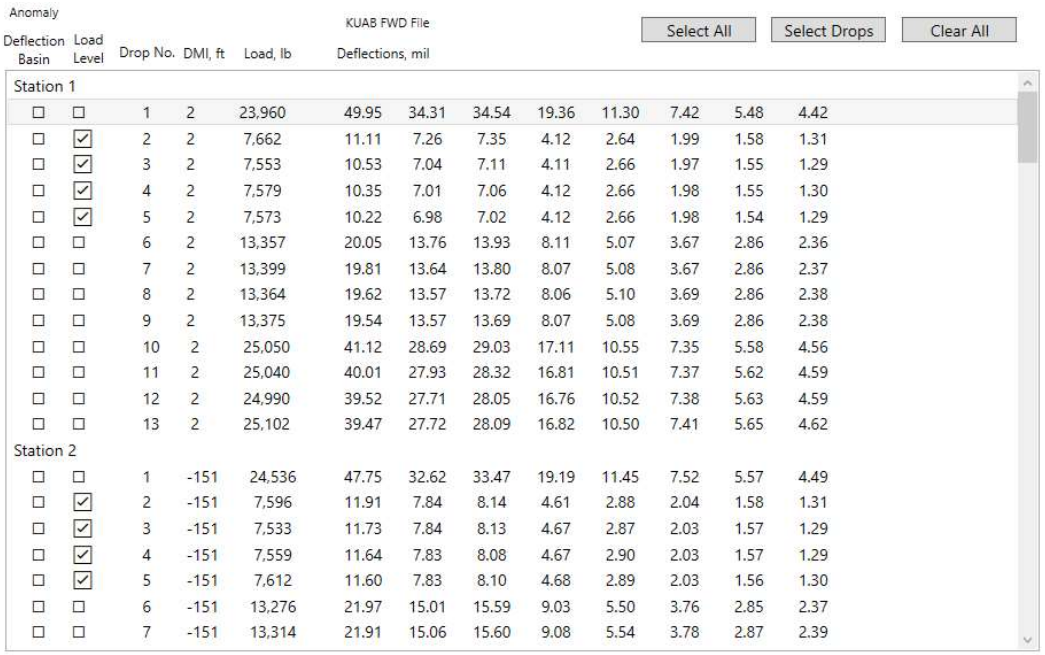
- FWD File Formats:

Equipment Manufacturer	Associated Data Format	File Extensions
Dynatest	R80, R32	.fwd, .hwd
	F20, F25	.fwd, .F20, .F25
	PDDX, XML	.ddx, .xml
KUAB	KUAB	.fwd, .hwd
JILS	M3, M5	.fwd, .hwd, .dat
PaveTest	PaveTest	.csv
CarlBro	CarlBro	.fwd, .txt
AASHTO	Pavement Deflection Data Exchange	.ddx

FWD Data

FWD File Information

- Anomaly Check: FWD drops are flagged if anomalies are identified.
  - Load levels are flagged if FWD plate load is smaller than 9,900 lbs. or larger than 60,000 lbs. (44 kN – 267 kN)
  - Deflection basins are flagged if the recorded deflections are one of the following:
    - Showing a non-decreasing sequence
    - Larger than 100 mils (2540 mc)
    - Zero

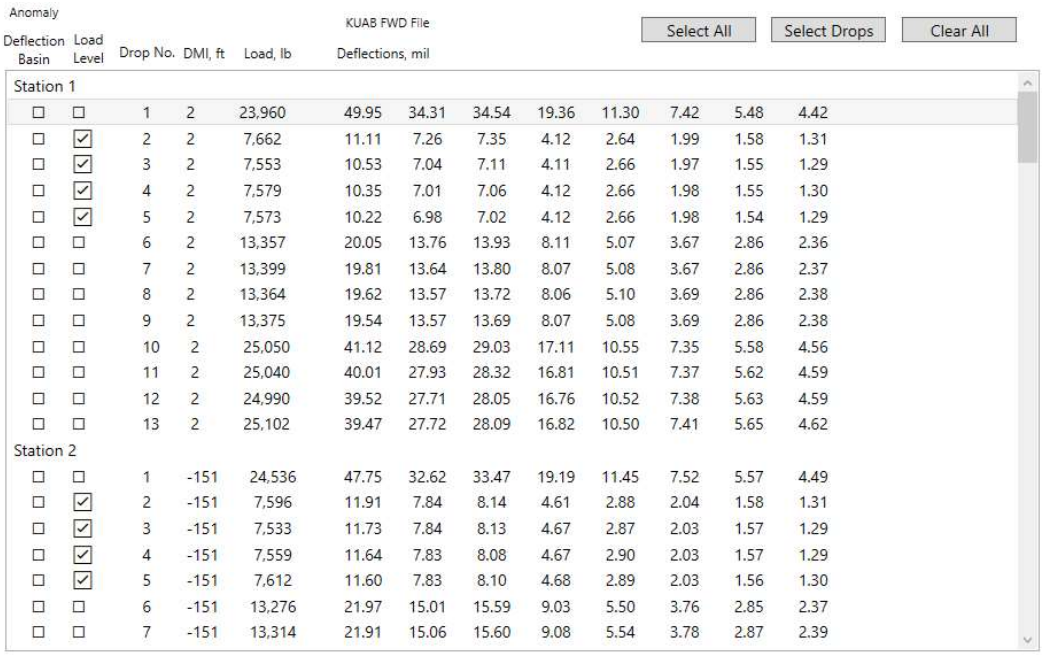


Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37
☐	☐	7	-151	13,314	21.91	15.06	15.60	9.08	5.54	3.78	2.87	2.39

FWD Data

FWD File Information

- Drop No: Number of FWD drops at each test point.
- DMI: the station or location of each FWD test point.
- Load: the applied FWD load at each drop.
- Deflections: measured deflections by each FWD sensor.
- Station: operator comments are identified and replaced with a station number.



Select All | Select Drops | Clear All

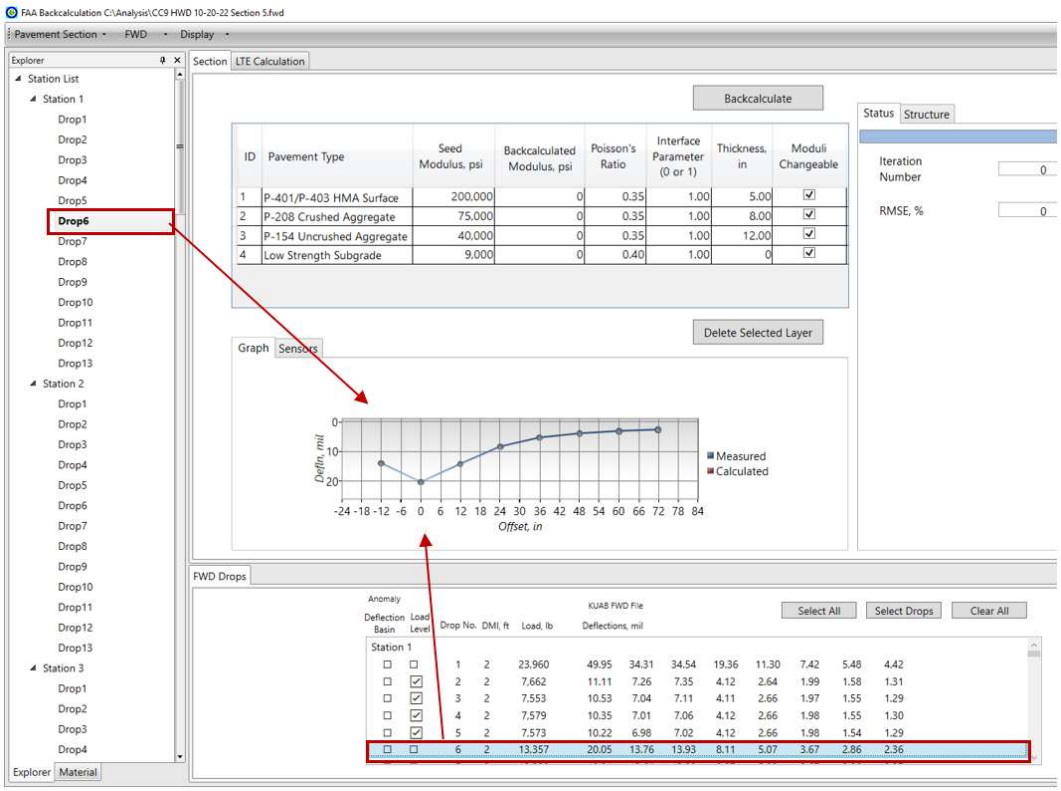
KUAB FWD File

Anomaly												
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37
☐	☐	7	-151	13,314	21.91	15.06	15.60	9.08	5.54	3.78	2.87	2.39

FWD Data

FWD File Information

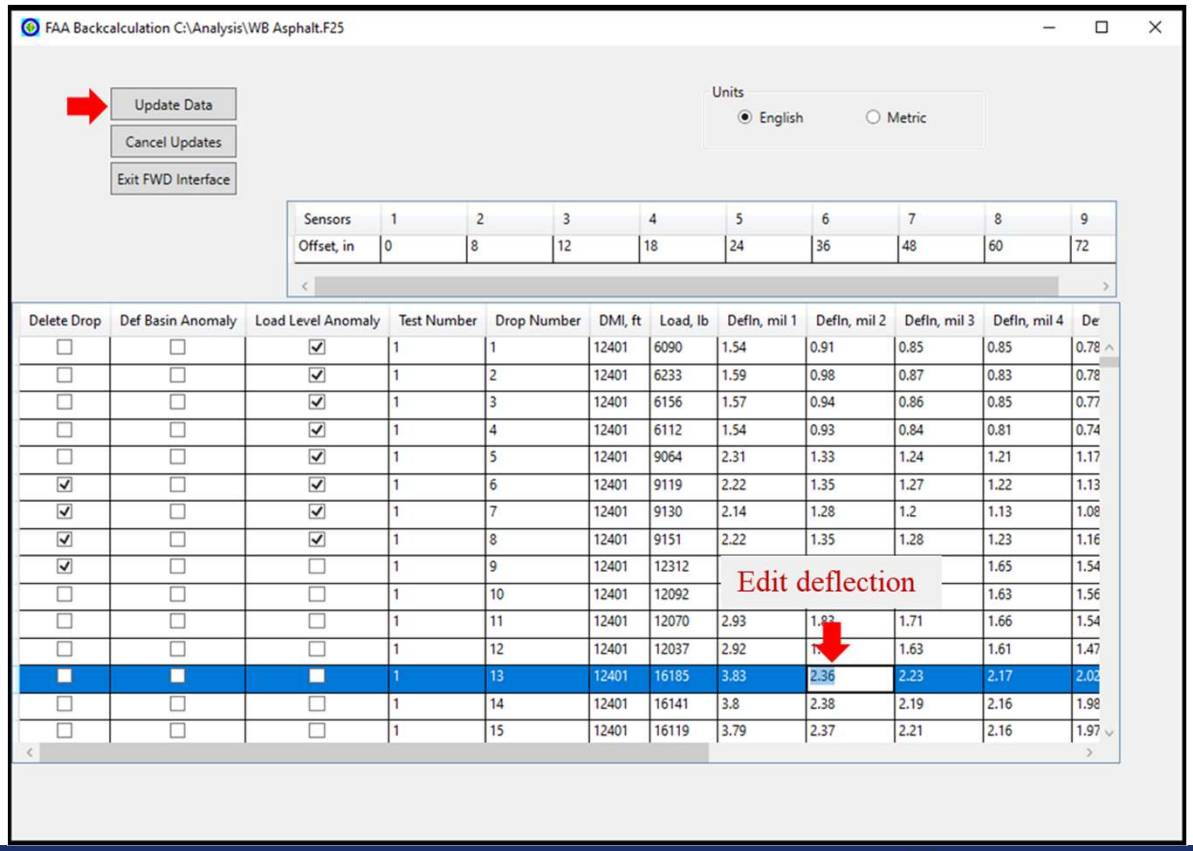
- Deflection information, deflection graph and sensor table will update as the user clicks on an FWD drop.



FWD Data

View FWD File

- Edit sensor data



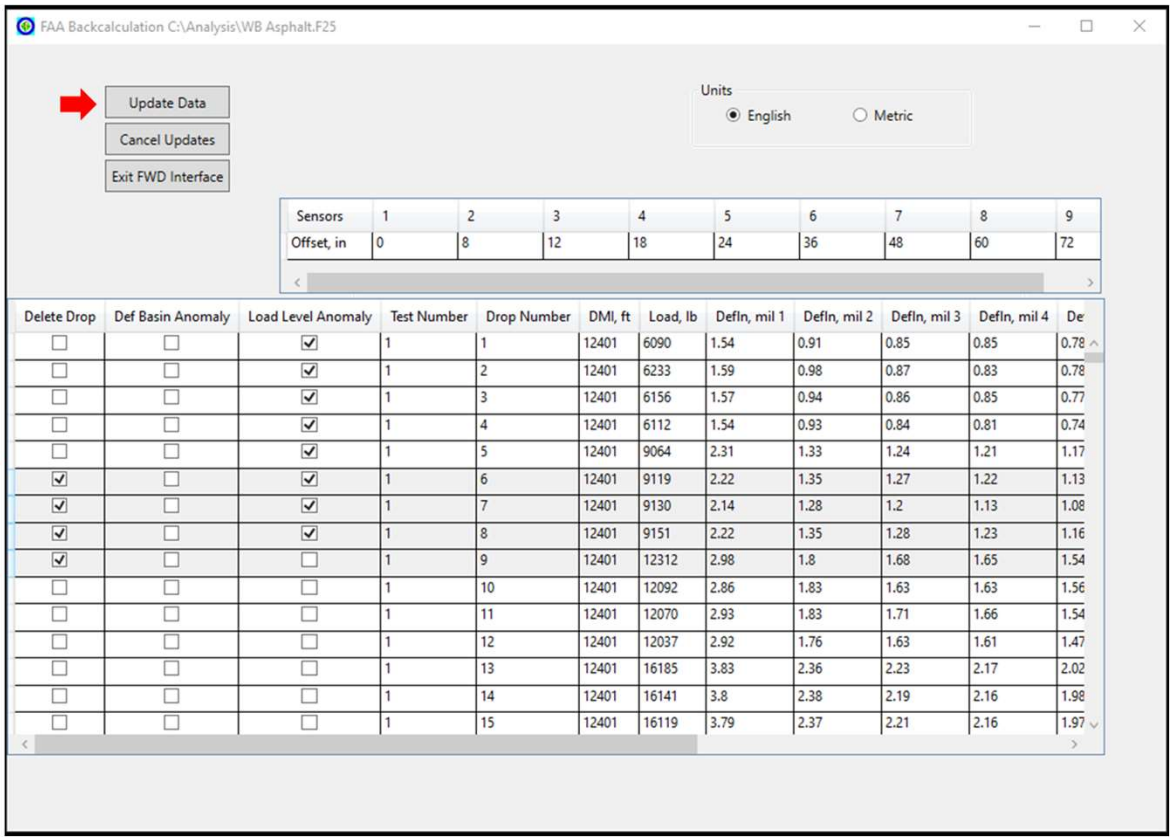
Sensors	1	2	3	4	5	6	7	8	9
Offset, in	0	8	12	18	24	36	48	60	72

Delete Drop	Def Basin Anomaly	Load Level Anomaly	Test Number	Drop Number	DMI, ft	Load, lb	Defln, mil 1	Defln, mil 2	Defln, mil 3	Defln, mil 4	De
☐	☐	☑	1	1	12401	6090	1.54	0.91	0.85	0.85	0.78
☐	☐	☑	1	2	12401	6233	1.59	0.98	0.87	0.83	0.78
☐	☐	☑	1	3	12401	6156	1.57	0.94	0.86	0.85	0.77
☐	☐	☑	1	4	12401	6112	1.54	0.93	0.84	0.81	0.74
☐	☐	☑	1	5	12401	9064	2.31	1.33	1.24	1.21	1.17
☑	☐	☑	1	6	12401	9119	2.22	1.35	1.27	1.22	1.13
☑	☐	☑	1	7	12401	9130	2.14	1.28	1.2	1.13	1.08
☑	☐	☑	1	8	12401	9151	2.22	1.35	1.28	1.23	1.16
☑	☐	☐	1	9	12401	12312				1.65	1.54
☐	☐	☐	1	10	12401	12092				1.63	1.56
☐	☐	☐	1	11	12401	12070	2.93	1.83	1.71	1.66	1.54
☐	☐	☐	1	12	12401	12037	2.92	1.8	1.63	1.61	1.47
☐	☐	☐	1	13	12401	16185	3.83	2.36	2.23	2.17	2.02
☐	☐	☐	1	14	12401	16141	3.8	2.38	2.19	2.16	1.98
☐	☐	☐	1	15	12401	16119	3.79	2.37	2.21	2.16	1.97

FWD Data

View FWD File

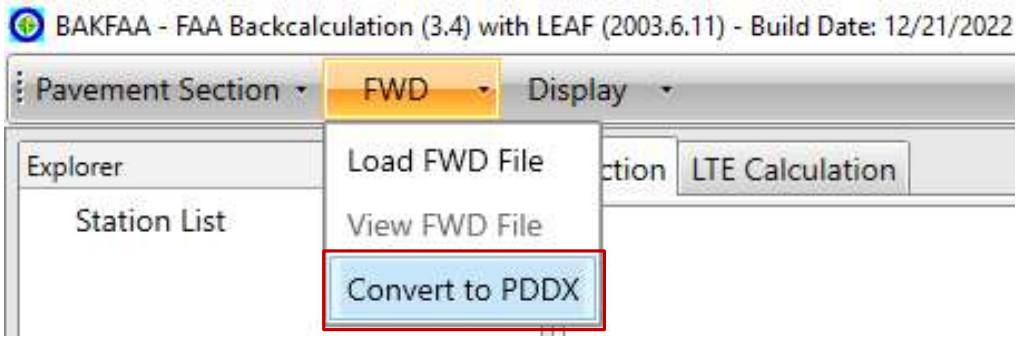
- Delete drops



FWD Data

Convert FWD File to PDDX Format

- Convert FWD file format supported by BAKFAA to Pavement Deflection Data Exchange (PDDX) format file
- PDDX format specification has a modular structure in which data elements of similar characteristics are grouped in individual sections.
- The PDDX files can be loaded to BAKFAA



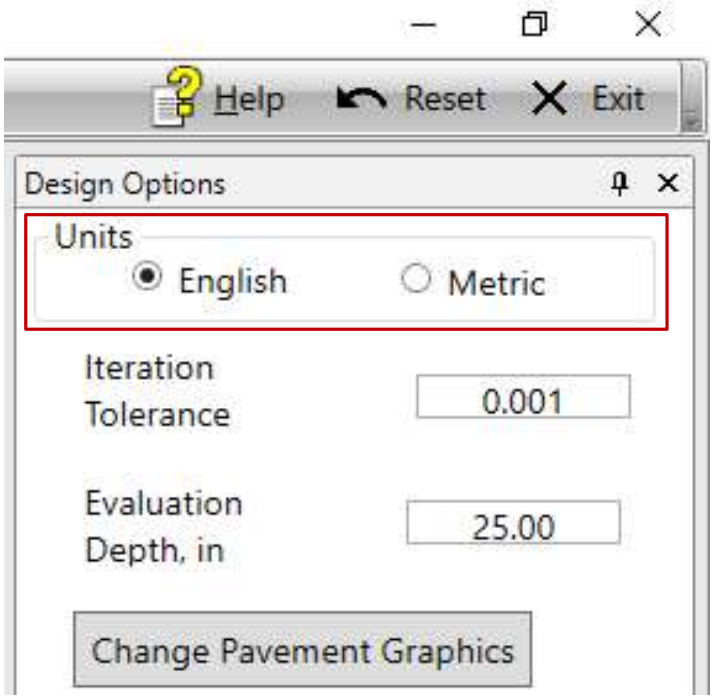
FWD Data

FWD File Parser

- A file processing class used to read FWD files.
- This class uses the file format specifications from FWD manufacturers and generates a set of properties to retrieve data required for functions in the backcalculation and LEAF analysis.
- The advantage of this design is that when a new data format is introduced by a manufacturer, only a file specification needs to be created for the new layout.
- BAKFAA will read the specification and parses the file without requiring any reprograming in the source code.

Units

- BAKFAA can display values in English or metric units
- Radio button under Design Options toggles between the options
- Selection updates all FWD and pavement information, and dictates the units used in the output file



Units

Entity	Unit	
	English	Metric
Pavement Layer Thickness	Inch (in.)	Millimeter (mm)
Distance	Feet (ft.)	Meter (m)
FWD Sensor Offset	Inch (in.)	Millimeter (mm)
FWD Sensor Deflection	Mil	Micron (mc)
FWD Plate Radius	Inch (in.)	Millimeter (mm)
FWD Plate Contact Area	Square inch (in^2)	Square millimeter (mm^2)
FWD Plate Contact Pressure	Pound per square inch (psi)	Megapascal (MPa)
Load	Pound (lbs.)	Kilonewton (kN)
Modulus	Pound per square inch (psi)	Megapascal (MPa)

Units

English

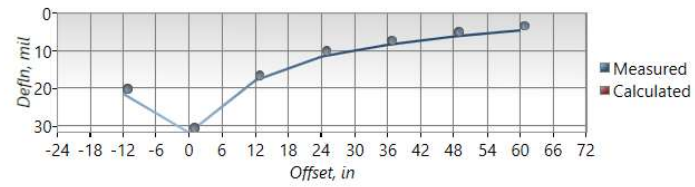
Section | LTE Calculation

Backcalculate

ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

Delete Selected Layer

Graph | Sensors



Metric

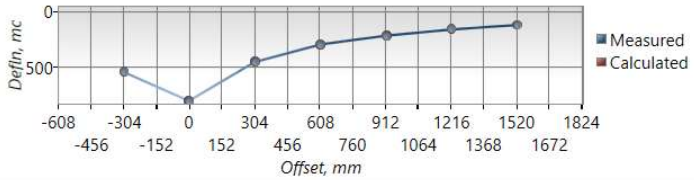
Section | LTE Calculation

Backcalculate

ID	Pavement Type	Seed Modulus, MPa	Backcalculated Modulus, MPa	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, mm	Moduli Changeable
1	P-401/P-403 HMA Surface	1,379	0	0.35	1.00	127	☑
2	P-208 Crushed Aggregate	517	0	0.35	1.00	203	☑
3	P-154 Uncrushed Aggregate	276	0	0.35	1.00	305	☑
4	Low Strength Subgrade	62	0	0.40	1.00	0	☑

Delete Selected Layer

Graph | Sensors



Units

English

Graph | Sensors

Exclude Sensor at Offset	☐	☐	☐	☐	☐	☐	☐
Sensor	1	2	3	4	5	6	7
Offset, in	0.0	12.0	24.0	36.0	48.0	60.0	72.0
Defln, mil	20.1	13.9	8.1	5.1	3.7	2.9	2.4
Calc,mil	20.2	13.5	8.4	5.3	3.6	2.7	2.2

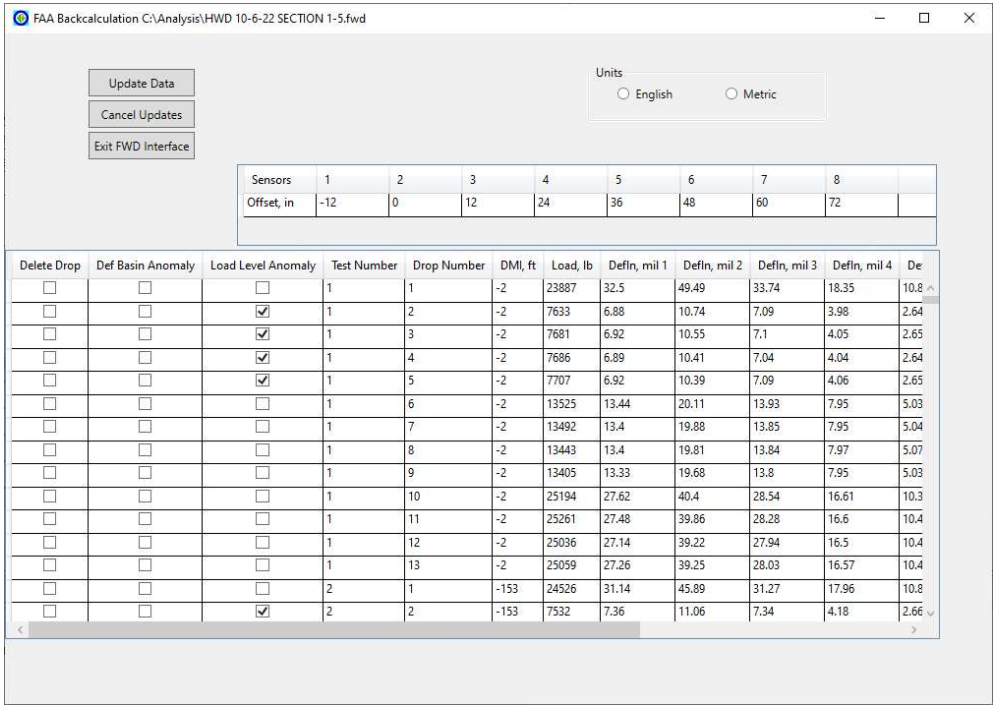
Metric

Graph | Sensors

Exclude Sensor at Offset	☐	☐	☐	☐	☐	☐	☐
Sensor	1	2	3	4	5	6	7
Offset, mm	0	305	610	914	1,219	1,524	1,829
Defln, mc	509	354	206	129	93	73	60
Calc, mc	513	343	213	136	92	69	55

Units

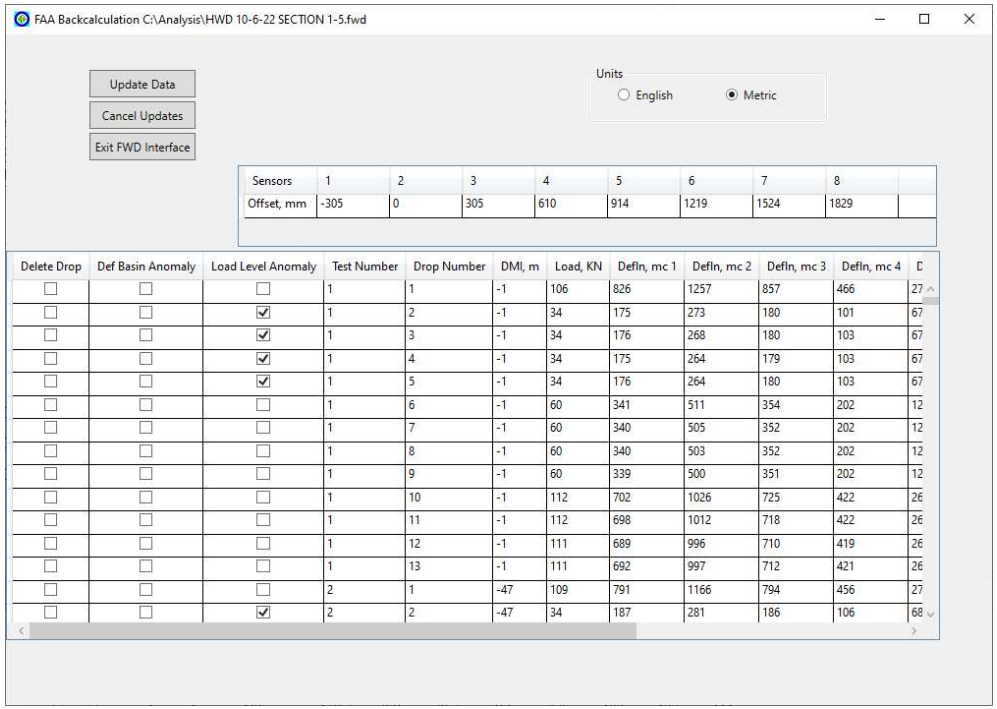
English



Sensors	1	2	3	4	5	6	7	8
Offset, in	-12	0	12	24	36	48	60	72

Delete Drop	Def Basin Anomaly	Load Level Anomaly	Test Number	Drop Number	DMI, ft	Load, lb	Defln, mil 1	Defln, mil 2	Defln, mil 3	Defln, mil 4	De
☐	☐	☐	1	1	-2	23887	32.5	49.49	33.74	18.35	10.8
☐	☐	☑	1	2	-2	7633	6.88	10.74	7.09	3.98	2.64
☐	☐	☑	1	3	-2	7681	6.92	10.55	7.1	4.05	2.65
☐	☐	☑	1	4	-2	7686	6.89	10.41	7.04	4.04	2.64
☐	☐	☑	1	5	-2	7707	6.92	10.39	7.09	4.06	2.65
☐	☐	☐	1	6	-2	13525	13.44	20.11	13.93	7.95	5.03
☐	☐	☐	1	7	-2	13492	13.4	19.88	13.85	7.95	5.04
☐	☐	☐	1	8	-2	13443	13.4	19.81	13.84	7.97	5.07
☐	☐	☐	1	9	-2	13405	13.33	19.68	13.8	7.95	5.03
☐	☐	☐	1	10	-2	25194	27.62	40.4	28.54	16.61	10.3
☐	☐	☐	1	11	-2	25261	27.48	39.86	28.28	16.6	10.4
☐	☐	☐	1	12	-2	25036	27.14	39.22	27.94	16.5	10.4
☐	☐	☐	1	13	-2	25059	27.26	39.25	28.03	16.57	10.4
☐	☐	☐	2	1	-153	24526	31.14	45.89	31.27	17.96	10.8
☐	☐	☑	2	2	-153	7532	7.36	11.06	7.34	4.18	2.66

Metric



Sensors	1	2	3	4	5	6	7	8
Offset, mm	-305	0	305	610	914	1219	1524	1829

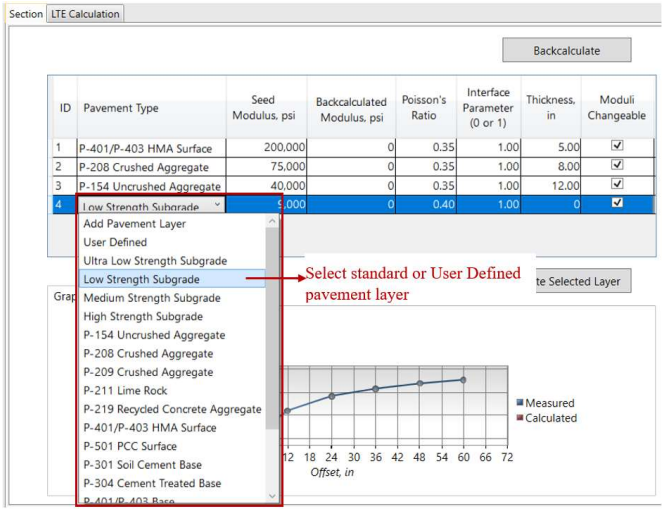
Delete Drop	Def Basin Anomaly	Load Level Anomaly	Test Number	Drop Number	DMI, m	Load, KN	Defln, mc 1	Defln, mc 2	Defln, mc 3	Defln, mc 4	D
☐	☐	☐	1	1	-1	106	826	1257	857	466	27
☐	☐	☑	1	2	-1	34	175	273	180	101	67
☐	☐	☑	1	3	-1	34	176	268	180	103	67
☐	☐	☑	1	4	-1	34	175	264	179	103	67
☐	☐	☑	1	5	-1	34	176	264	180	103	67
☐	☐	☐	1	6	-1	60	341	511	354	202	12
☐	☐	☐	1	7	-1	60	340	505	352	202	12
☐	☐	☐	1	8	-1	60	340	503	352	202	12
☐	☐	☐	1	9	-1	60	339	500	351	202	12
☐	☐	☐	1	10	-1	112	702	1026	725	422	26
☐	☐	☐	1	11	-1	112	698	1012	718	422	26
☐	☐	☐	1	12	-1	111	689	996	710	419	26
☐	☐	☐	1	13	-1	111	692	997	712	421	26
☐	☐	☐	2	1	-47	109	791	1166	794	456	27
☐	☐	☑	2	2	-47	34	187	281	186	106	68

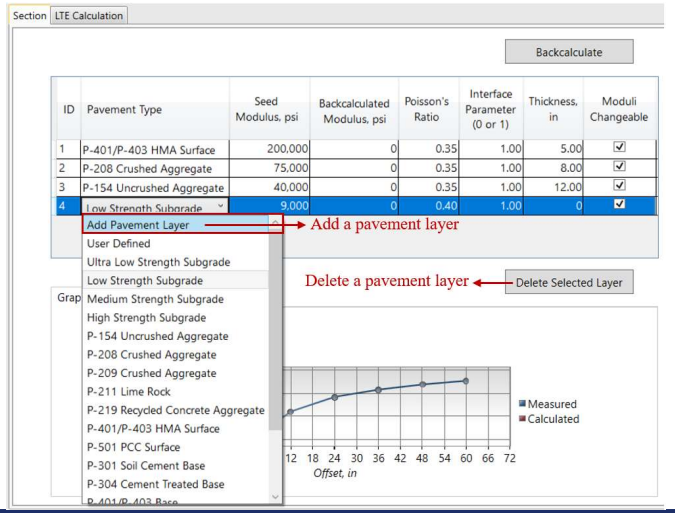
BAKFAA Inputs

BAKFAA INPUTS

Pavement Layer Properties

- Pavement Layer Type:
  - Minimum 3 layers and maximum 6 layers
  - Select layer type from drop down menu.
  - Library of standard FAA materials (AC 150/5370-10)
  - Four subgrade types with different strengths
  - "User Defined" layer type
  - Add or remove a layer from any location in the structure.
  - Last layer is a semi-infinite layer

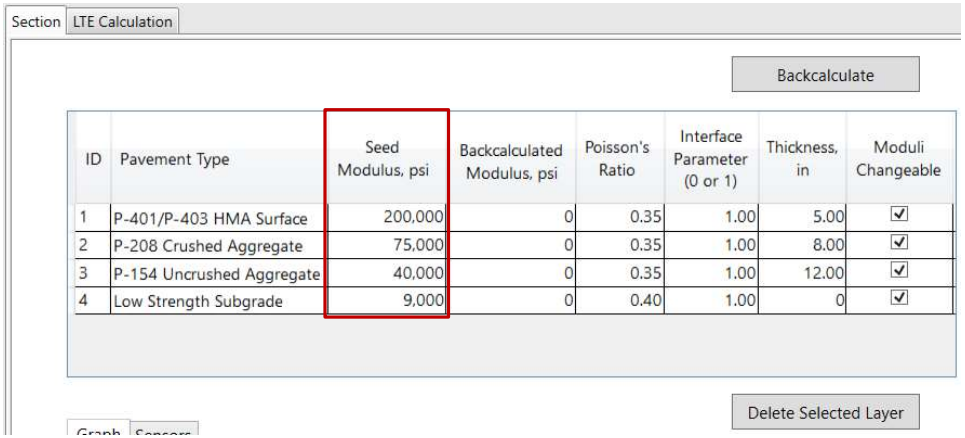




BAKFAA INPUTS

Pavement Layer Properties

- Seed Modulus:
  - Initial modulus value used in backcalculation iterative procedure
  - BAKFAA has default seed modulus values for FAA standard materials
  - Default seed modulus values are editable, but BAKFAA will automatically change the pavement type to "User Defined"

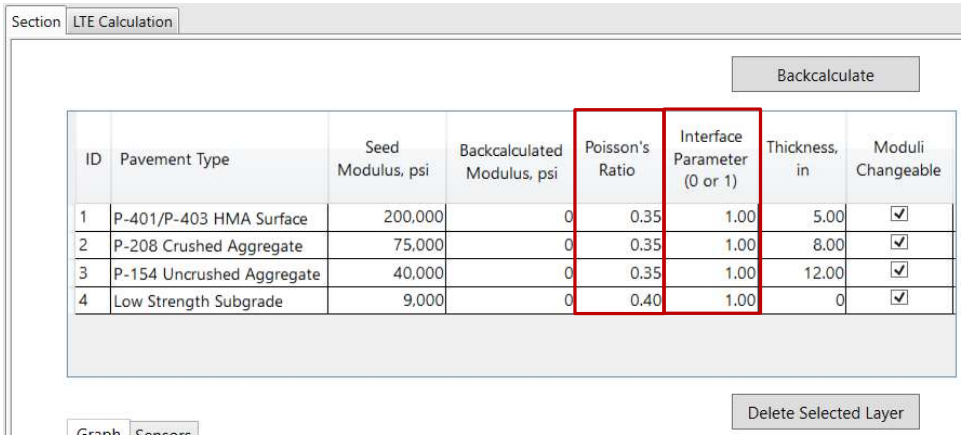


ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

BAKFAA INPUTS

Pavement Layer Properties

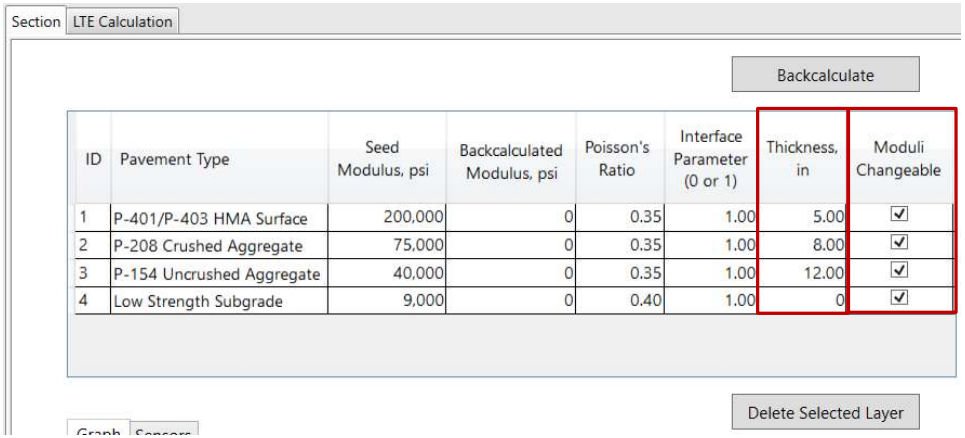
- Poisson's Ratio:
  - BAKFAA has default Poisson's ratio values for FAA standard materials
  - Numerical value between 0.0 to 0.5
- Interface Parameter:
  - Accepts values of 0 or 1
  - 0 indicates no bond
  - 1 indicates fully bonded layers



BAKFAA INPUTS

Pavement Layer Properties

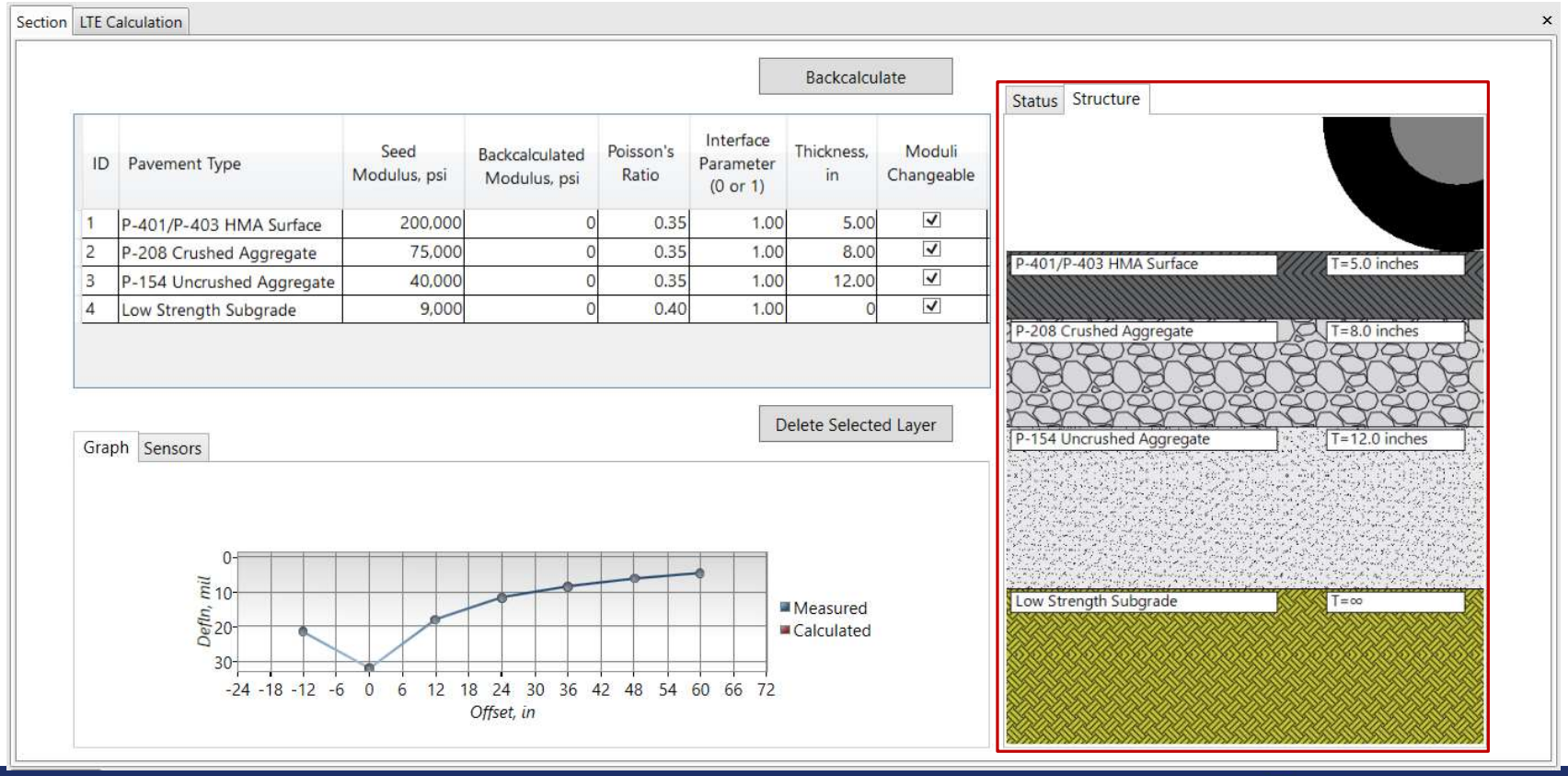
- Thickness:
  - BAKFAA requires a non-zero thickness for each pavement layer
  - The thickness of the last layer in a pavement structure must be zero to represent a semi-infinite layer
- Moduli Changeable:
  - Determines whether the layer modulus will change in the backcalculation analysis
  - Checkbox toggled off indicates the layer modulus will remain fixed in the backcalculation analysis



ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-401/P-403 HMA Surface	200,000	0	0.35	1.00	5.00	☑
2	P-208 Crushed Aggregate	75,000	0	0.35	1.00	8.00	☑
3	P-154 Uncrushed Aggregate	40,000	0	0.35	1.00	12.00	☑
4	Low Strength Subgrade	9,000	0	0.40	1.00	0	☑

BAKFAA INPUTS

Pavement Layer Structure



BAKFAA INPUTS

Backcalculation Input

- Iteration Tolerance:
  - Dimensionless parameter that controls the stoppage criteria in the iterative process of backcalculation analysis
  - At the end of each iteration

$$Err = \sum_{i=1}^{N} (MD_i - CD_i)^2$$

MD_i and CD_i : measured and calculated deflections at sensor i
N : the number of FWD sensors

$$rtol = 2 \times \frac{(Err)_j - (Err)_{j+1}}{(Err)_j + (Err)_{j+1}}$$

$(Err)_j$ and $(Err)_{j+1}$ are the errors at two consecutive iterations j and $j+1$

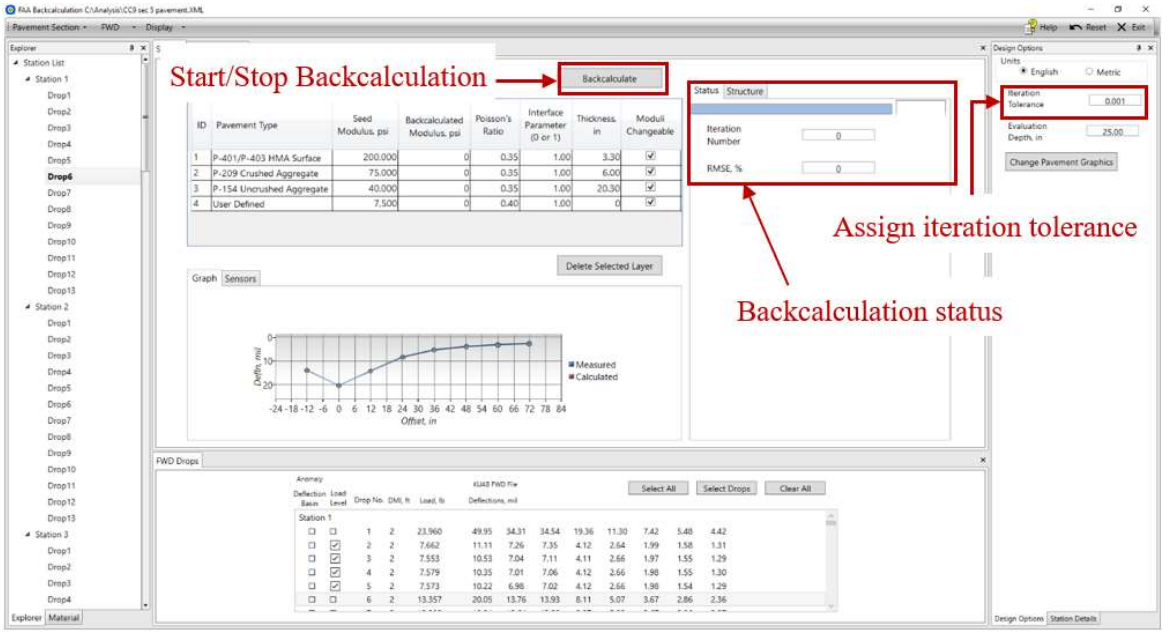
  - BAKFAA stops backcalculation when the *rtol* is less than the user defined Iteration Tolerance
  - Typical range is 0.001-0.1. Iterate through this range to achieve desired accuracy.

Backcalculation Analysis

Backcalculation

Analysis Mode

- Single Analysis Mode: when a single FWD drop is selected. Results will be shown in the program.
- Batch Analysis Mode: when more than one drop is selected. Results will be saved in .csv file.



Backcalculation

Select FWD File

- Select individual drops

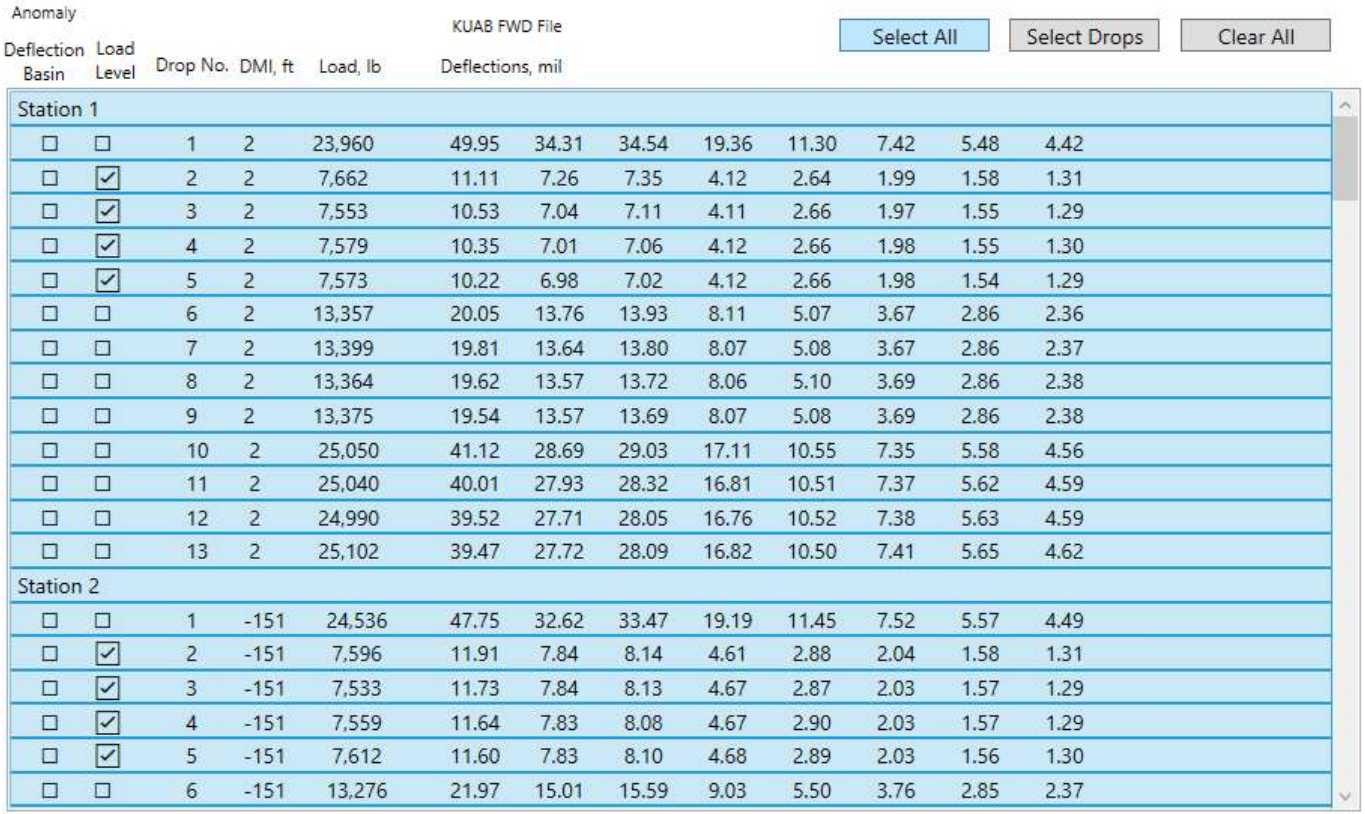


Anomaly Deflection Basin	Anomaly Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62

Backcalculation

Select FWD File

- Select All button



KUAB FWD File

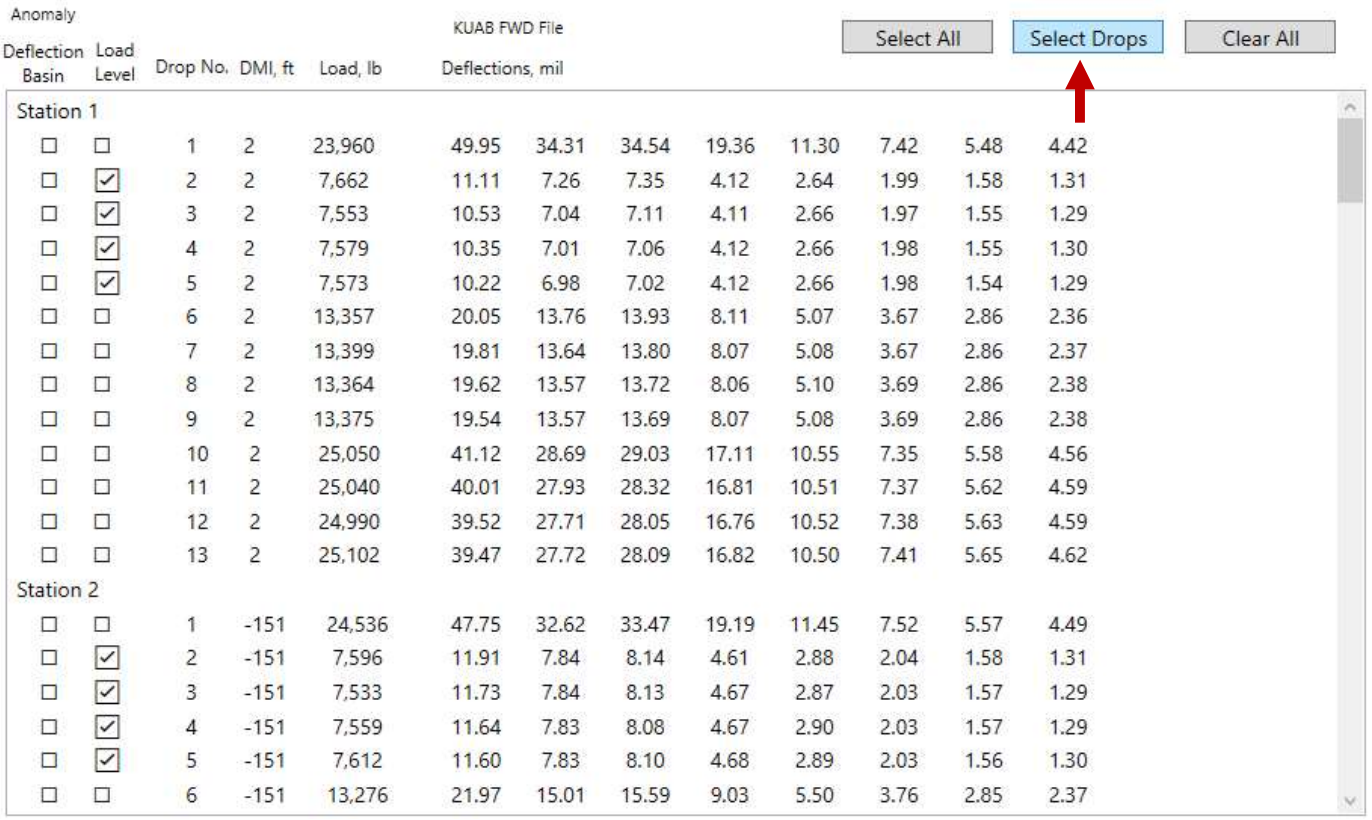
Select All | Select Drops | Clear All

Anomaly Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37

Backcalculation

Select FWD File

- Select Drops button



Anomaly

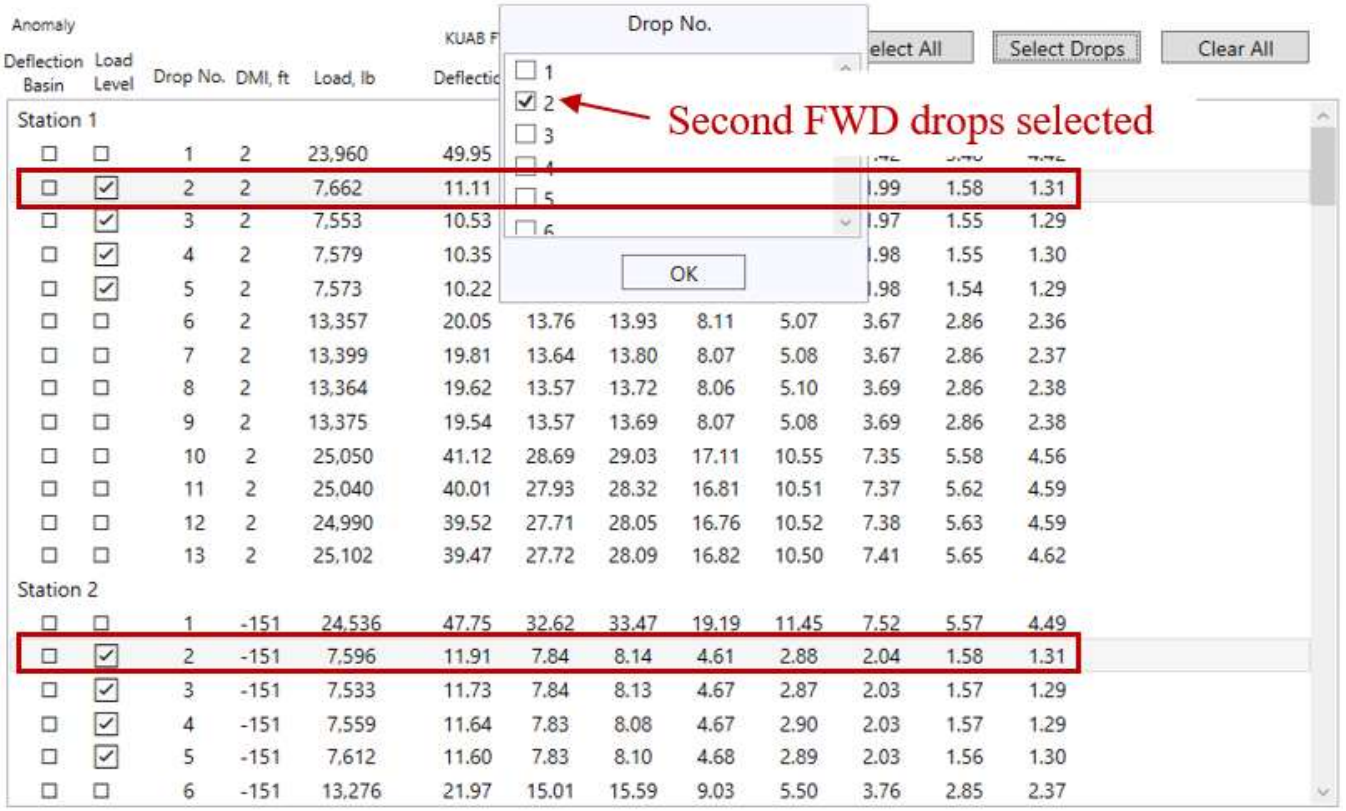
KUAB FWD File

Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil							
Station 1												
☐	☐	1	2	23,960	49.95	34.31	34.54	19.36	11.30	7.42	5.48	4.42
☐	☑	2	2	7,662	11.11	7.26	7.35	4.12	2.64	1.99	1.58	1.31
☐	☑	3	2	7,553	10.53	7.04	7.11	4.11	2.66	1.97	1.55	1.29
☐	☑	4	2	7,579	10.35	7.01	7.06	4.12	2.66	1.98	1.55	1.30
☐	☑	5	2	7,573	10.22	6.98	7.02	4.12	2.66	1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37

Backcalculation

Select FWD File

- Select Drops button
  - Selection menu

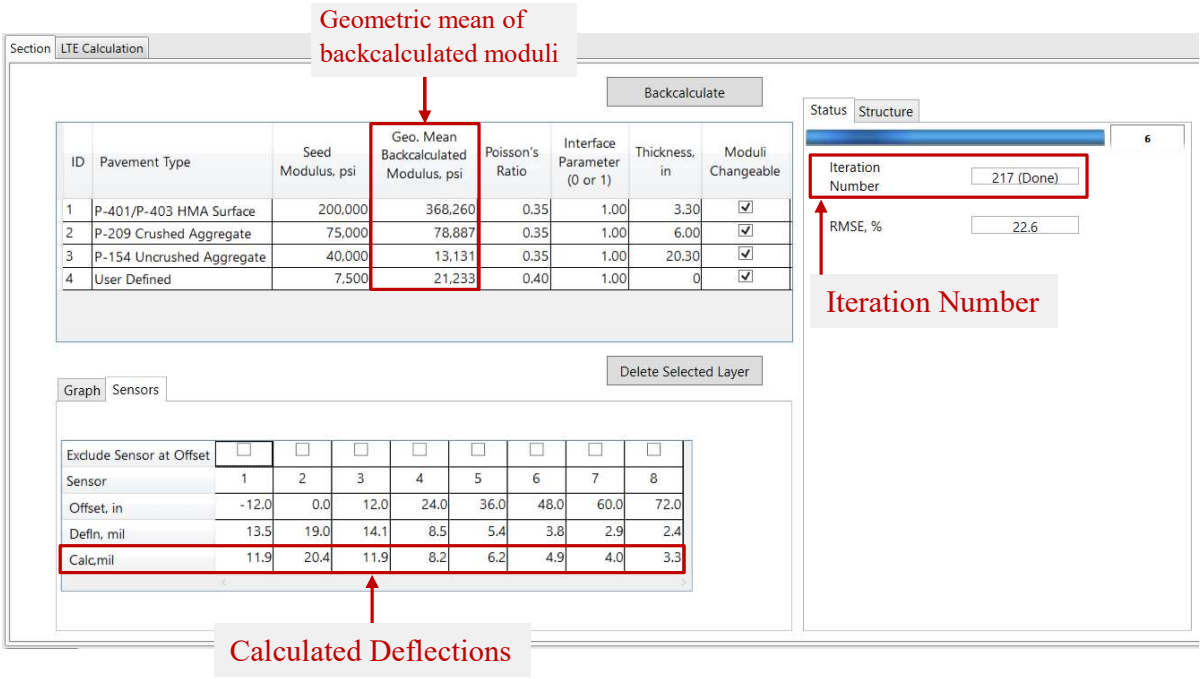


Anomaly Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	KUAB F Deflectic							
Station 1												
☐	☐	1	2	23,960	49.95							
☐	☑	2	2	7,662	11.11						1.99	1.58	1.31
☐	☑	3	2	7,553	10.53						1.97	1.55	1.29
☐	☑	4	2	7,579	10.35						1.98	1.55	1.30
☐	☑	5	2	7,573	10.22						1.98	1.54	1.29
☐	☐	6	2	13,357	20.05	13.76	13.93	8.11	5.07	3.67	2.86	2.36
☐	☐	7	2	13,399	19.81	13.64	13.80	8.07	5.08	3.67	2.86	2.37
☐	☐	8	2	13,364	19.62	13.57	13.72	8.06	5.10	3.69	2.86	2.38
☐	☐	9	2	13,375	19.54	13.57	13.69	8.07	5.08	3.69	2.86	2.38
☐	☐	10	2	25,050	41.12	28.69	29.03	17.11	10.55	7.35	5.58	4.56
☐	☐	11	2	25,040	40.01	27.93	28.32	16.81	10.51	7.37	5.62	4.59
☐	☐	12	2	24,990	39.52	27.71	28.05	16.76	10.52	7.38	5.63	4.59
☐	☐	13	2	25,102	39.47	27.72	28.09	16.82	10.50	7.41	5.65	4.62
Station 2												
☐	☐	1	-151	24,536	47.75	32.62	33.47	19.19	11.45	7.52	5.57	4.49
☐	☑	2	-151	7,596	11.91	7.84	8.14	4.61	2.88	2.04	1.58	1.31
☐	☑	3	-151	7,533	11.73	7.84	8.13	4.67	2.87	2.03	1.57	1.29
☐	☑	4	-151	7,559	11.64	7.83	8.08	4.67	2.90	2.03	1.57	1.29
☐	☑	5	-151	7,612	11.60	7.83	8.10	4.68	2.89	2.03	1.56	1.30
☐	☐	6	-151	13,276	21.97	15.01	15.59	9.03	5.50	3.76	2.85	2.37

Backcalculation

Analysis Output

- Backcalculated Moduli:
  - In batch mode: geometric mean ("Geo. Mean")
- Calculated Deflections:
  - In batch mode: displayed only for the last drop selected
- Iteration Number:
  - In batch mode: displayed only for the last drop selected



Backcalculation

Analysis Output

- Root Mean Square Error (RMSE): between measured and calculated deflections

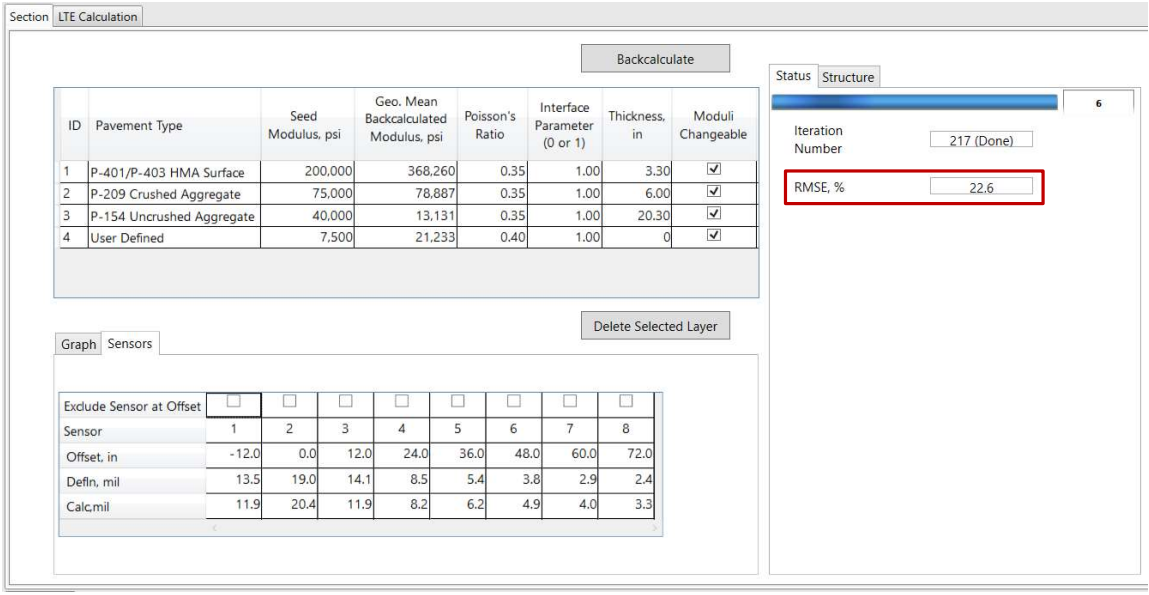
$$RMSE = 100 \times \left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{d_i^c - d_i^m}{d_i^m}\right)^2\right]^{1/2}$$

where:

d_i^c : Calculated deflection for ith sensor

d_i^m : Measured deflection for ith sensor

N: Number of sensors



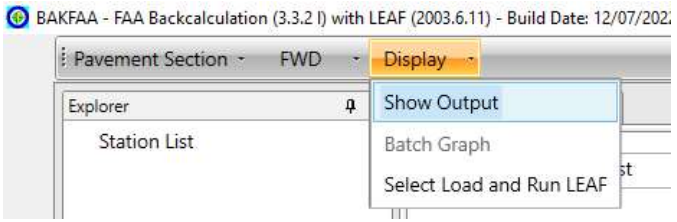
Backcalculation

Analysis Output

- .CSV Output File content:
  - Pavement structure (pavement type, seed modulus, Poisson's ratio, interface parameter, thickness, and moduli changeable)
  - FWD (DMI, drop no., load, and operator comments or remarks)
  - Iteration tolerance
  - Backcalculated modulus
  - RMSE
  - Number of iterations completed
  - LTE values based on the user selection
  - Measured deflections for each selected FWD drop
  - Calculated deflections for each selected FWD drop

Backcalculation

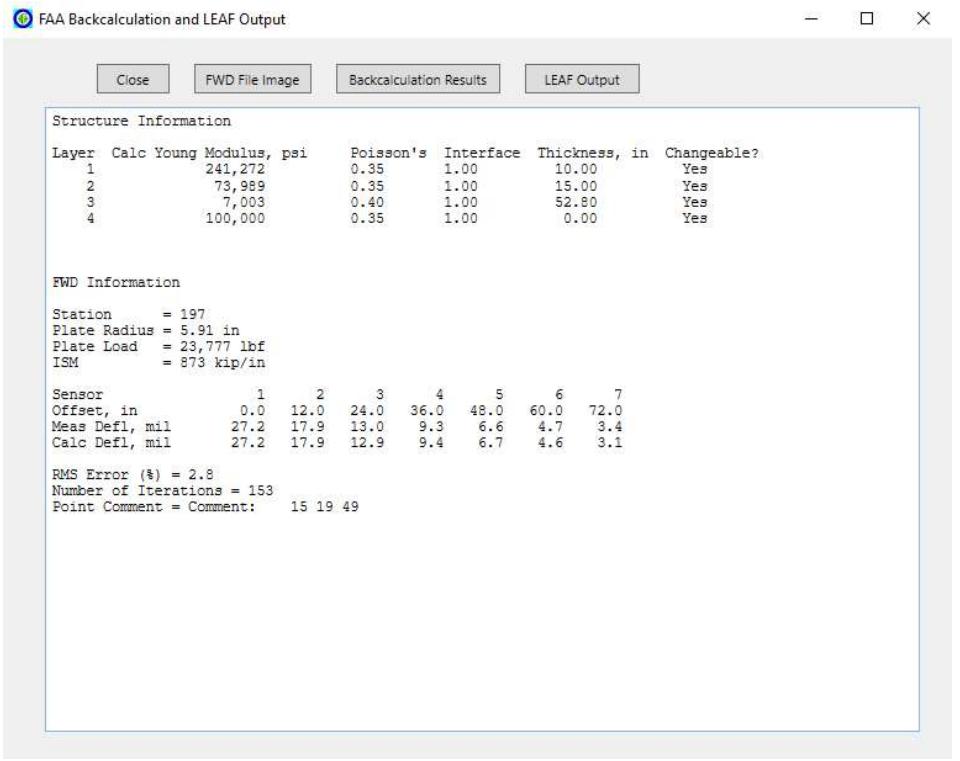
Show Output



- Show Output – Single Analysis Mode
- Displays backcalculation results for that drop, including:
  - Calculated deflections
  - RMSE
  - Iteration number
  - FWD information
  - Impulse Stiffness Modulus ($ISM=L/D_0$)

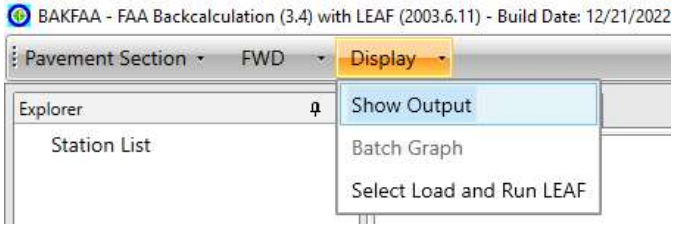
L is the applied FWD load;

D_0 is the deflection under FWD load plate.



Backcalculation

Show Output

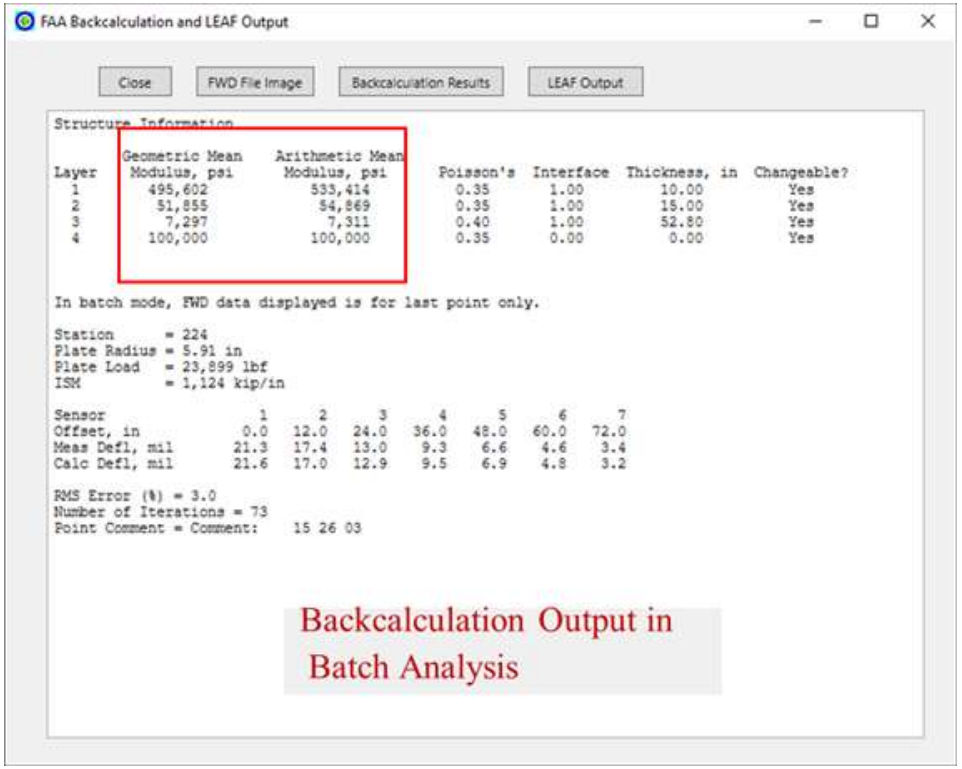


- Show Output – Batch Analysis Mode
  - Backcalculation outputs, FWD information, and ISM are displayed for the last drop selected
  - Arithmetic and geometric means of backcalculated modulus of each layer

$$Arithmetic\ Mean\ (E_1) = \frac{1}{n}(E_{11} + E_{12} + \cdots + E_{1n})$$

$$Geometric\ Mean\ (E_1) = (E_{11} \times E_{12} \times \ldots \times E_{1n})^{1/n}$$

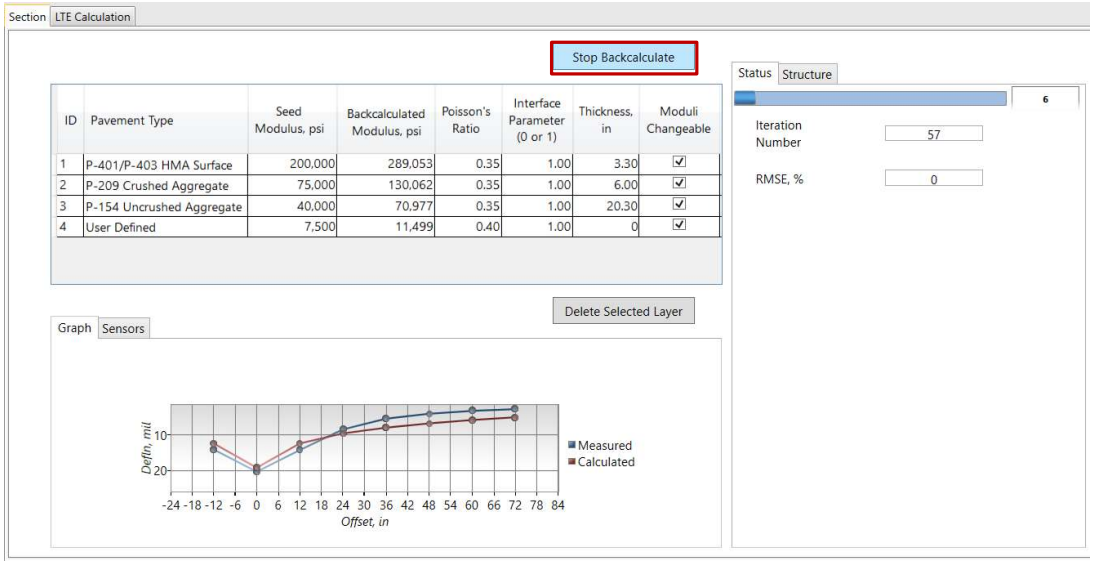
n is the number of test points selected for backcalculation; E_{11}, E_{12} and E_{1n} are the backcalculated modulus of layer 1 at test points 1, 2 and n.



Backcalculation

Stop Backcalculation

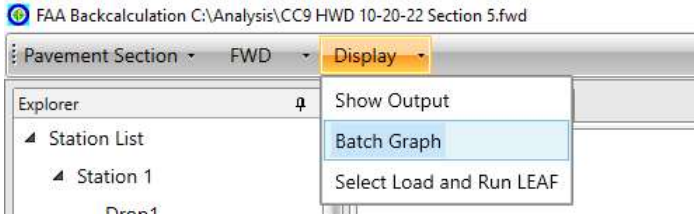
- Stop Backcalculation
- Aborts in-progress backcalculation
- Backcalculation results can still be found up to the point of termination:
  - In “Show Output” for single mode
  - In the .csv file for batch mode

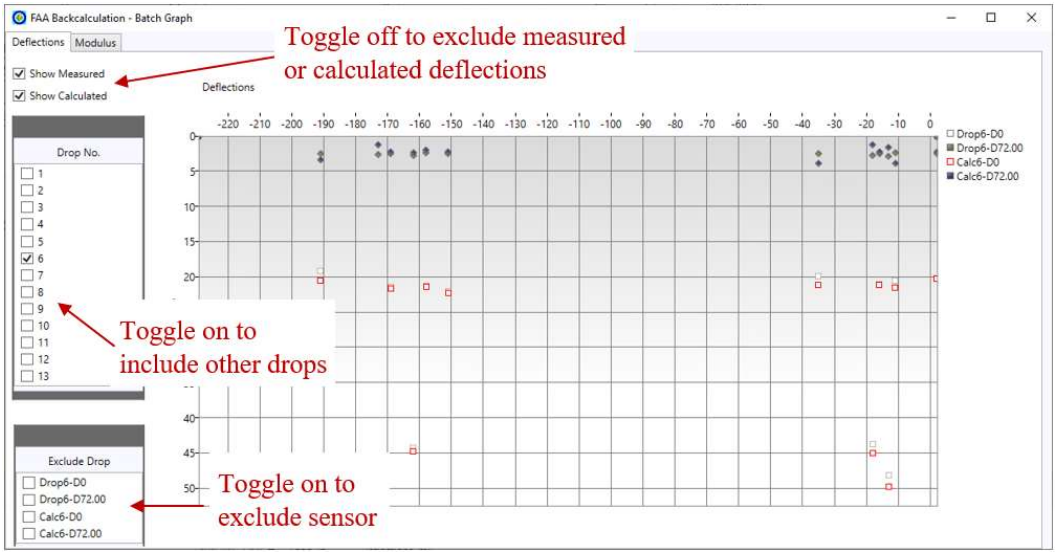


Backcalculation

Batch Graph

- Batch Graph
- Deflections tab: graphs measured and calculated deflections for two sensors:
  - Sensor under the FWD load plate (D_0)
  - The outermost sensor (D_{72})
- "Show Measured" displays measured deflections.
- "Show Calculated" displays backcalculated deflections.
- X-axis is DMI location of selected FWD test points.

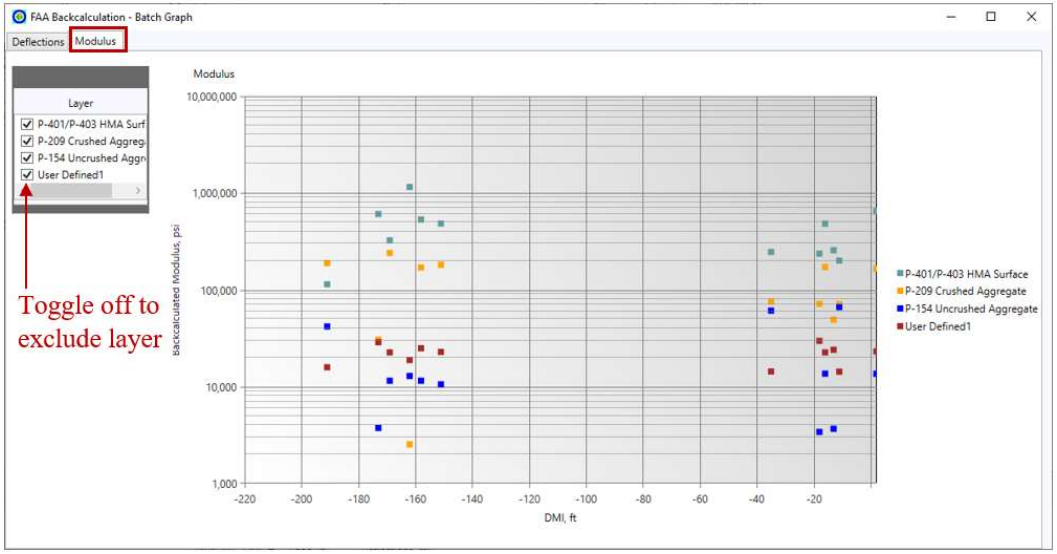




Backcalculation

Batch Graph

- Batch Graph
- Modulus tab: displays backcalculated modulus values in logarithmic scale.
- X-axis is DMI location of selected FWD test points.
- User can toggle off any layer of pavement in the backcalculated modulus graph.

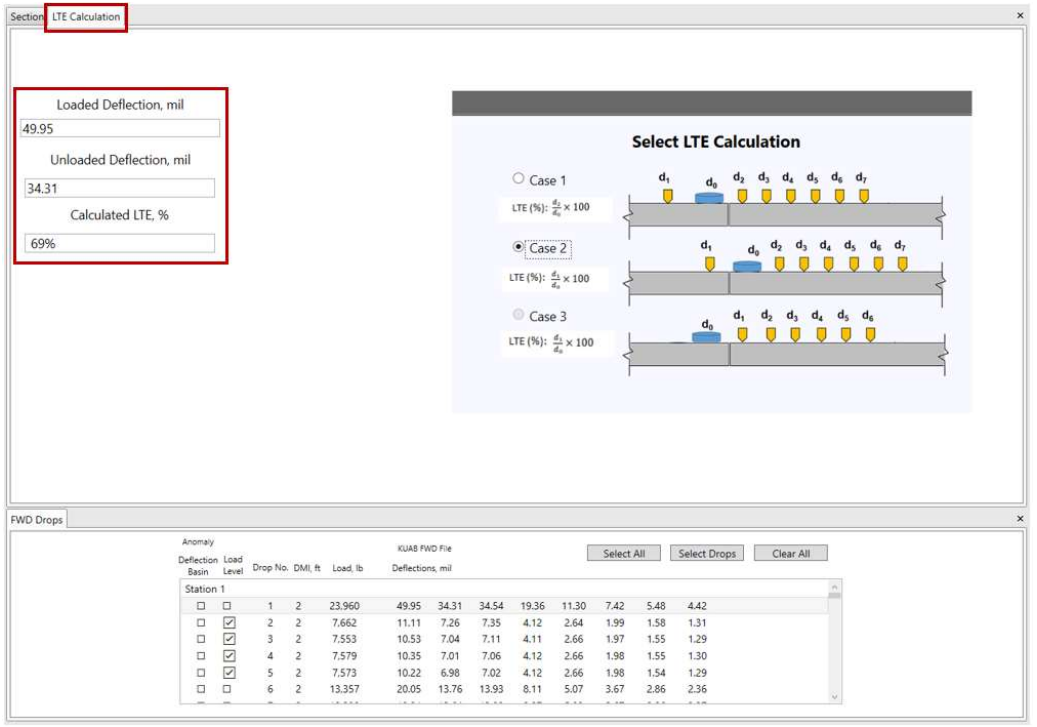


Load Transfer Efficiency (LTE) Calculation

LTE Calculation

Calculate LTE between Two Jointed Concrete Slabs

- Select between case 1, 2, or 3 based on FWD sensor configuration and FWD test layout at slab joints.
  - BAKFAA disables case 1 and 2 for FWD tests with no negative offset sensor (-12 in. offset) or when the user excludes the negative offset sensor.
  - BAKFAA disables case 3 for FWD tests with a negative offset sensor.



Layered Elastic Analysis Program (LEAF)

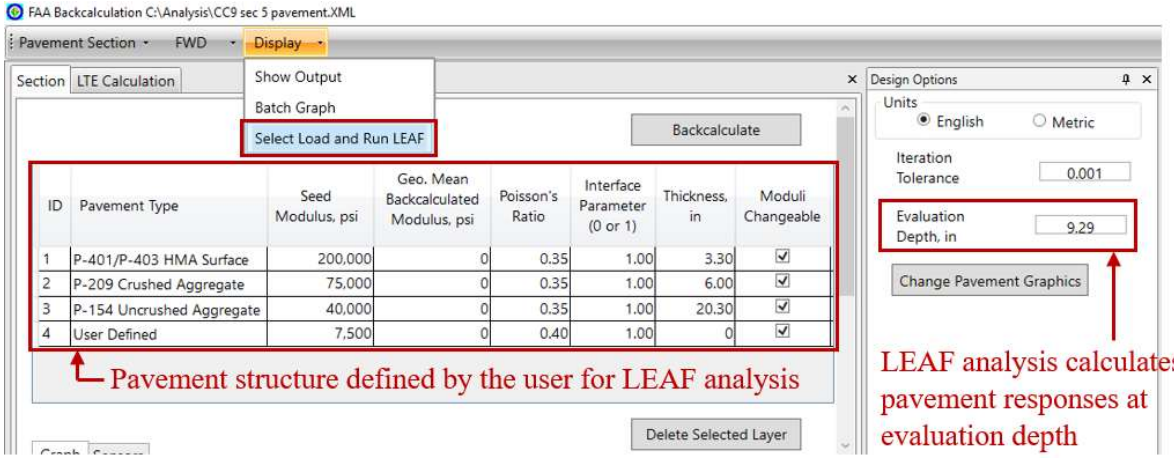
Run LEAF

- LEAF is built into BAKFAA.
- Compute pavement responses (stress, strain, and deflection) under defined load types at different evaluation points within the pavement structure.
- LEAF is based on the layered elastic model.
- The model is based on the Boussinesq mathematical model which assumes:
  - Pavement layers extend infinitely in the horizontal direction.
  - Bottom layer extends infinitely downward.
  - Materials are not stressed beyond their elastic ranges.

Run LEAF

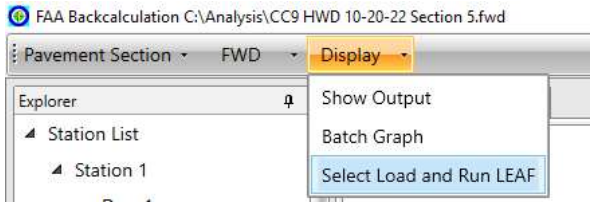
Inputs Required for LEAF Analysis

- Inputs required for LEAF analysis:
  - Modulus of elasticity (either seed modulus or backcalculated modulus)
  - Poisson's ratio
  - Layer thickness
  - Layer interface condition
  - Depth of evaluation point
- Continuity in responses across pavement layers depends on their interface condition.
  - Depth of evaluation must represent the target layer

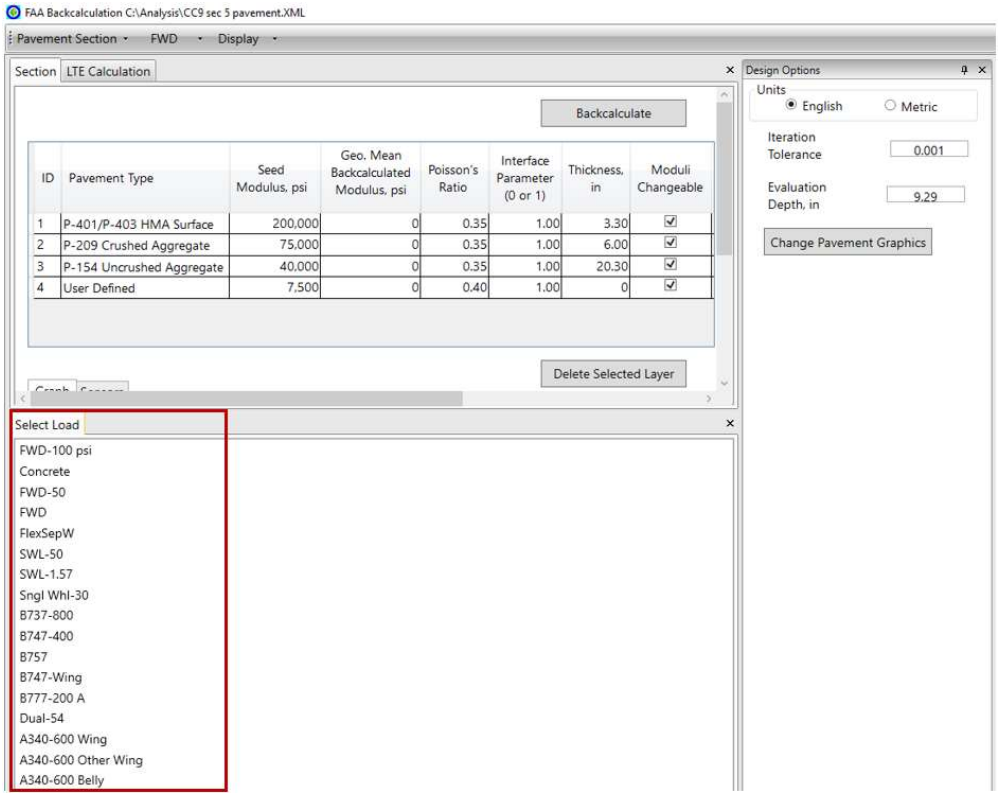


Run LEAF

Select Load and Gear Configuration

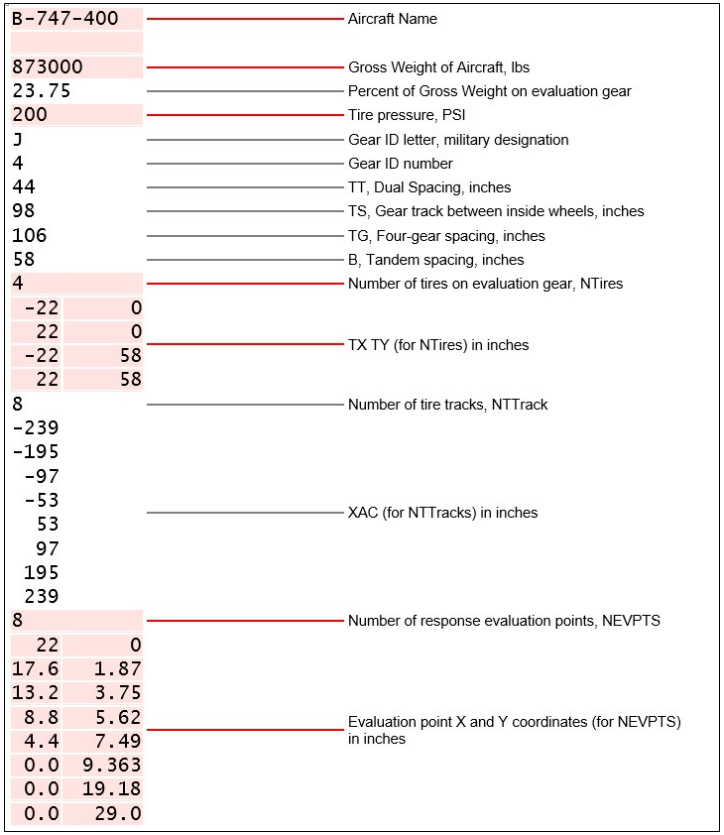


- Select load from library
  - Generic gear geometry
  - Defined aircraft
  - Defined FWD loading configurations
  - Library information is in LEAFAircraft.Ext file located in BAKFAA directory (C:\Program Files\FAA\BAKFAA)
- User can modify LEAFAircraft.Ext file or add user-defined landing gear geometry or FWD configuration.
- User can modify coordinates of evaluation points



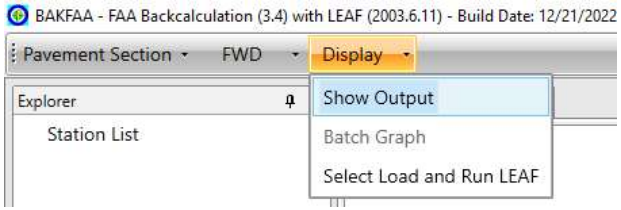
Run LEAF

Library Example (B-747-400)

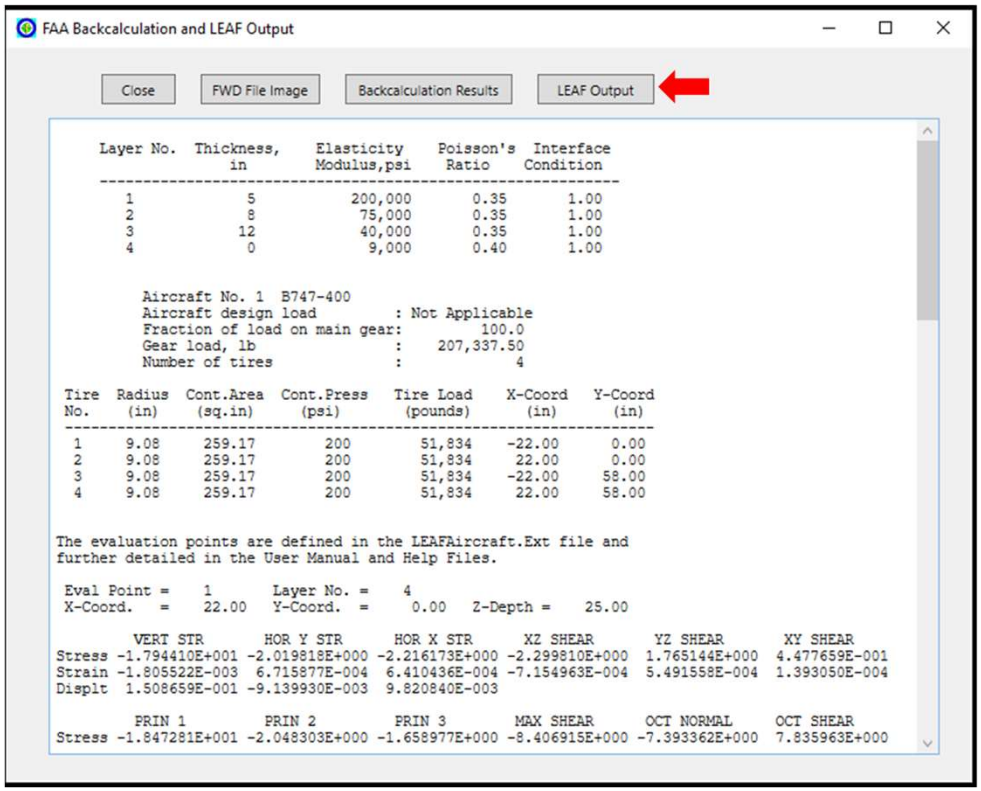


Run LEAF

LEAF Outputs



- Show Output – LEAF Output
- Pavement layer structure table
- Aircraft table
- Tires data table
- Pavement responses at evaluation points including eval point no., layer no. (determined by evaluation depth), X-coord, Y-coord, and Z-depth




```
Layer No.  Thickness,    Elasticity    Poisson's  Interface
              in        Modulus,psi     Ratio     Condition
-------------------------------------------------------------
    1           5          200,000       0.35       1.00
    2           8           75,000       0.35       1.00
    3          12           40,000       0.35       1.00
    4           0            9,000       0.40       1.00


      Aircraft No. 1  B747-400
      Aircraft design load         : Not Applicable
      Fraction of load on main gear:         100.0
      Gear load, lb                :    207,337.50
      Number of tires              :             4

Tire  Radius  Cont.Area  Cont.Press   Tire Load    X-Coord   Y-Coord
No.     (in)    (sq.in)       (psi)      (pounds)     (in)      (in)
-------------------------------------------------------------------
 1     9.08     259.17         200        51,834    -22.00      0.00
 2     9.08     259.17         200        51,834     22.00      0.00
 3     9.08     259.17         200        51,834    -22.00     58.00
 4     9.08     259.17         200        51,834     22.00     58.00

The evaluation points are defined in the LEAFAircraft.Ext file and
further detailed in the User Manual and Help Files.

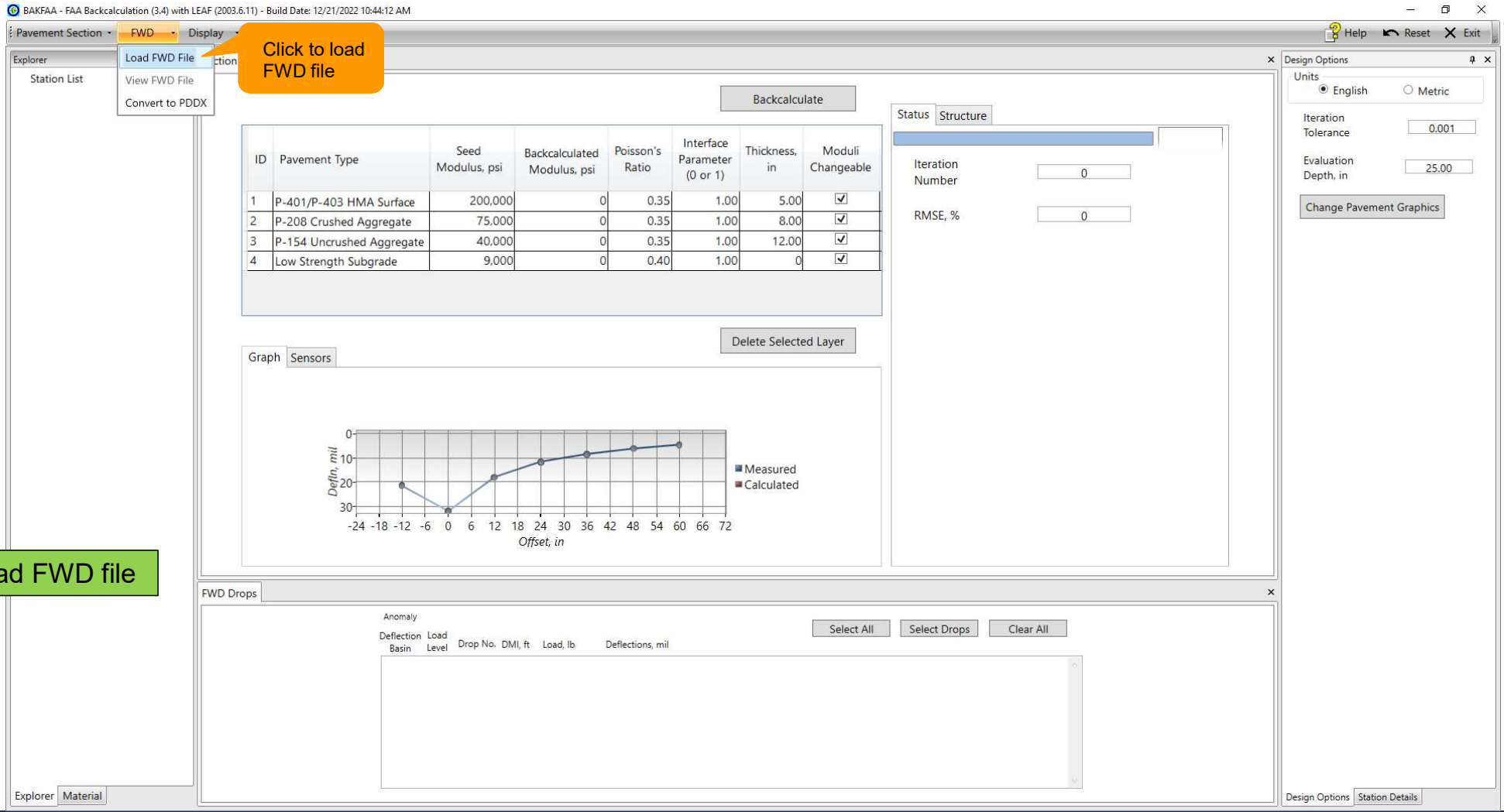
 Eval Point =    1       Layer No. =    4
 X-Coord.   =   22.00    Y-Coord.  =    0.00   Z-Depth =   25.00

         VERT STR       HOR Y STR      HOR X STR      XZ SHEAR       YZ SHEAR       XY SHEAR
Stress -1.794410E+001 -2.019818E+000 -2.216173E+000 -2.299810E+000  1.765144E+000  4.477659E-001
Strain -1.805522E-003  6.715877E-004  6.410436E-004 -7.154963E-004  5.491558E-004  1.393050E-004
Displt  1.508659E-001 -9.139930E-003  9.820840E-003

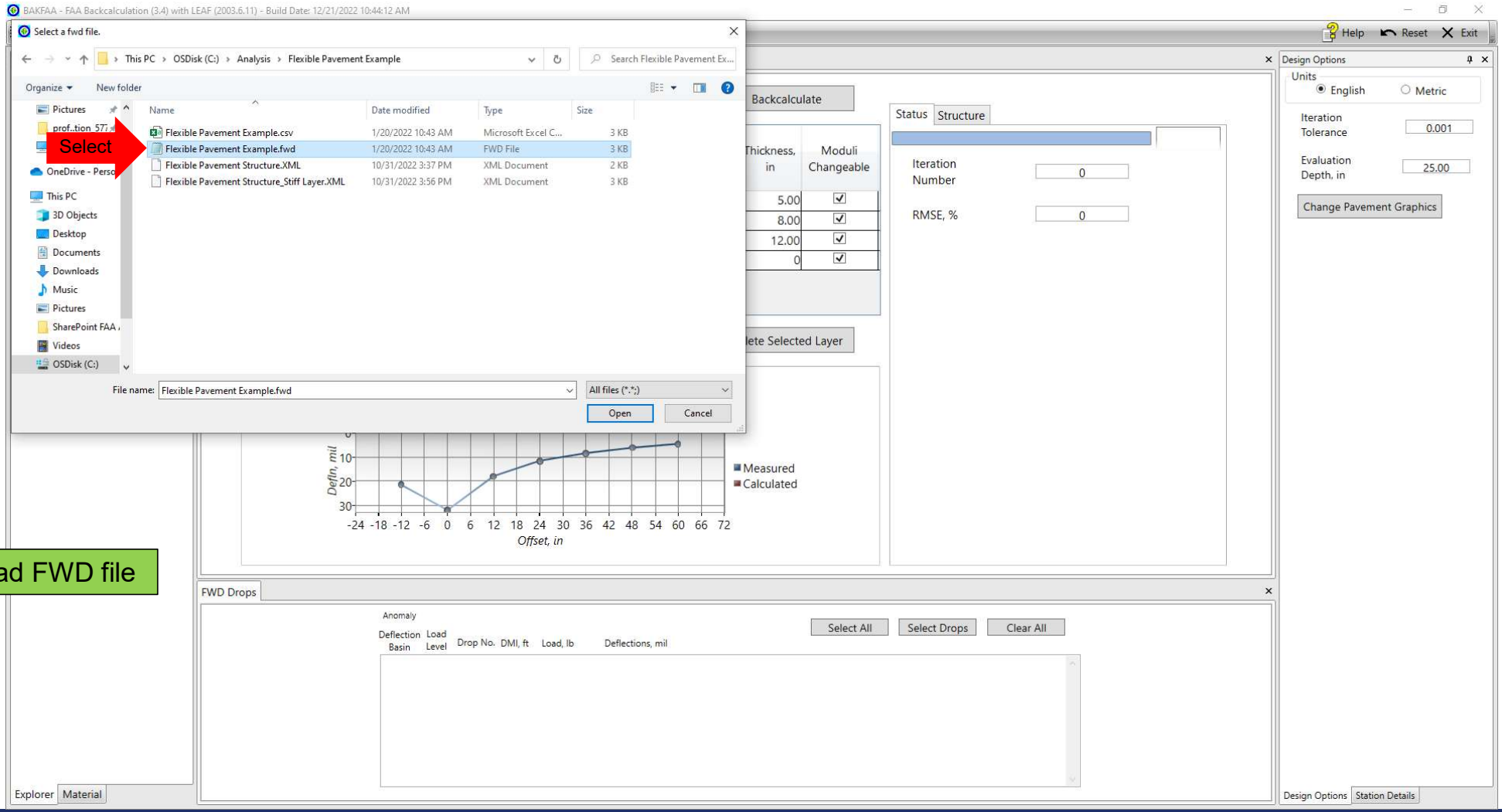
         PRIN 1         PRIN 2         PRIN 3        MAX SHEAR      OCT NORMAL     OCT SHEAR
Stress -1.847281E+001 -2.048303E+000 -1.658977E+000 -8.406915E+000 -7.393362E+000  7.835963E+000
```

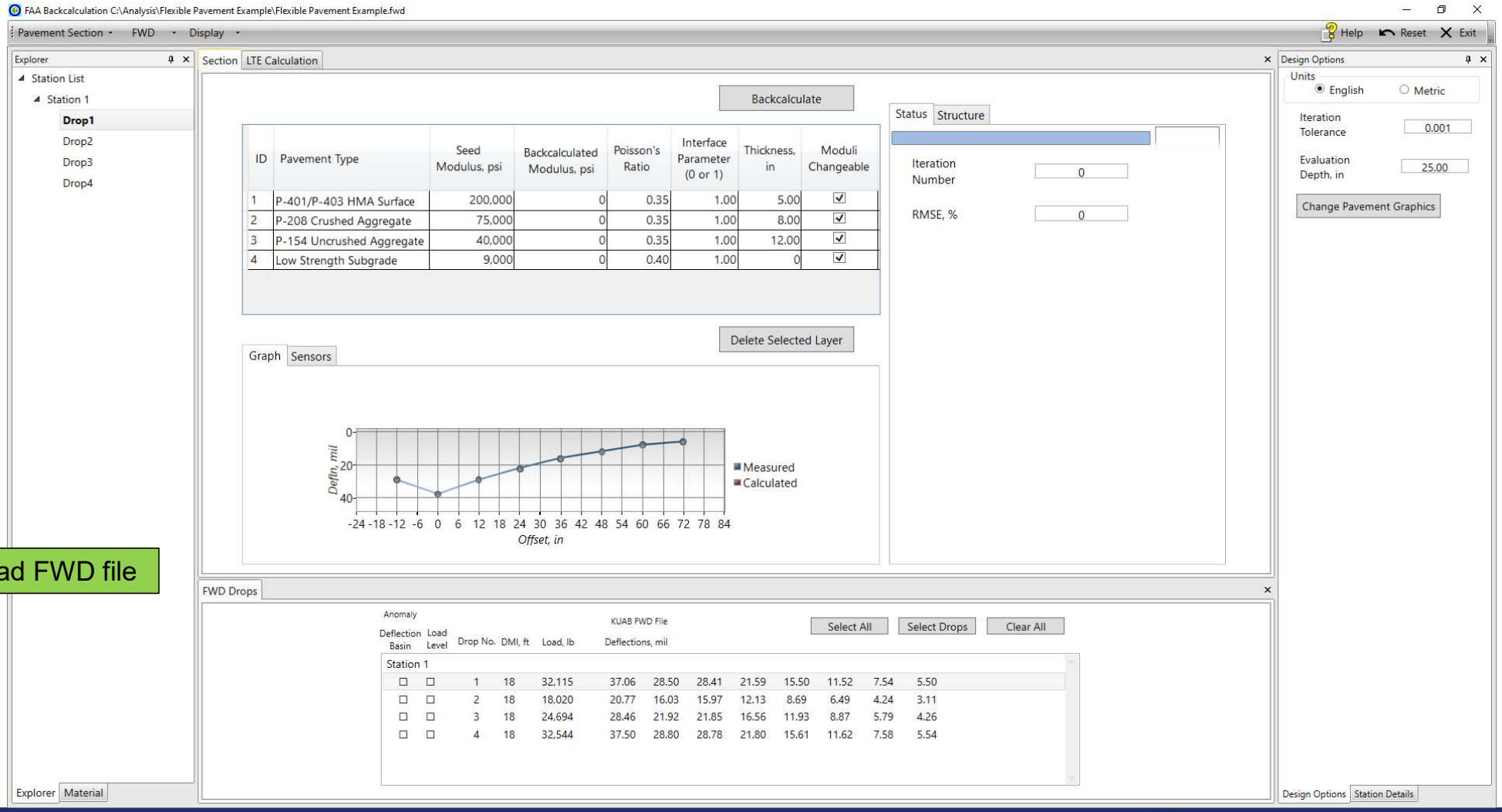
Software Demo

Flexible Pavement Example



Load FWD file

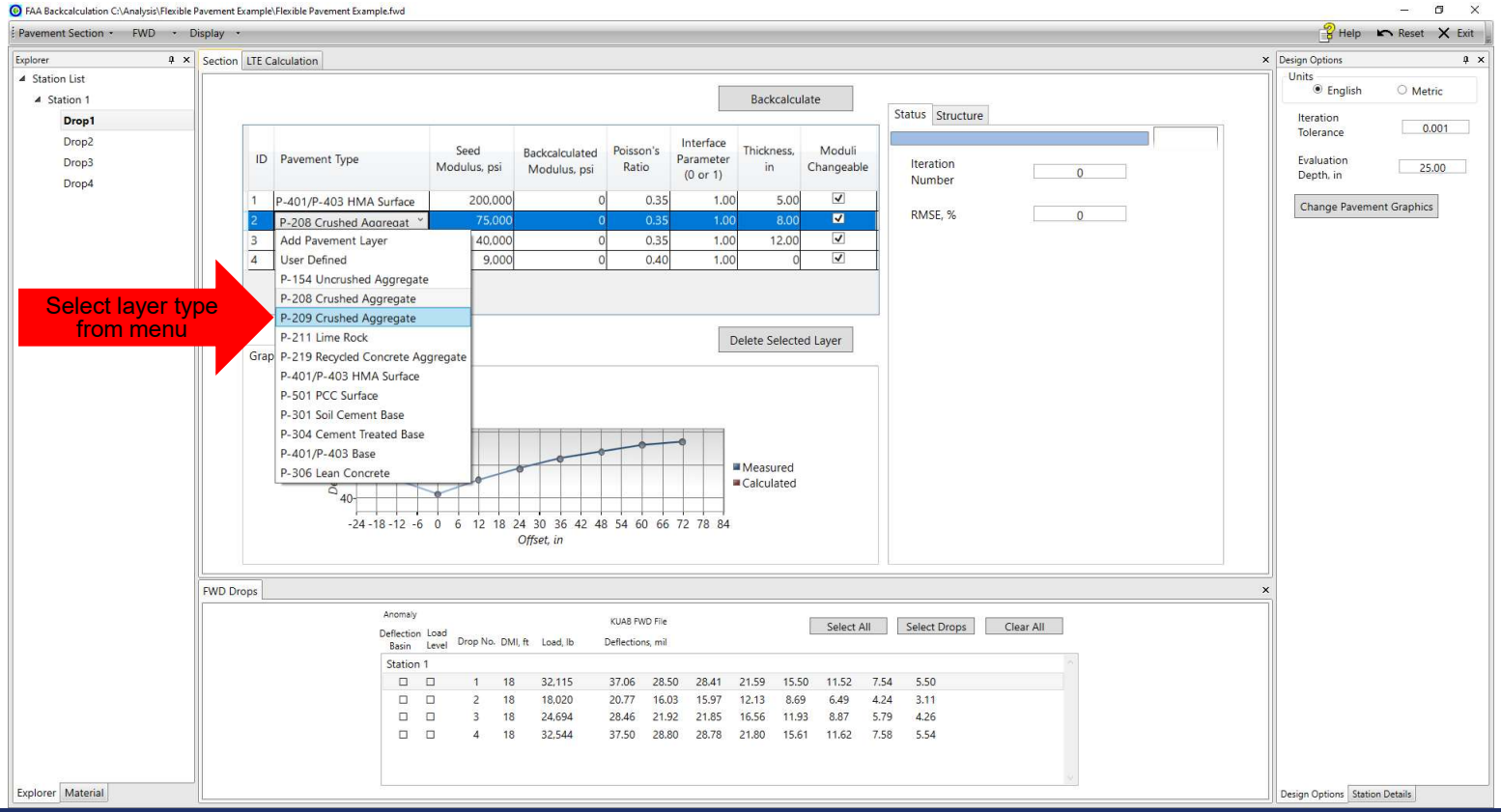

Select
Load FWD file



Load FWD file

Assign pavement structure

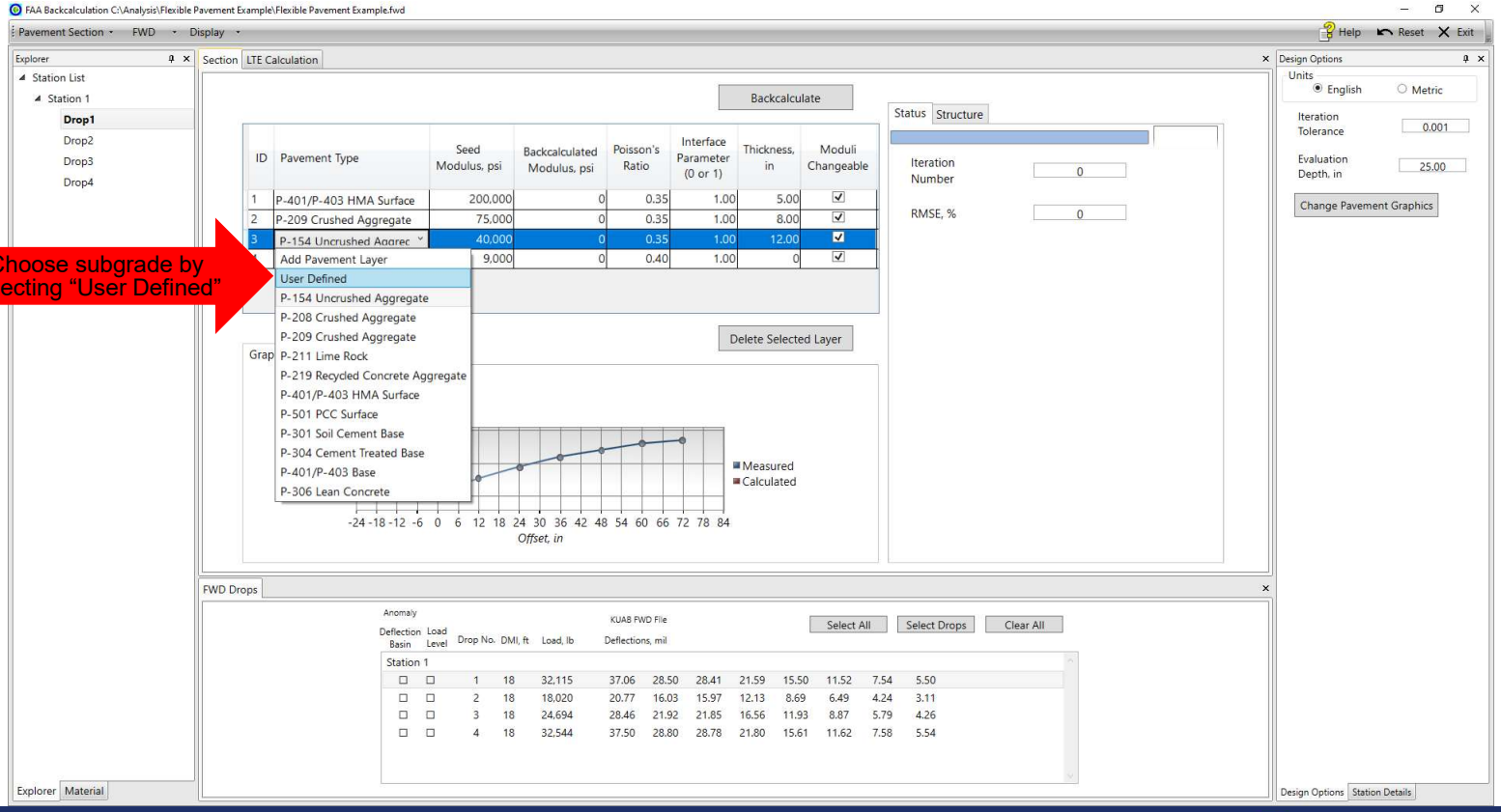
Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	



Assign pavement structure

Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	

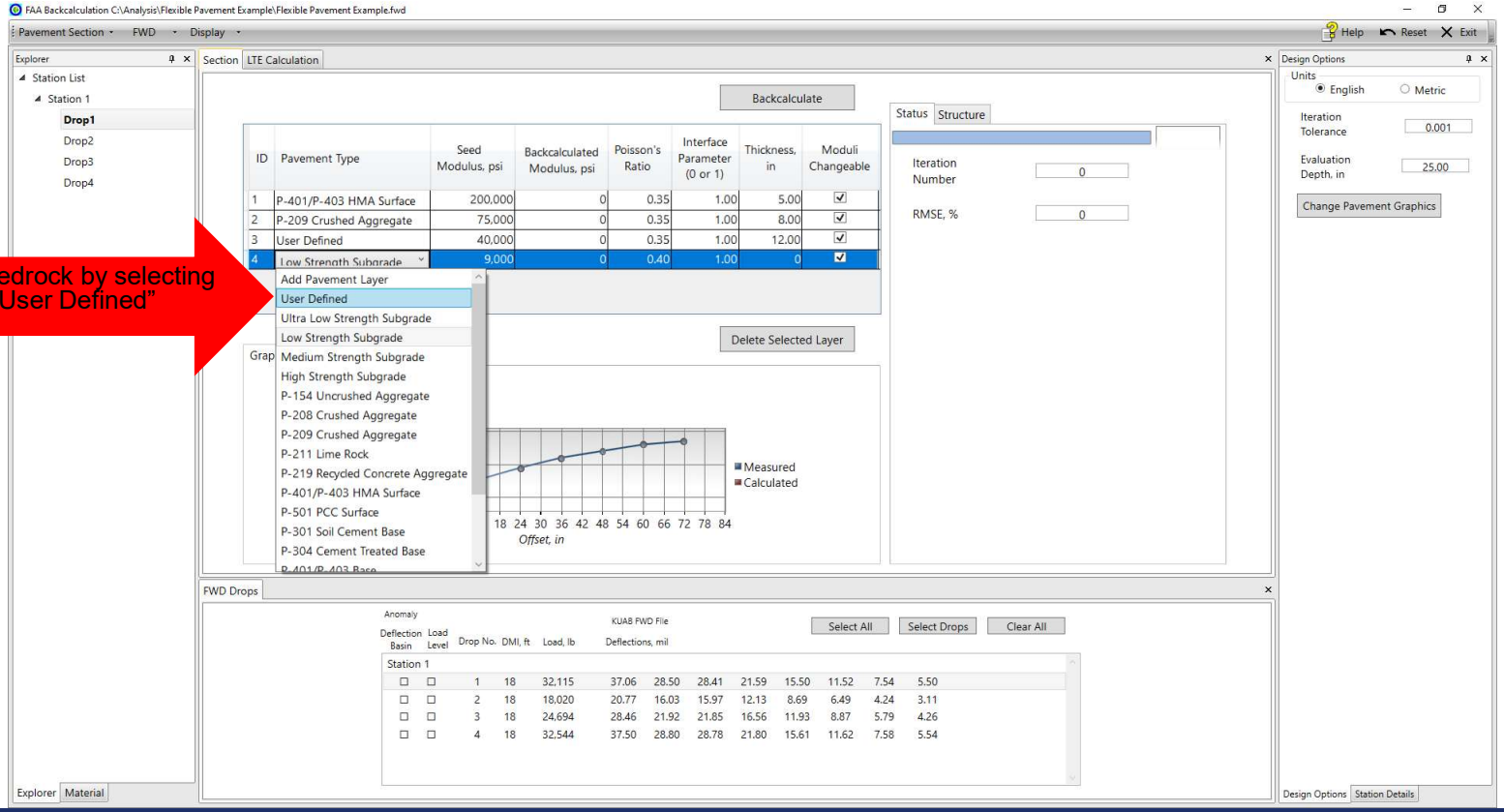
Choose subgrade by selecting “User Defined”



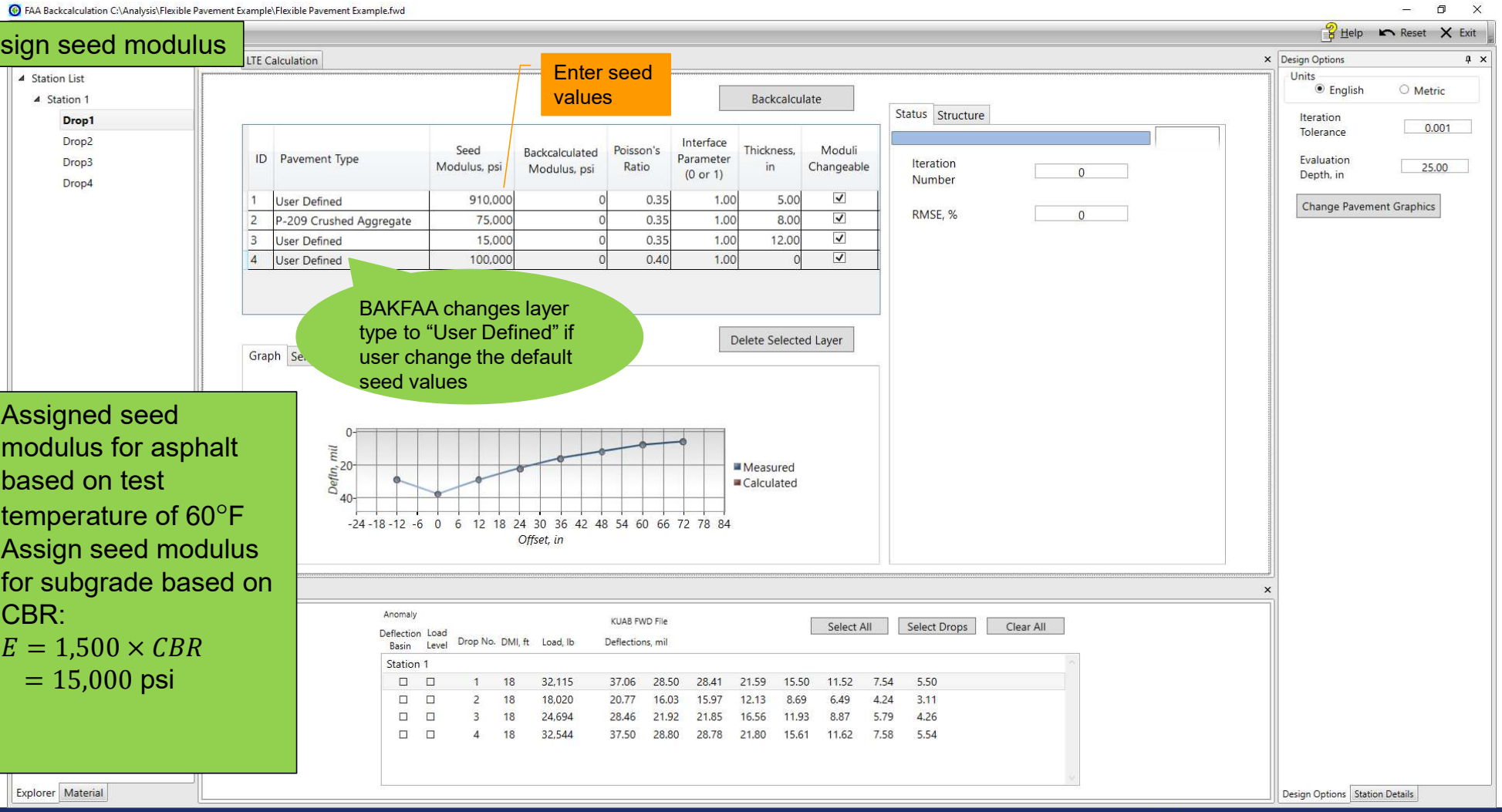
Assign pavement structure

Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	

Add bedrock by selecting "User Defined"



Assign seed modulus



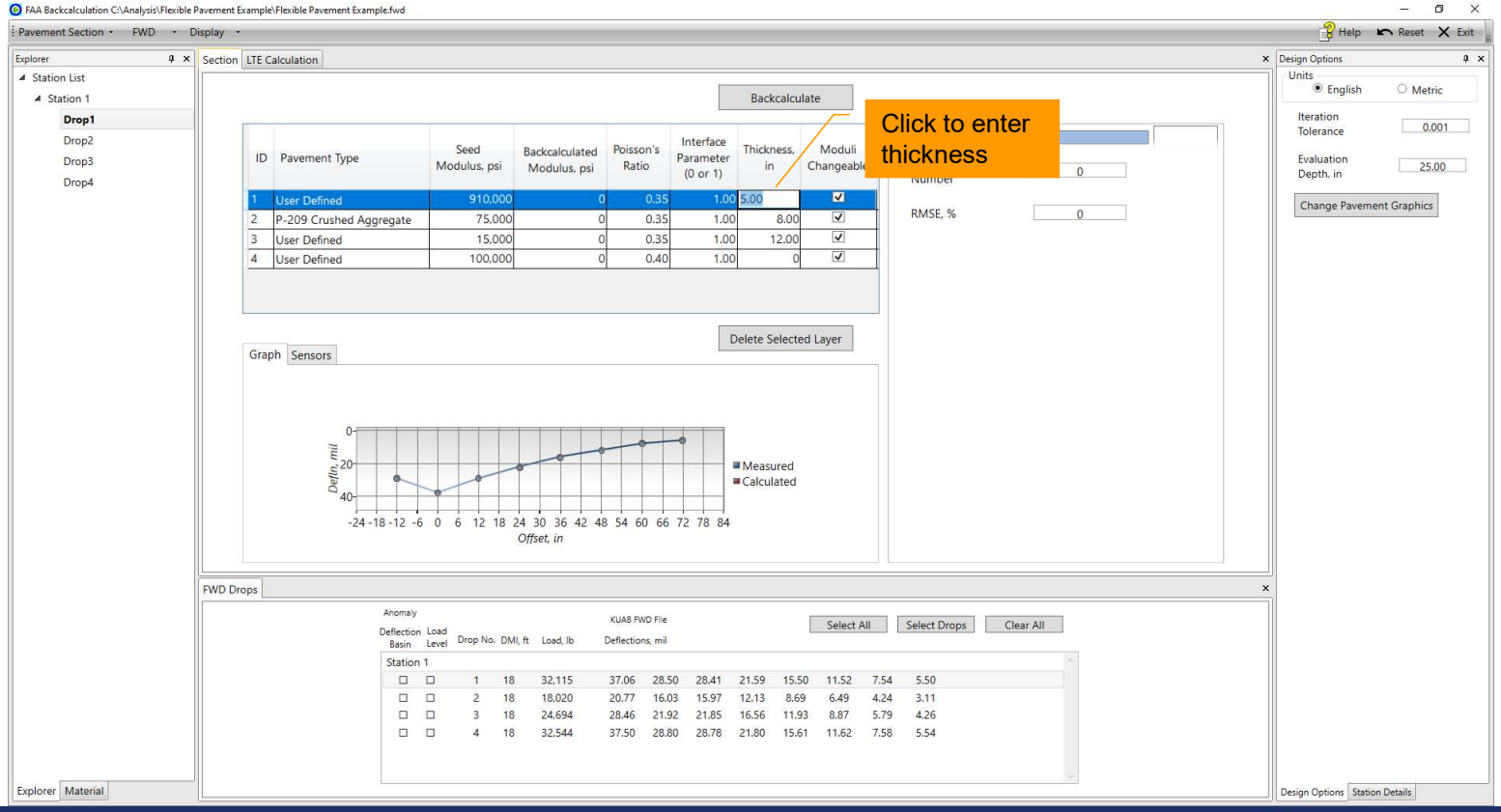
- Assigned seed modulus for asphalt based on test temperature of 60°F
- Assign seed modulus for subgrade based on CBR:

  $E = 1{,}500 \times CBR$

  $= 15{,}000$ psi

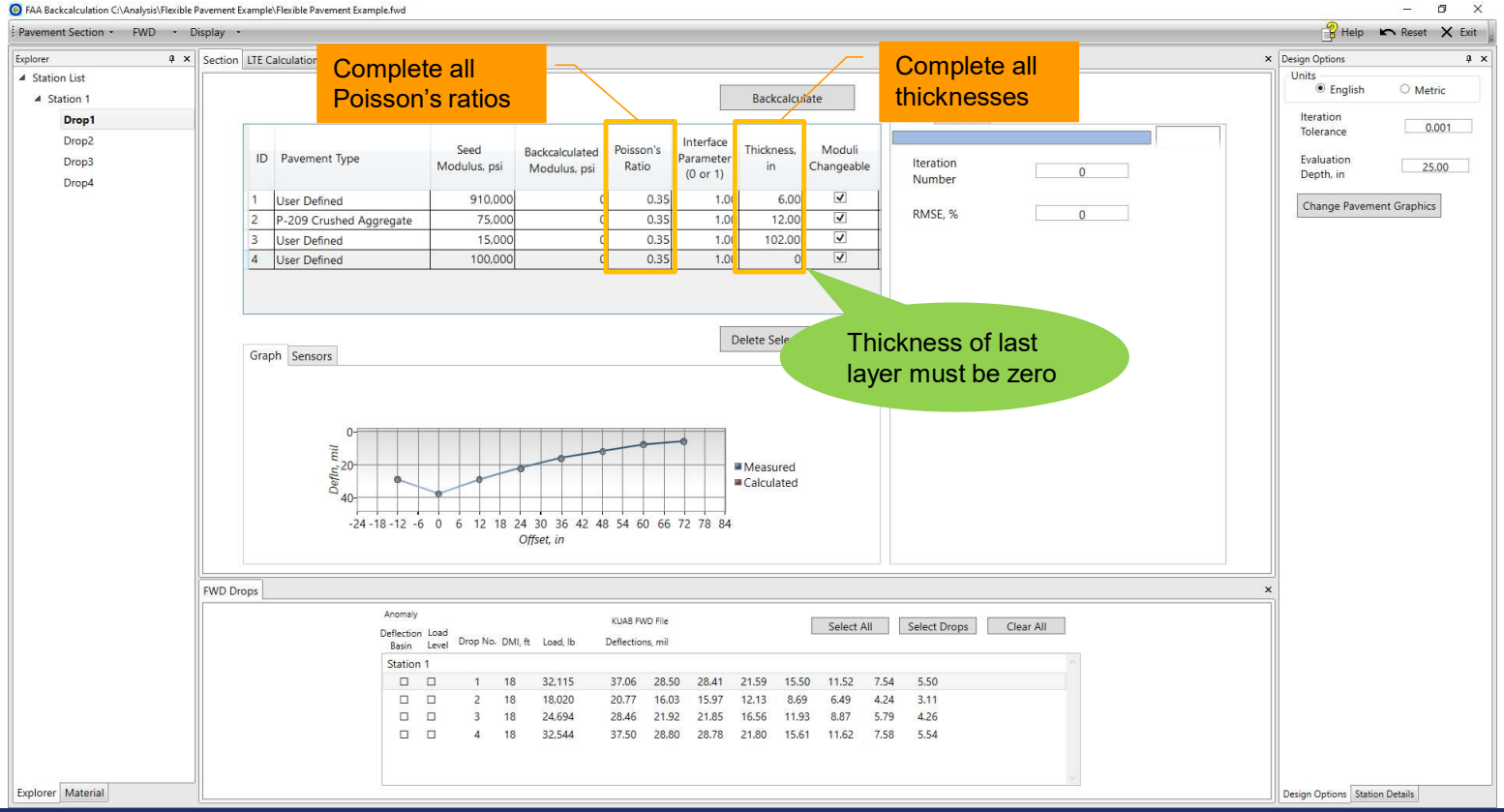
Assign layer thickness

Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	

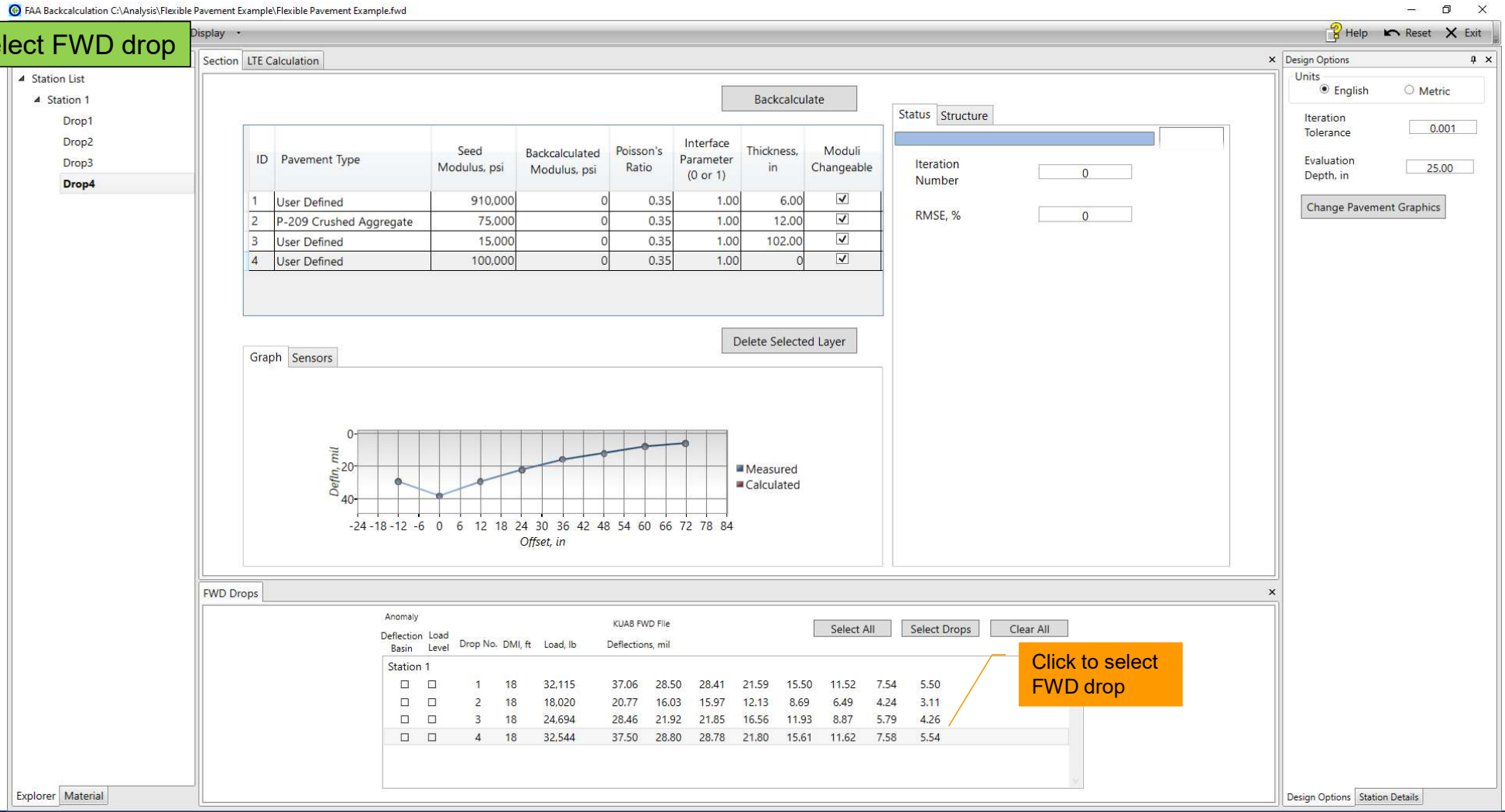


Assign layer thickness

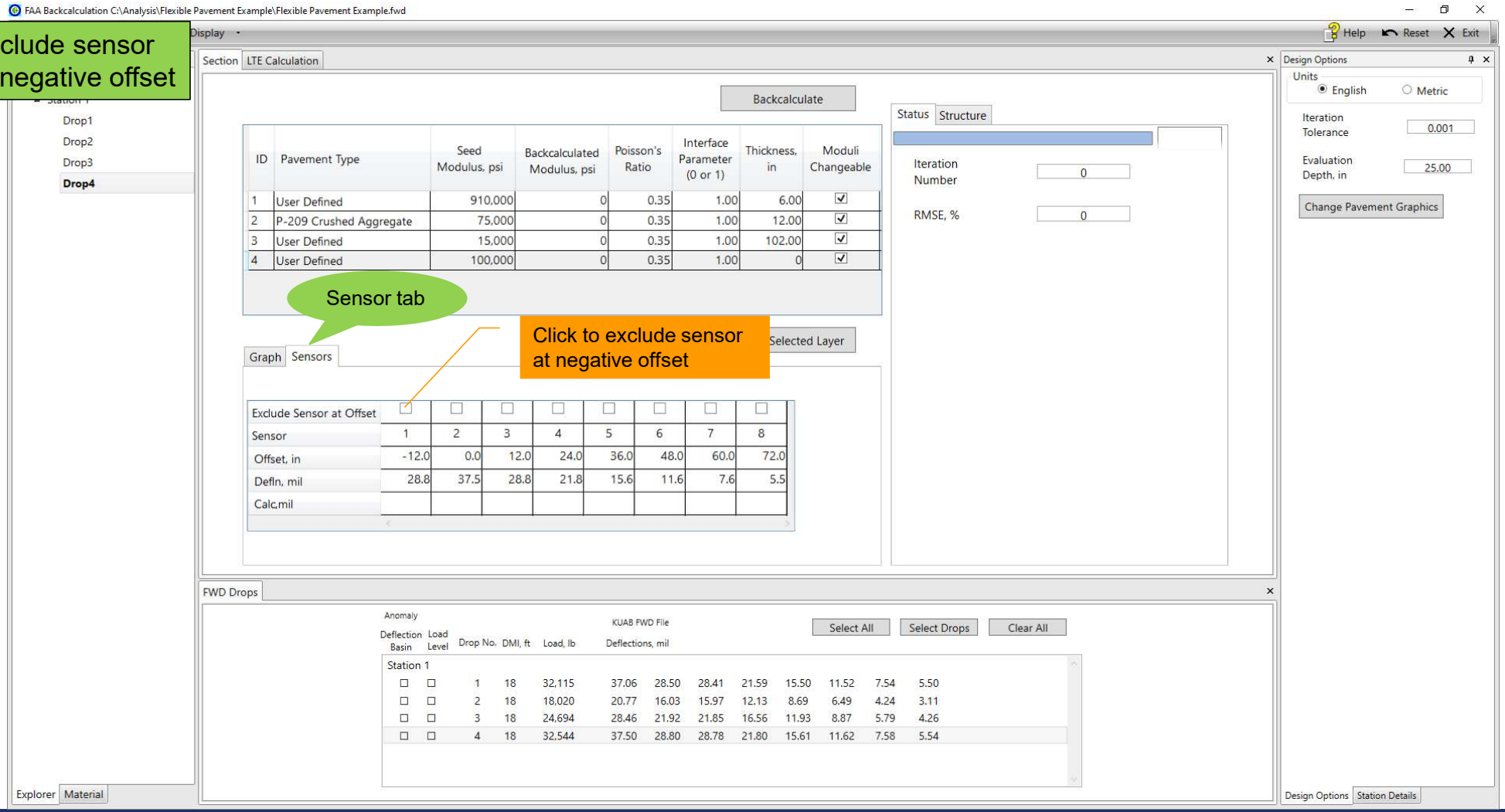
Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	



Select FWD drop



Exclude sensor at negative offset

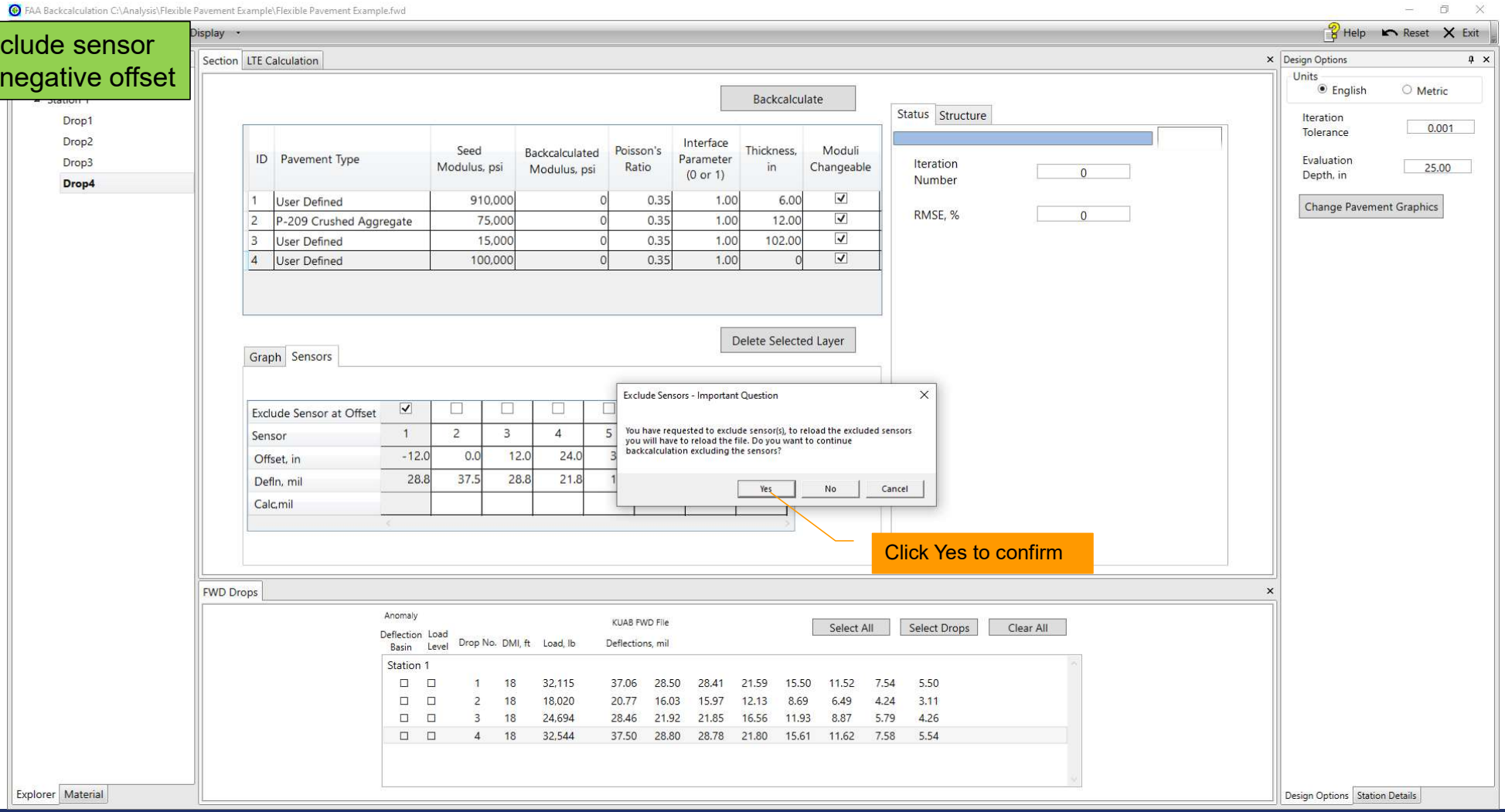


ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	User Defined	910,000	0	0.35	1.00	6.00	☑
2	P-209 Crushed Aggregate	75,000	0	0.35	1.00	12.00	☑
3	User Defined	15,000	0	0.35	1.00	102.00	☑
4	User Defined	100,000	0	0.35	1.00	0	☑

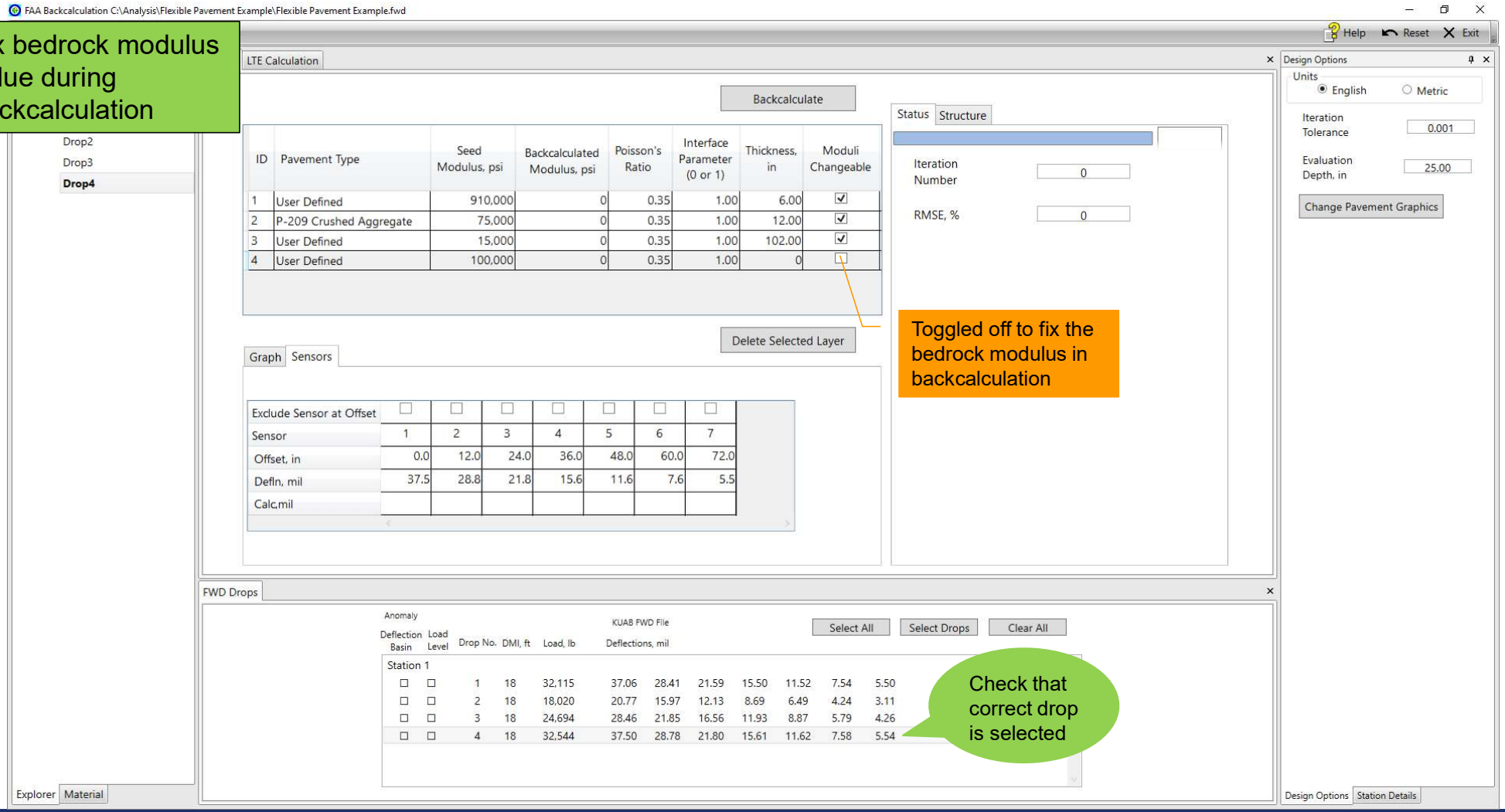
Exclude Sensor at Offset	☐	☐	☐	☐	☐	☐	☐	☐
Sensor	1	2	3	4	5	6	7	8
Offset, in	-12.0	0.0	12.0	24.0	36.0	48.0	60.0	72.0
Defln, mil	28.8	37.5	28.8	21.8	15.6	11.6	7.6	5.5
Calc,mil								

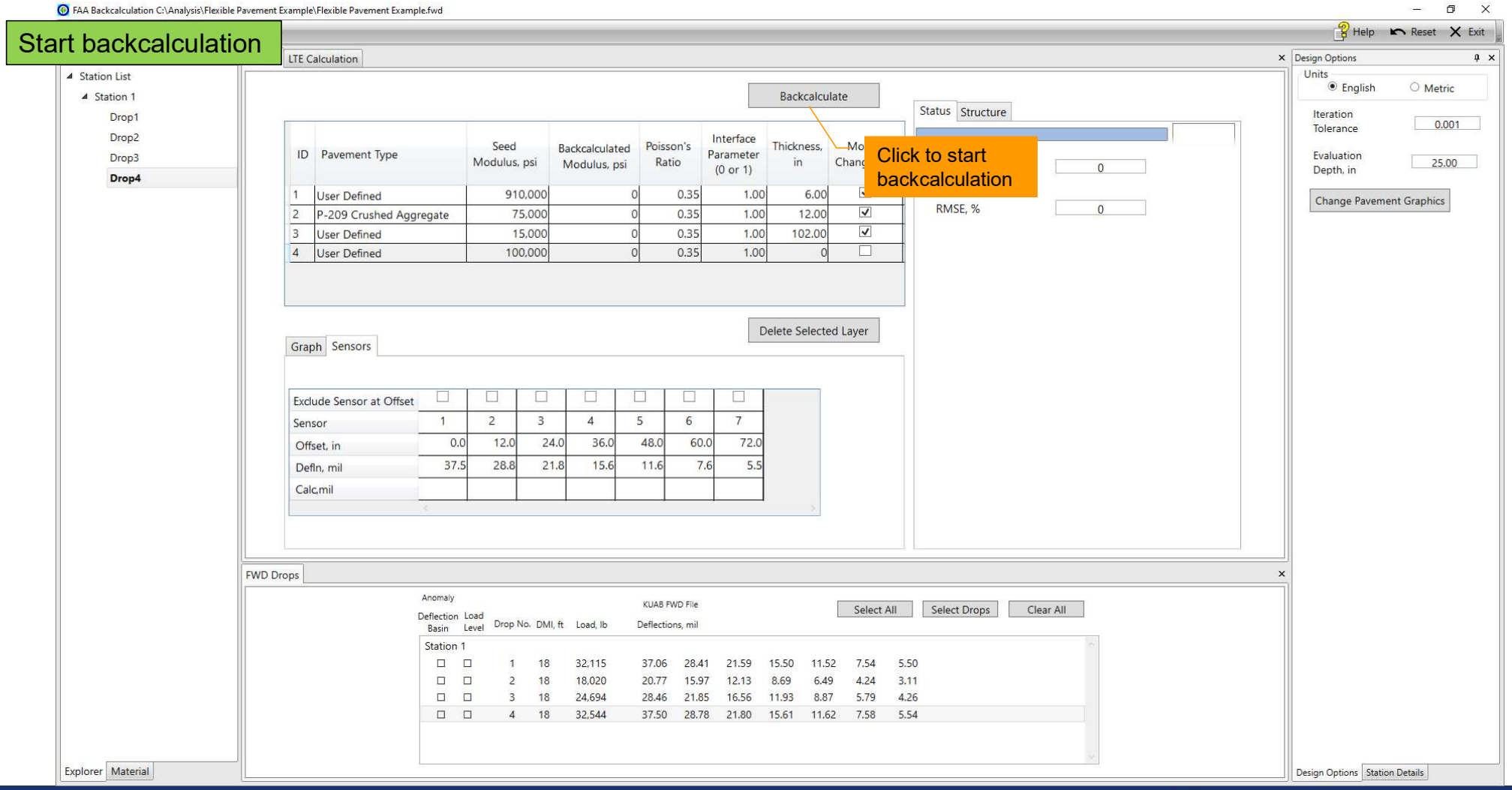
Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	Deflections, mil						
Station 1											
☐	☐	1	18	32,115	37.06	28.50	28.41	21.59	15.50	11.52	7.54	5.50
☐	☐	2	18	18,020	20.77	16.03	15.97	12.13	8.69	6.49	4.24	3.11
☐	☐	3	18	24,694	28.46	21.92	21.85	16.56	11.93	8.87	5.79	4.26
☐	☐	4	18	32,544	37.50	28.80	28.78	21.80	15.61	11.62	7.58	5.54

Exclude sensor at negative offset

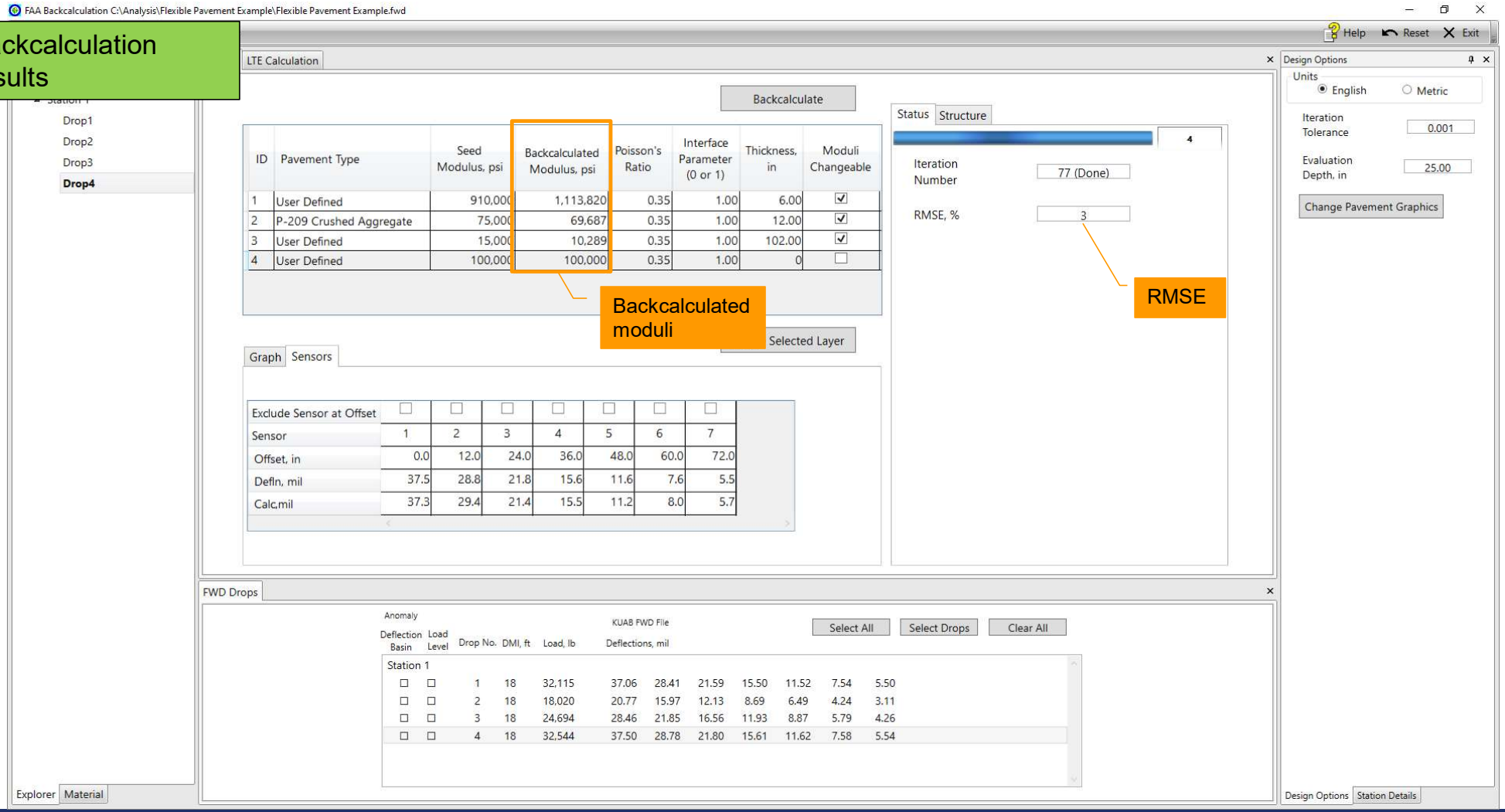


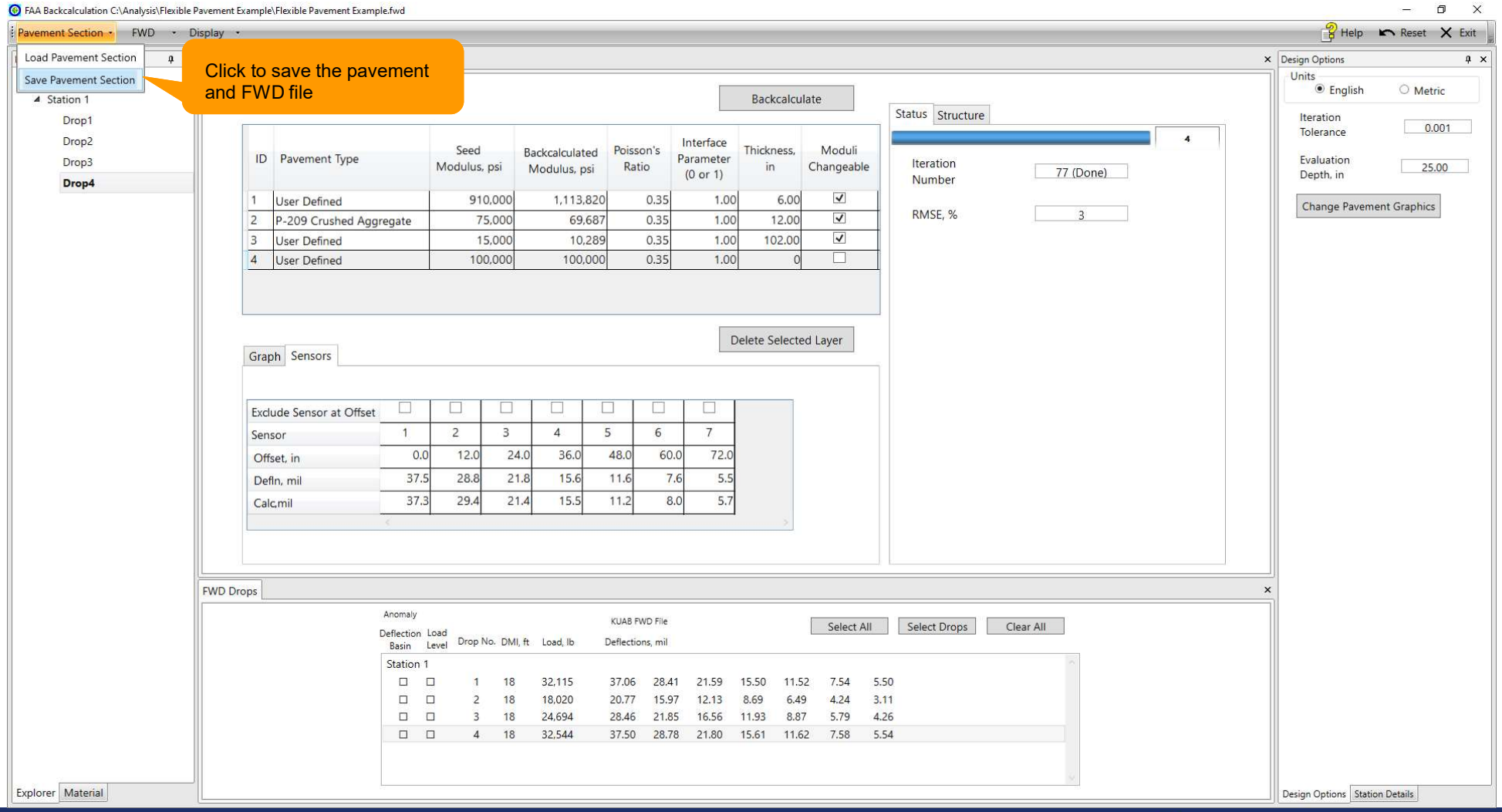
Fix bedrock modulus value during backcalculation

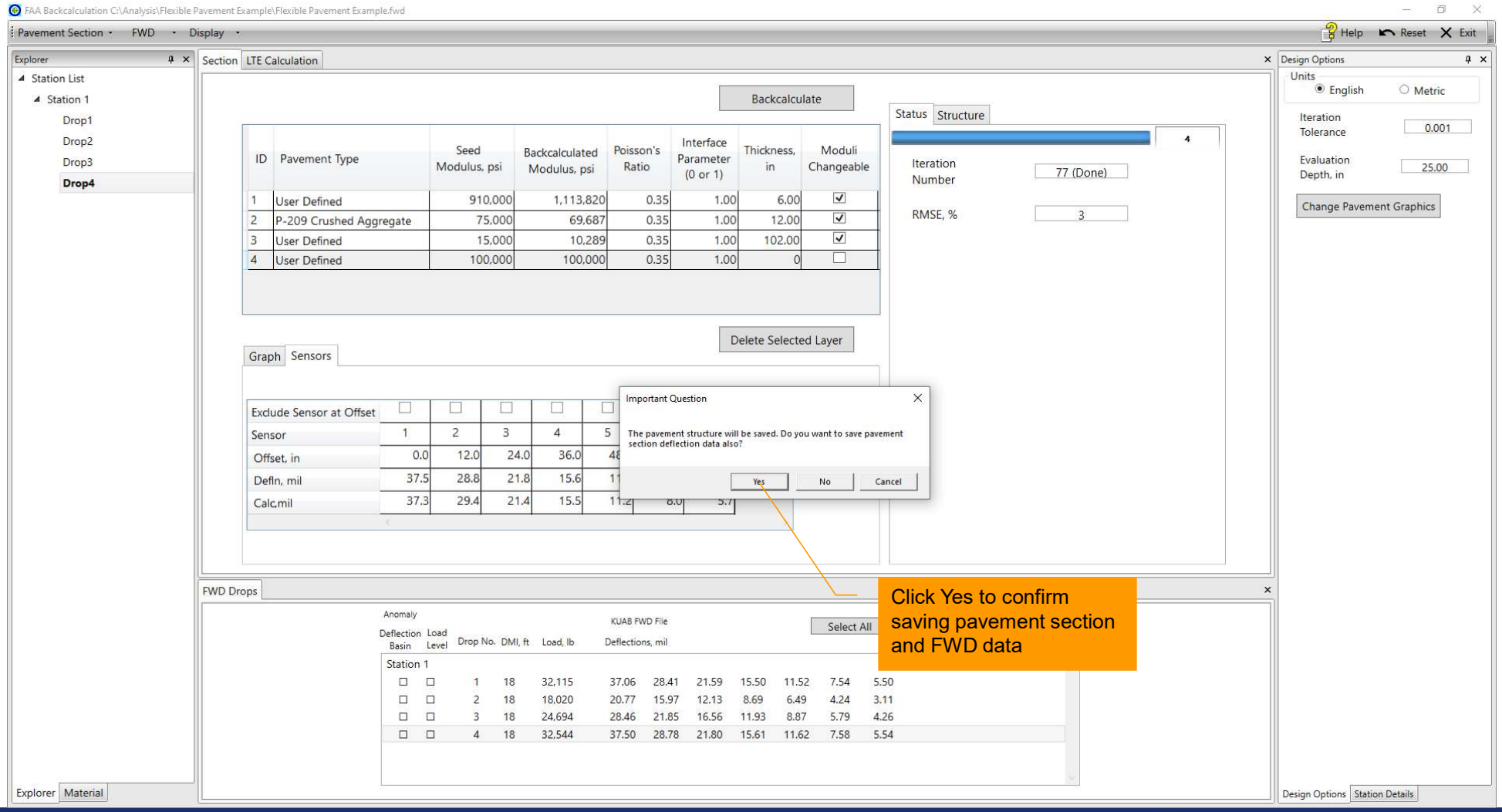


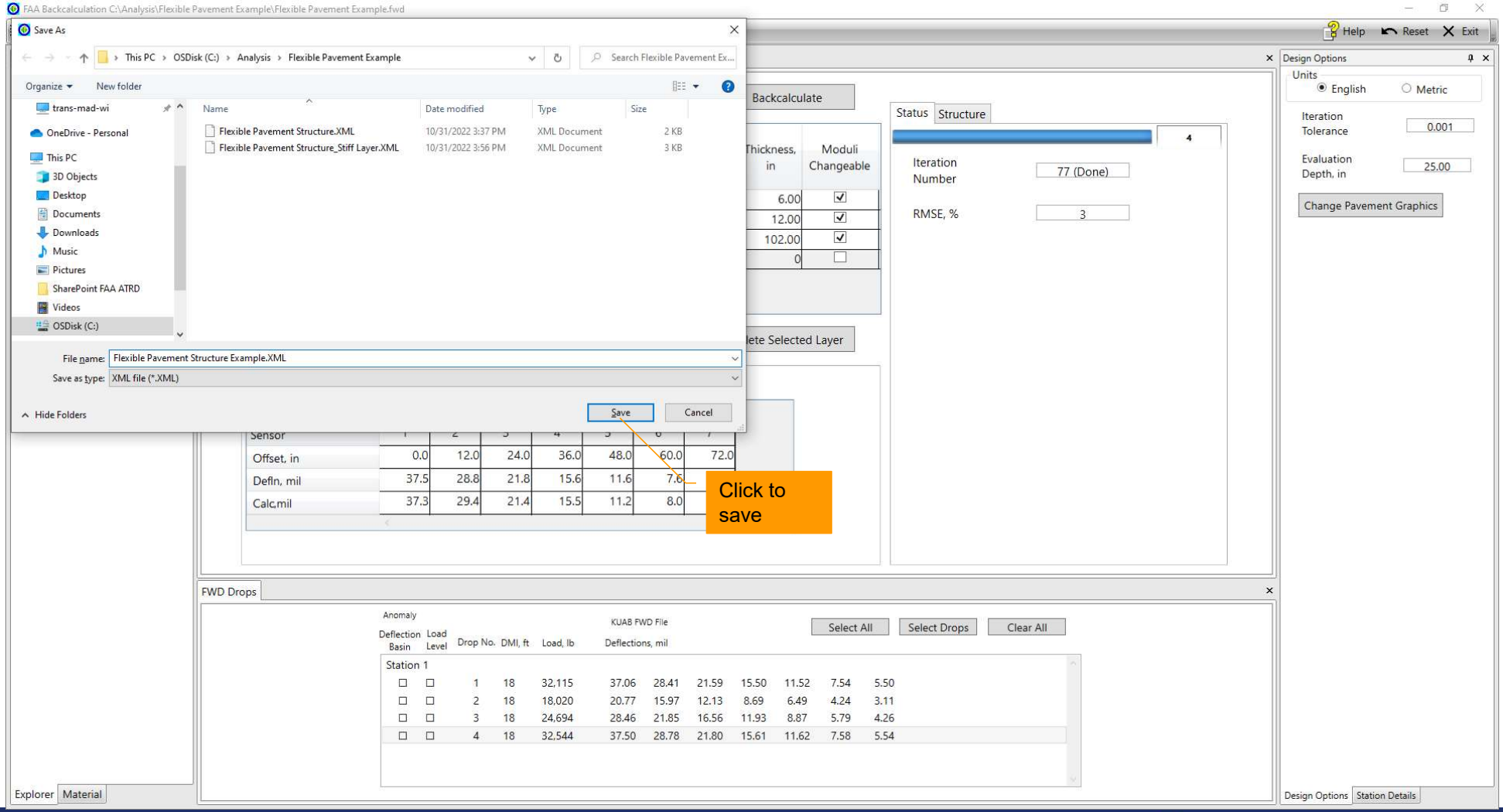

Start backcalculation
FAA Backcalculation C:\Analysis\Flexible Pavement Example\Flexible Pavement Example.fwd
Help
Reset
Exit
Station List
Station 1
Drop1
Drop2
Drop3
Drop4
LTE Calculation
Backcalculate
Click to start backcalculation
ID
Pavement Type
Seed Modulus, psi
Backcalculated Modulus, psi
Poisson's Ratio
Interface Parameter (0 or 1)
Thickness, in
1 User Defined 910,000 0 0.35 1.00 6.00
2 P-209 Crushed Aggregate 75,000 0 0.35 1.00 12.00
3 User Defined 15,000 0 0.35 1.00 102.00
4 User Defined 100,000 0 0.35 1.00 0
Status
Structure
RMSE, %
Delete Selected Layer
Graph
Sensors
Exclude Sensor at Offset
Sensor 1 2 3 4 5 6 7
Offset, in 0.0 12.0 24.0 36.0 48.0 60.0 72.0
Defln, mil 37.5 28.8 21.8 15.6 11.6 7.6 5.5
Calc,mil
Design Options
Units
English
Metric
Iteration Tolerance 0.001
Evaluation Depth, in 25.00
Change Pavement Graphics
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
KUAB FWD File Deflections, mil
Select All
Select Drops
Clear All
Station 1
1 18 32,115 37.06 28.41 21.59 15.50 11.52 7.54 5.50
2 18 18,020 20.77 15.97 12.13 8.69 6.49 4.24 3.11
3 18 24,694 28.46 21.85 16.56 11.93 8.87 5.79 4.26
4 18 32,544 37.50 28.78 21.80 15.61 11.62 7.58 5.54
Explorer
Material
Design Options
Station Details

Backcalculation results




FAA Backcalculation C:\Analysis\Flexible Pavement Example\Flexible Pavement Example.fwd
Pavement Section
FWD
Display
Help
Reset
Exit
Load Pavement Section
Save Pavement Section
Click to save the pavement and FWD file
Station 1
Drop1
Drop2
Drop3
Drop4
Backcalculate
ID
Pavement Type
Seed Modulus, psi
Backcalculated Modulus, psi
Poisson's Ratio
Interface Parameter (0 or 1)
Thickness, in
Moduli Changeable
1 User Defined 910,000 1,113,820 0.35 1.00 6.00
2 P-209 Crushed Aggregate 75,000 69,687 0.35 1.00 12.00
3 User Defined 15,000 10,289 0.35 1.00 102.00
4 User Defined 100,000 100,000 0.35 1.00 0
Status
Structure
Iteration Number
77 (Done)
RMSE, %
3
Delete Selected Layer
Graph
Sensors
Exclude Sensor at Offset
Sensor 1 2 3 4 5 6 7
Offset, in 0.0 12.0 24.0 36.0 48.0 60.0 72.0
Defln, mil 37.5 28.8 21.8 15.6 11.6 7.6 5.5
Calc,mil 37.3 29.4 21.4 15.5 11.2 8.0 5.7
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
KUAB FWD File
Deflections, mil
Select All
Select Drops
Clear All
Station 1
1 18 32,115 37.06 28.41 21.59 15.50 11.52 7.54 5.50
2 18 18,020 20.77 15.97 12.13 8.69 6.49 4.24 3.11
3 18 24,694 28.46 21.85 16.56 11.93 8.87 5.79 4.26
4 18 32,544 37.50 28.78 21.80 15.61 11.62 7.58 5.54
Explorer
Material
Design Options
Station Details

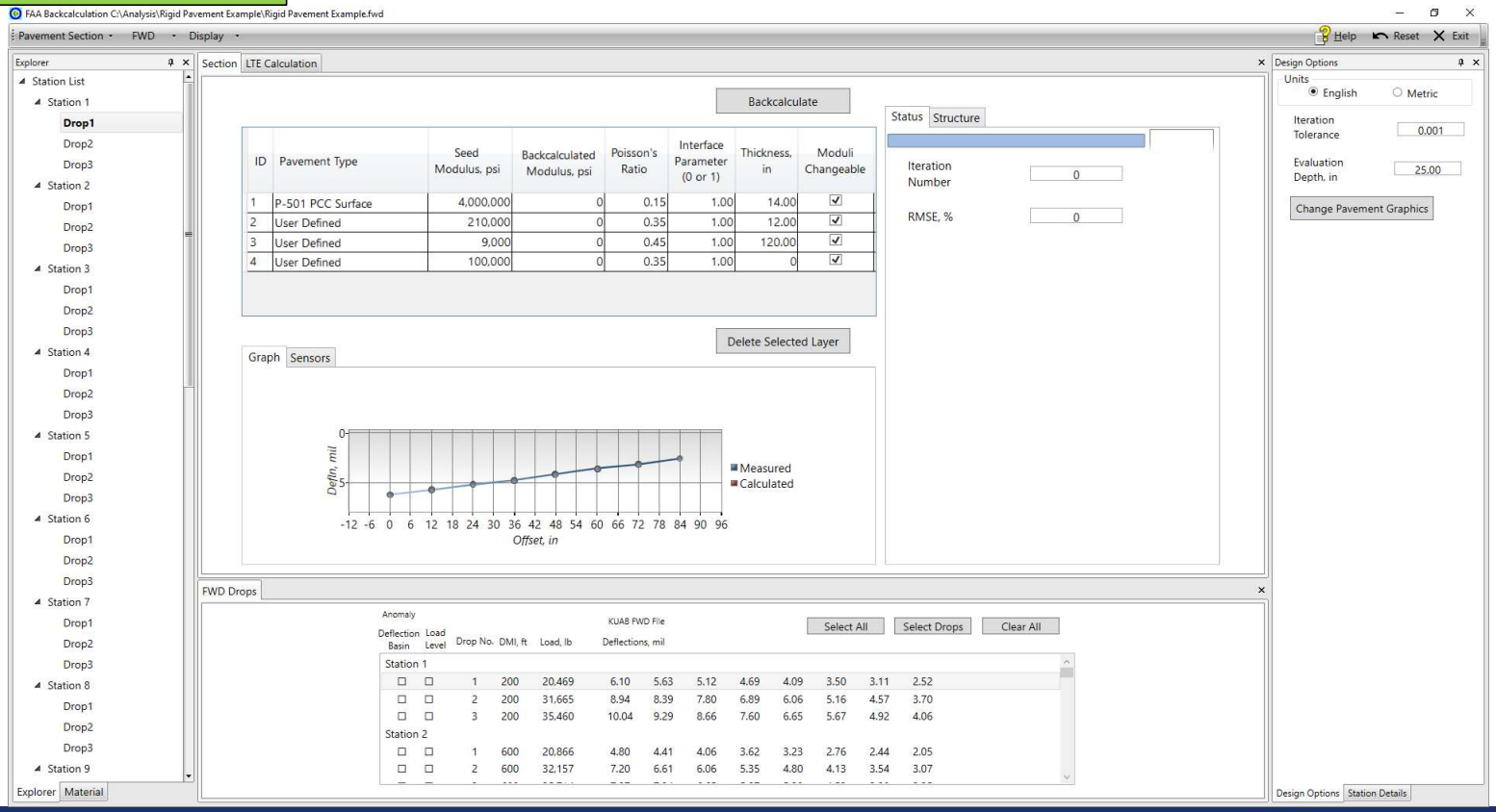

FAA Backcalculation C:\Analysis\Flexible Pavement Example\Flexible Pavement Example.fwd
Pavement Section
FWD
Display
Help
Reset
Exit
Explorer
Station List
Station 1
Drop1
Drop2
Drop3
Drop4
Section
LTE Calculation
Backcalculate
ID
Pavement Type
Seed Modulus, psi
Backcalculated Modulus, psi
Poisson's Ratio
Interface Parameter (0 or 1)
Thickness, in
Moduli Changeable
1 User Defined 910,000 1,113,820 0.35 1.00 6.00
2 P-209 Crushed Aggregate 75,000 69,687 0.35 1.00 12.00
3 User Defined 15,000 10,289 0.35 1.00 102.00
4 User Defined 100,000 100,000 0.35 1.00 0
Delete Selected Layer
Graph
Sensors
Exclude Sensor at Offset
Sensor 1 2 3 4 5
Offset, in 0.0 12.0 24.0 36.0
Defln, mil 37.5 28.8 21.8 15.6
Calc,mil 37.3 29.4 21.4 15.5
Status
Structure
Iteration Number
77 (Done)
RMSE, %
3
Important Question
The pavement structure will be saved. Do you want to save pavement section deflection data also?
Yes
No
Cancel
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
KUAB FWD File Deflections, mil
Select All
Station 1
1 18 32,115 37.06 28.41 21.59 15.50 11.52 7.54 5.50
2 18 18,020 20.77 15.97 12.13 8.69 6.49 4.24 3.11
3 18 24,694 28.46 21.85 16.56 11.93 8.87 5.79 4.26
4 18 32,544 37.50 28.78 21.80 15.61 11.62 7.58 5.54
Explorer
Material
Design Options
Station Details
Click Yes to confirm saving pavement section and FWD data


FAA Backcalculation C:\Analysis\Flexible Pavement Example\Flexible Pavement Example.fwd
Save As
This PC › OSDisk (C:) › Analysis › Flexible Pavement Example
Search Flexible Pavement Ex...
Organize
New folder
Name
Date modified
Type
Size
Flexible Pavement Structure.XML
10/31/2022 3:37 PM
XML Document
2 KB
Flexible Pavement Structure_Stiff Layer.XML
10/31/2022 3:56 PM
XML Document
3 KB
trans-mad-wi
OneDrive - Personal
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
SharePoint FAA ATRD
Videos
OSDisk (C:)
File name: Flexible Pavement Structure Example.XML
Save as type: XML file (*.XML)
Hide Folders
Save
Cancel
Help
Reset
Exit
Backcalculate
Thickness, in
Moduli Changeable
6.00
12.00
102.00
0
ete Selected Layer
Status
Structure
Iteration Number
77 (Done)
RMSE, %
3
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
Sensor
Offset, in
Defln, mil
Calc,mil
0.0
12.0
24.0
36.0
48.0
60.0
72.0
37.5
28.8
21.8
15.6
11.6
7.6
37.3
29.4
21.4
15.5
11.2
8.0
Click to save
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
KUAB FWD File
Deflections, mil
Select All
Select Drops
Clear All
Station 1
1 18 32,115 37.06 28.41 21.59 15.50 11.52 7.54 5.50
2 18 18,020 20.77 15.97 12.13 8.69 6.49 4.24 3.11
3 18 24,694 28.46 21.85 16.56 11.93 8.87 5.79 4.26
4 18 32,544 37.50 28.78 21.80 15.61 11.62 7.58 5.54
Explorer
Material
Design Options
Station Details

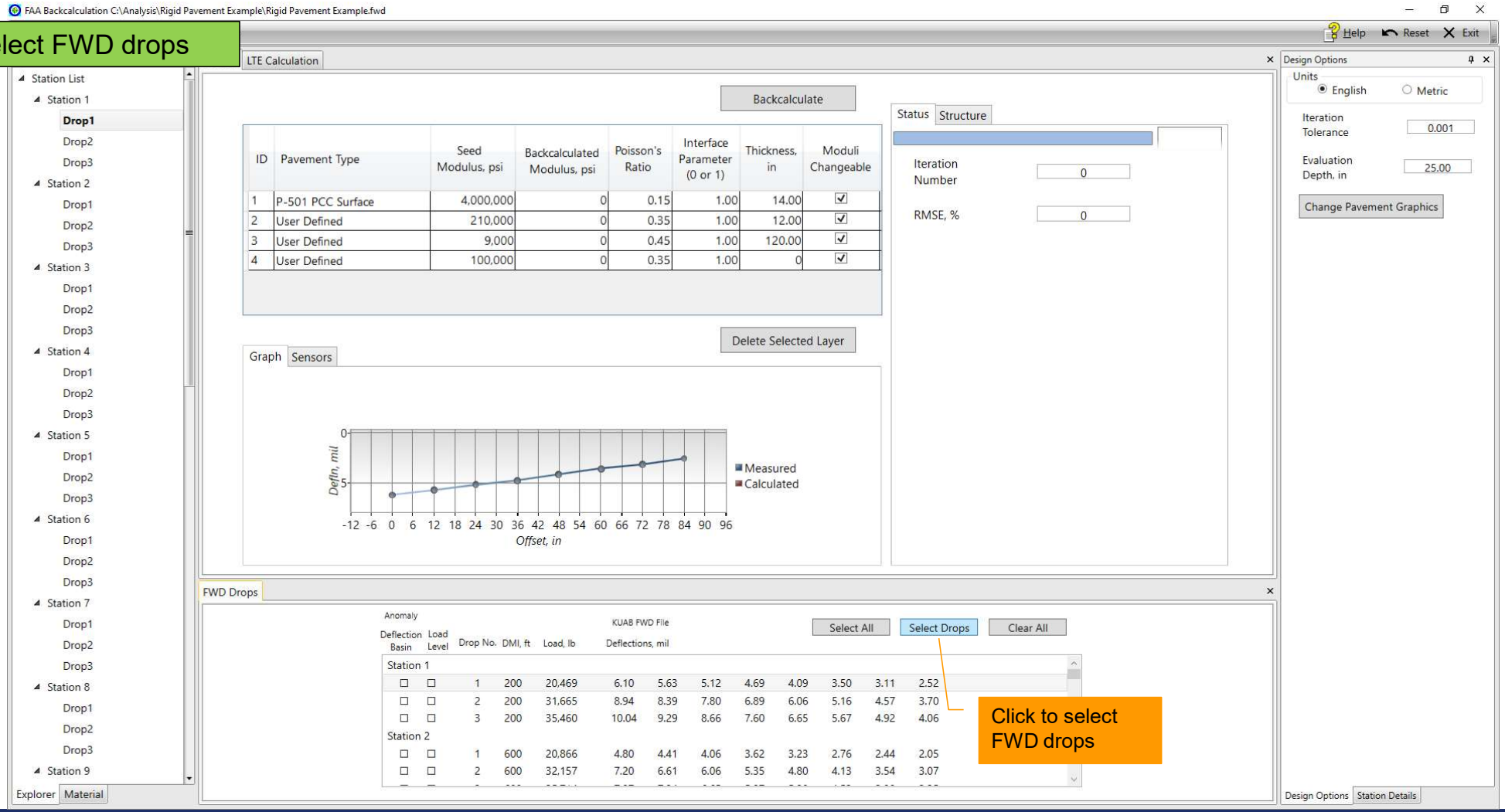
Rigid Pavement Example

- Load pavement section
- Load FWD file
- Assign layers and thickness
- Assign seed modulus

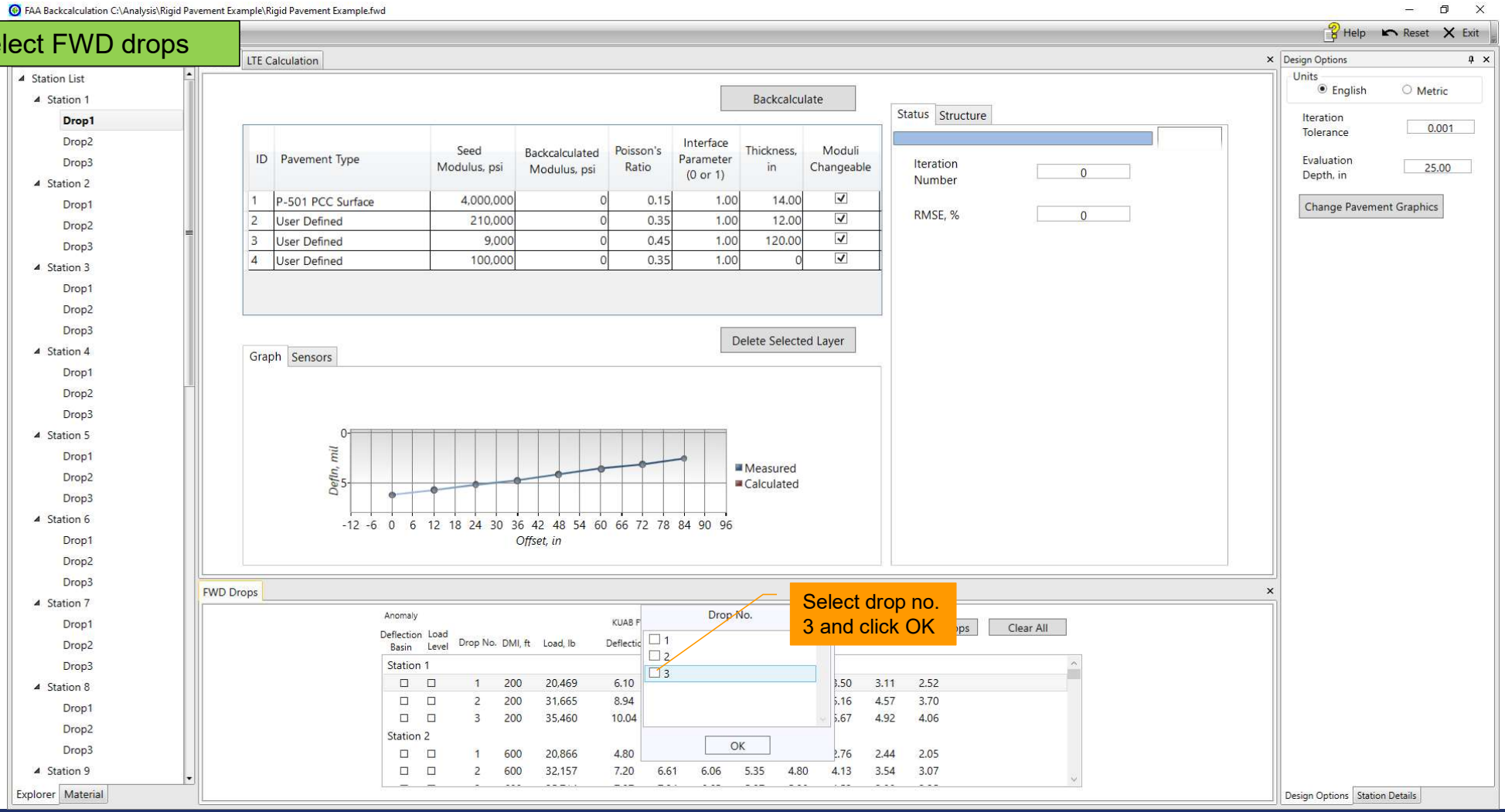
Material Type	Thickness (in)
P-501 cement concrete pavement	14
P-156 cement treated subgrade	12
Subgrade with a CBR of 6	



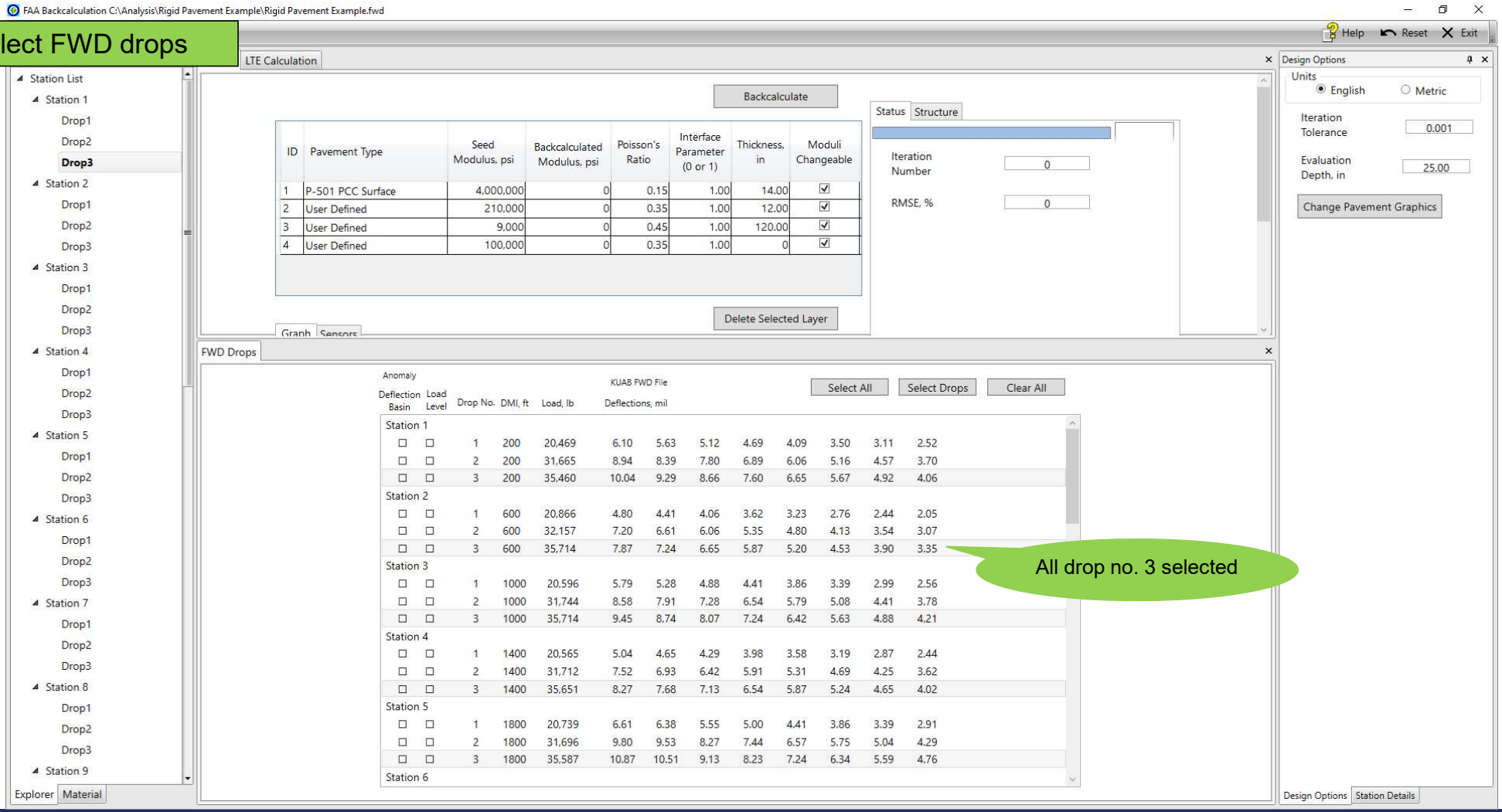
Select FWD drops



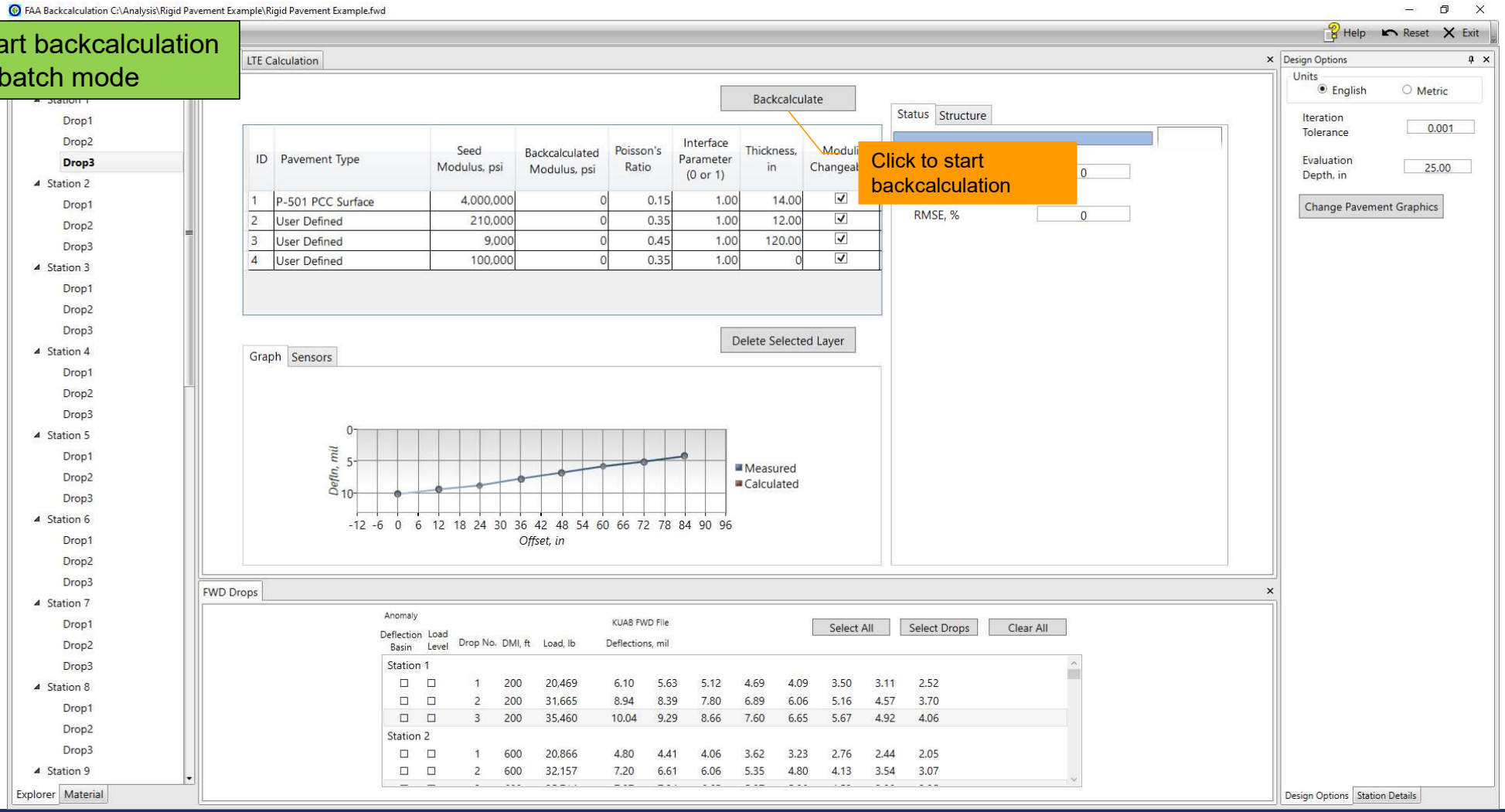
Select FWD drops



Select FWD drops



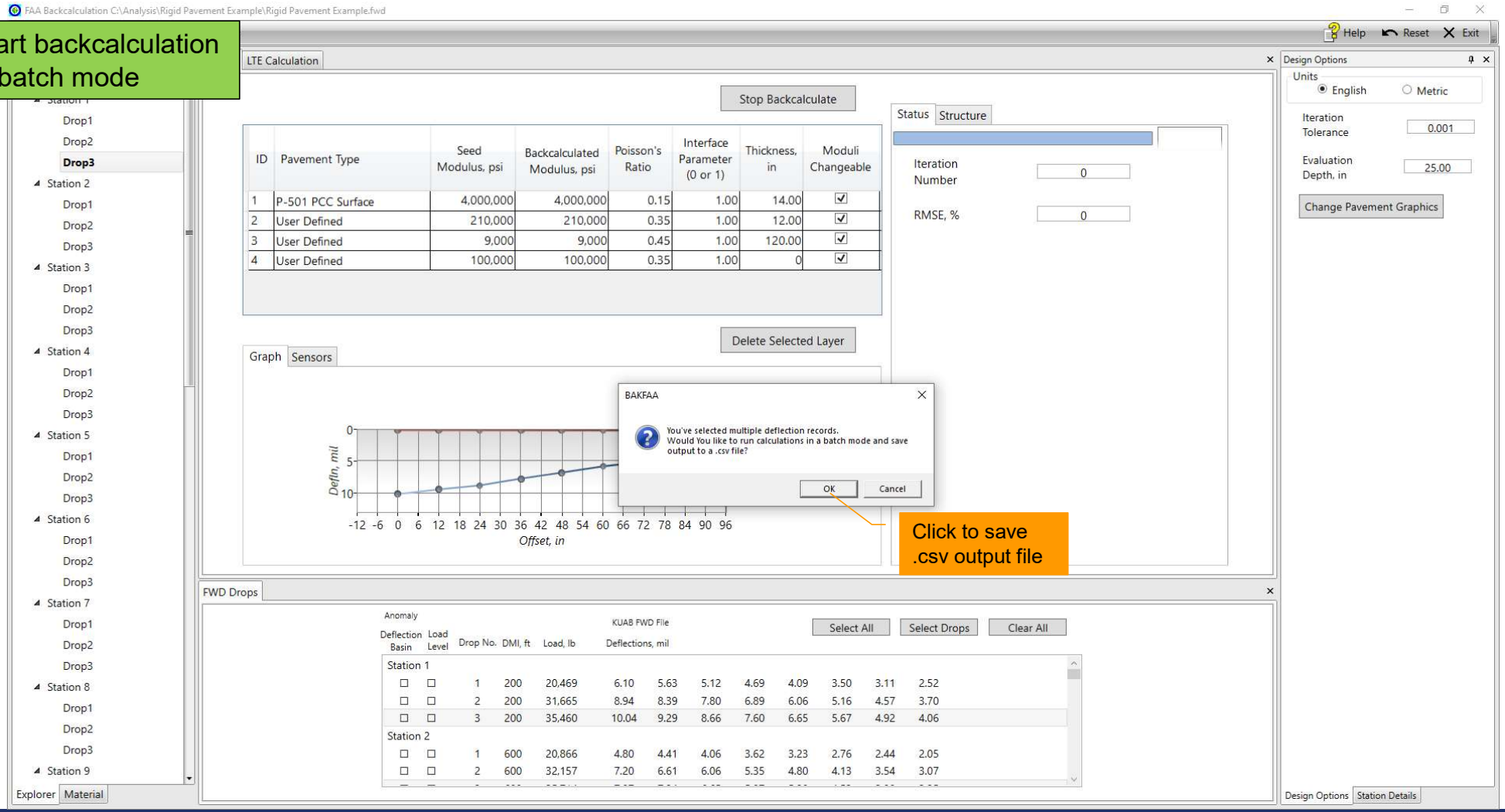
Start backcalculation in batch mode

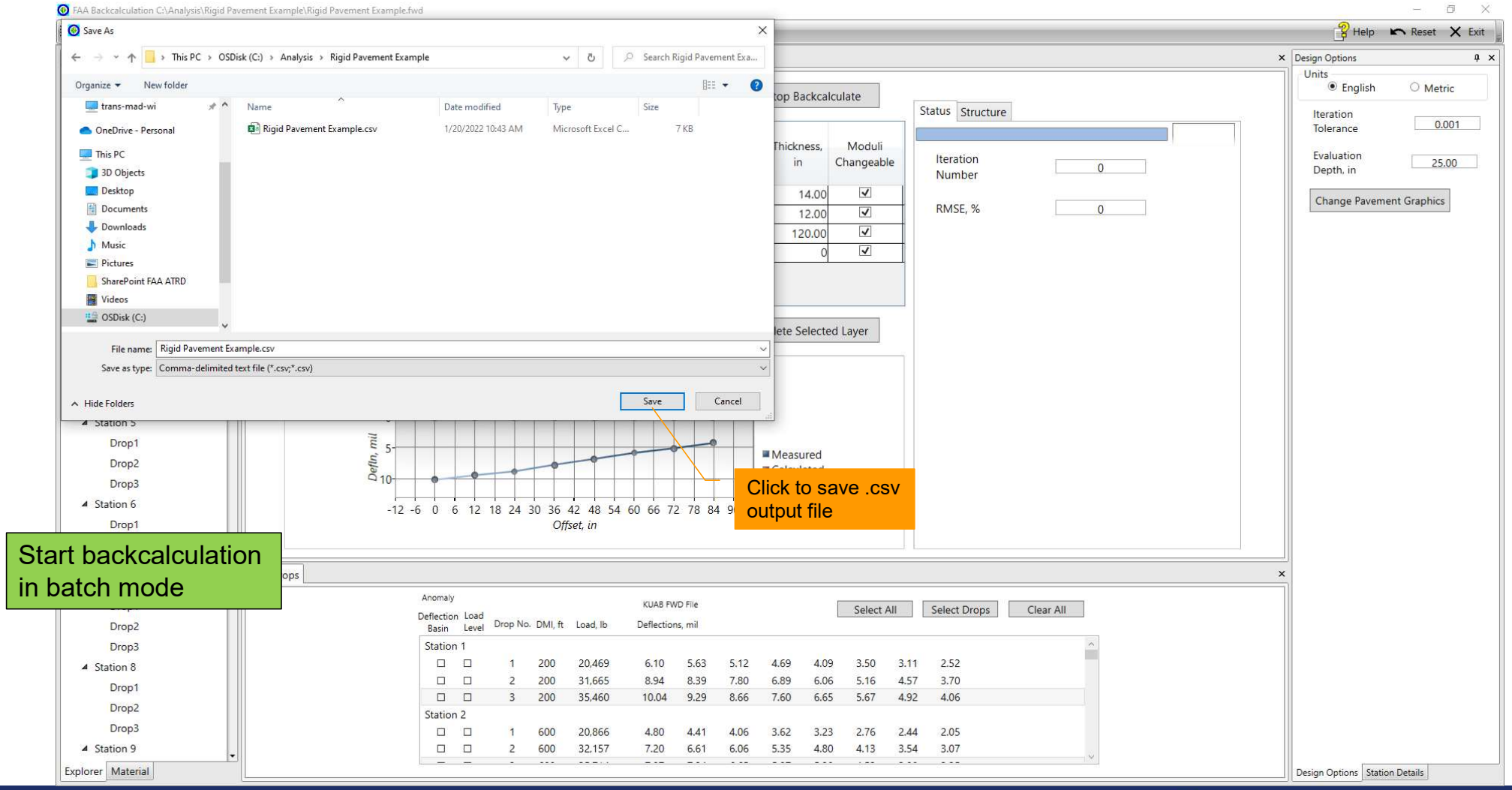


ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeat
1	P-501 PCC Surface	4,000,000	0	0.15	1.00	14.00	☑
2	User Defined	210,000	0	0.35	1.00	12.00	☑
3	User Defined	9,000	0	0.45	1.00	120.00	☑
4	User Defined	100,000	0	0.35	1.00	0	☑

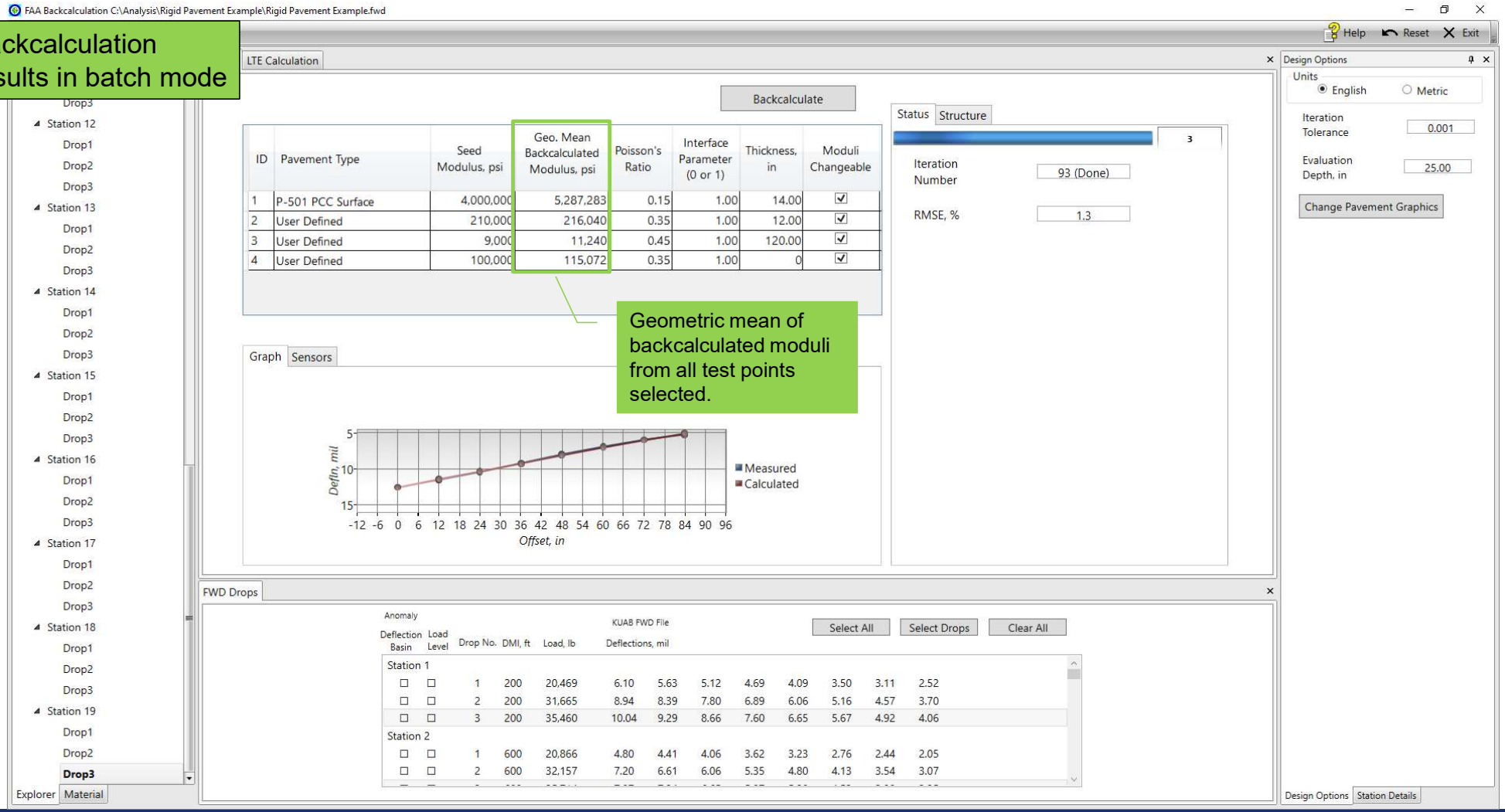
Anomaly Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	KUAB FWD File Deflections, mil							
Station 1												
☐	☐	1	200	20,469	6.10	5.63	5.12	4.69	4.09	3.50	3.11	2.52
☐	☐	2	200	31,665	8.94	8.39	7.80	6.89	6.06	5.16	4.57	3.70
☐	☐	3	200	35,460	10.04	9.29	8.66	7.60	6.65	5.67	4.92	4.06
Station 2												
☐	☐	1	600	20,866	4.80	4.41	4.06	3.62	3.23	2.76	2.44	2.05
☐	☐	2	600	32,157	7.20	6.61	6.06	5.35	4.80	4.13	3.54	3.07

Start backcalculation in batch mode




Start backcalculation in batch mode
Click to save .csv output file
Save As
File name: Rigid Pavement Example.csv
Save as type: Comma-delimited text file (*.csv;*.csv)
Save
Cancel
Hide Folders
Help
Reset
Exit
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
Status
Structure
Iteration Number
RMSE, %
Measured
Offset, in
Select All
Select Drops
Clear All
Explorer
Material
Design Options
Station Details

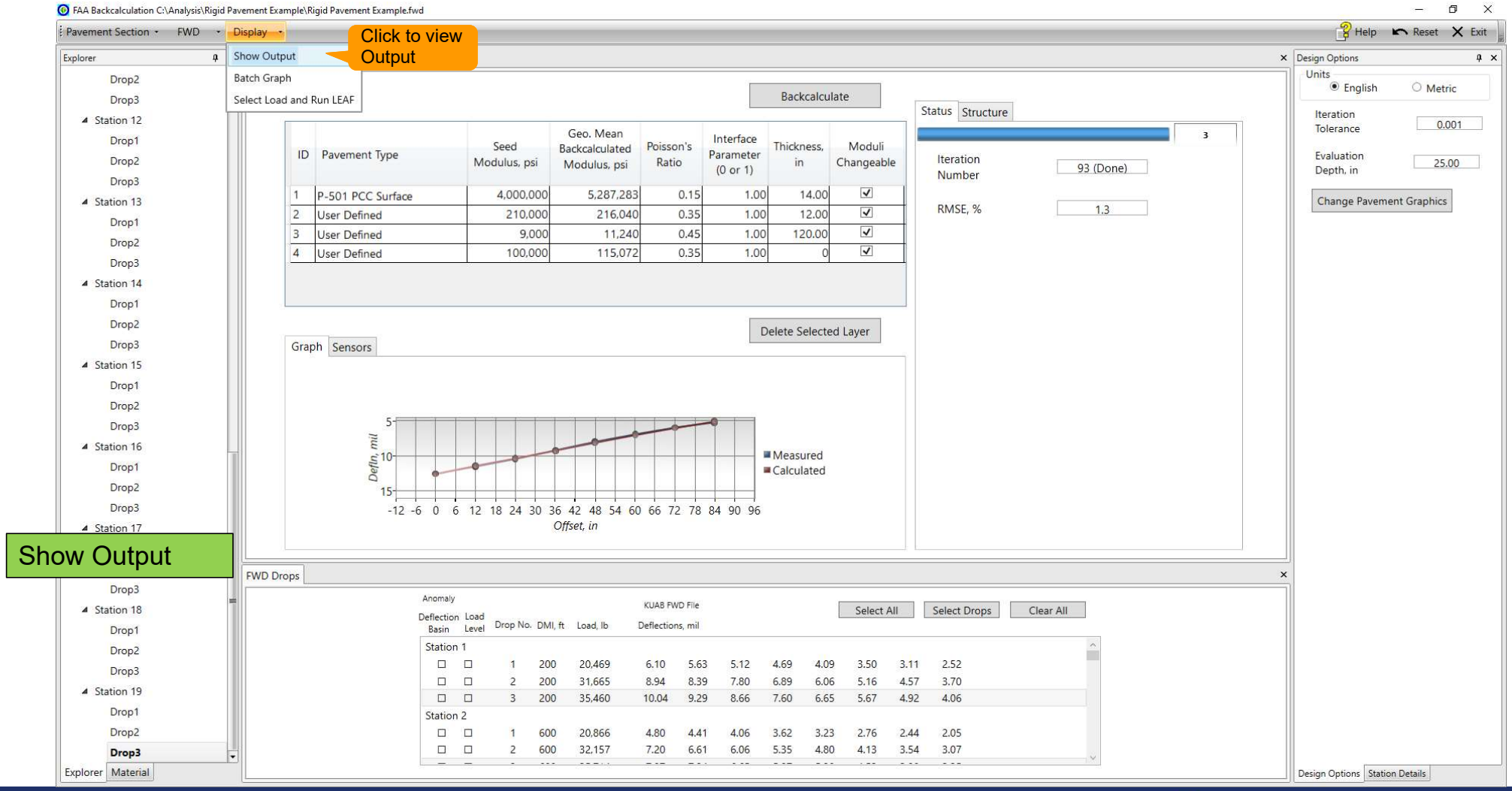
Backcalculation results in batch mode



ID	Pavement Type	Seed Modulus, psi	Geo. Mean Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	P-501 PCC Surface	4,000,000	5,287,283	0.15	1.00	14.00	☑
2	User Defined	210,000	216,040	0.35	1.00	12.00	☑
3	User Defined	9,000	11,240	0.45	1.00	120.00	☑
4	User Defined	100,000	115,072	0.35	1.00	0	☑

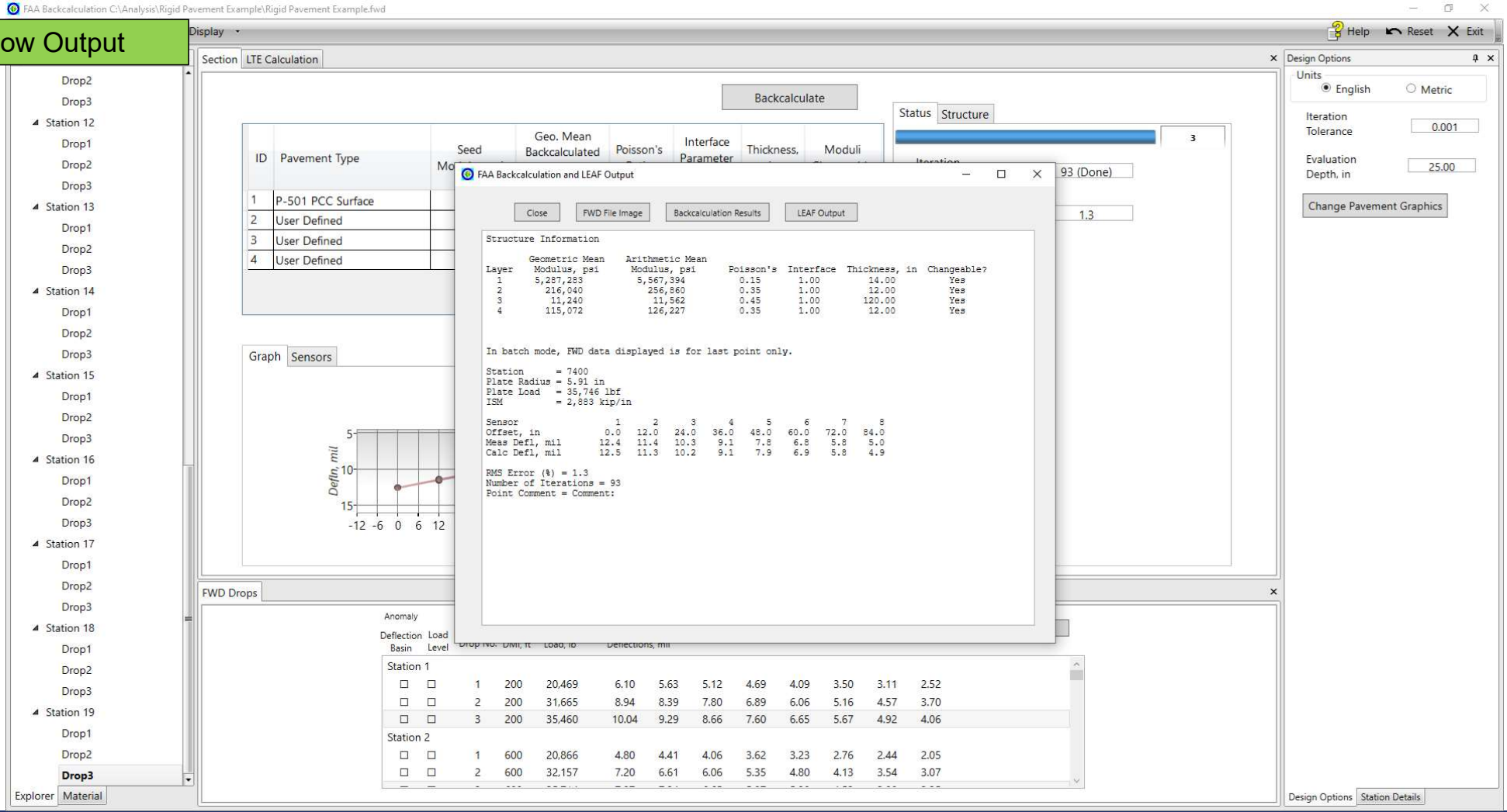
Geometric mean of backcalculated moduli from all test points selected.

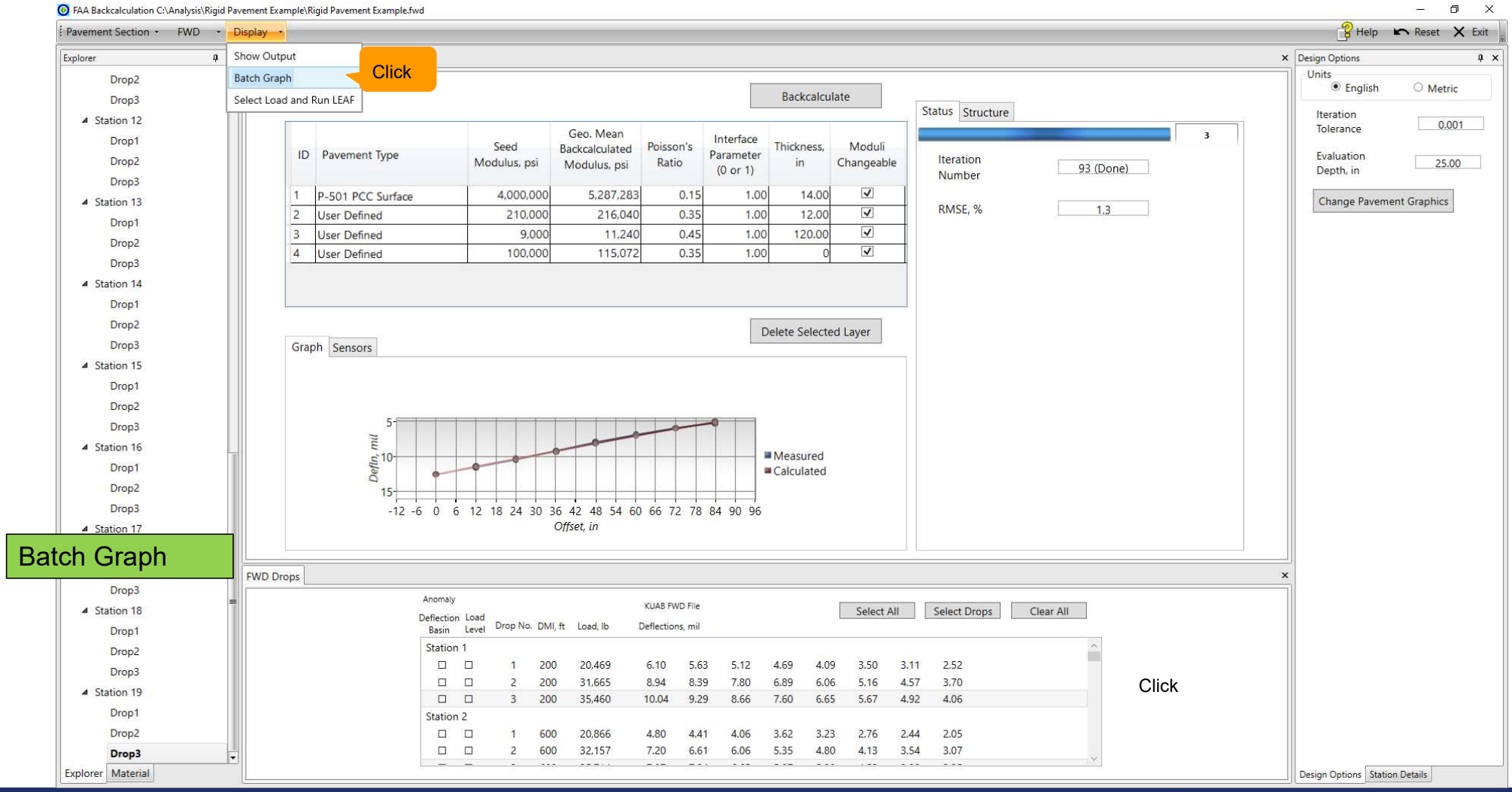
Anomaly: Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	KUAB FWD File: Deflections, mil							
Station 1												
☐	☐	1	200	20,469	6.10	5.63	5.12	4.69	4.09	3.50	3.11	2.52
☐	☐	2	200	31,665	8.94	8.39	7.80	6.89	6.06	5.16	4.57	3.70
☐	☐	3	200	35,460	10.04	9.29	8.66	7.60	6.65	5.67	4.92	4.06
Station 2												
☐	☐	1	600	20,866	4.80	4.41	4.06	3.62	3.23	2.76	2.44	2.05
☐	☐	2	600	32,157	7.20	6.61	6.06	5.35	4.80	4.13	3.54	3.07


FAA Backcalculation C:\Analysis\Rigid Pavement Example\Rigid Pavement Example.fwd
Pavement Section
FWD
Display
Show Output
Batch Graph
Select Load and Run LEAF
Click to view Output
Help
Reset
Exit
Explorer
Drop2
Drop3
Station 12
Drop1
Drop2
Drop3
Station 13
Drop1
Drop2
Drop3
Station 14
Drop1
Drop2
Drop3
Station 15
Drop1
Drop2
Drop3
Station 16
Drop1
Drop2
Drop3
Station 17
Drop3
Station 18
Drop1
Drop2
Drop3
Station 19
Drop1
Drop2
Drop3
Explorer
Material
Backcalculate
ID
Pavement Type
Seed Modulus, psi
Geo. Mean Backcalculated Modulus, psi
Poisson's Ratio
Interface Parameter (0 or 1)
Thickness, in
Moduli Changeable
1 P-501 PCC Surface 4,000,000 5,287,283 0.15 1.00 14.00
2 User Defined 210,000 216,040 0.35 1.00 12.00
3 User Defined 9,000 11,240 0.45 1.00 120.00
4 User Defined 100,000 115,072 0.35 1.00 0
Delete Selected Layer
Graph
Sensors
Defln, mil
Offset, in
Measured
Calculated
Status
Structure
Iteration Number
93 (Done)
RMSE, %
1.3
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
Design Options
Station Details
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
KUAB FWD File
Deflections, mil
Select All
Select Drops
Clear All
Station 1
1 200 20,469 6.10 5.63 5.12 4.69 4.09 3.50 3.11 2.52
2 200 31,665 8.94 8.39 7.80 6.89 6.06 5.16 4.57 3.70
3 200 35,460 10.04 9.29 8.66 7.60 6.65 5.67 4.92 4.06
Station 2
1 600 20,866 4.80 4.41 4.06 3.62 3.23 2.76 2.44 2.05
2 600 32,157 7.20 6.61 6.06 5.35 4.80 4.13 3.54 3.07

Show Output

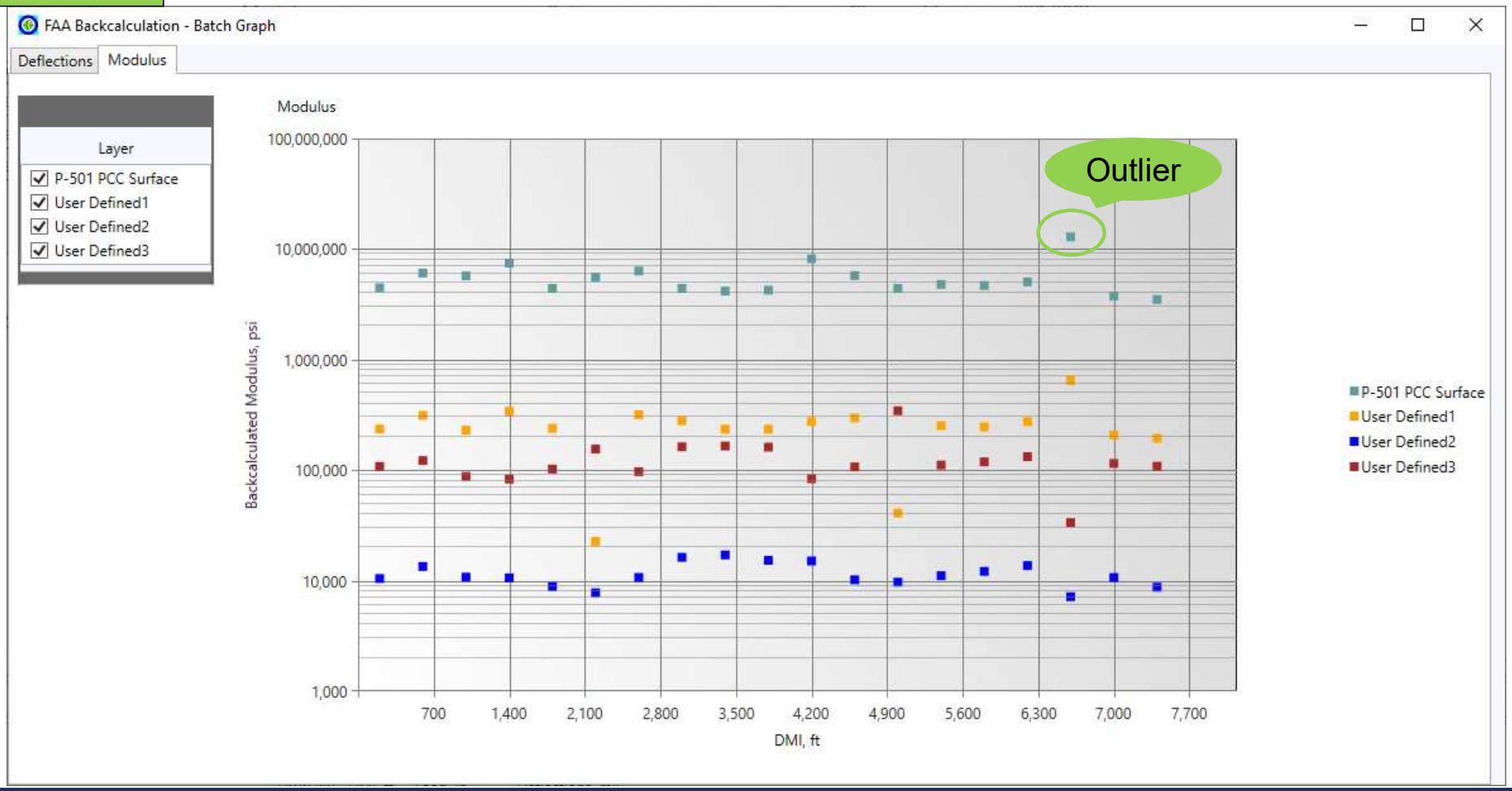
Show Output

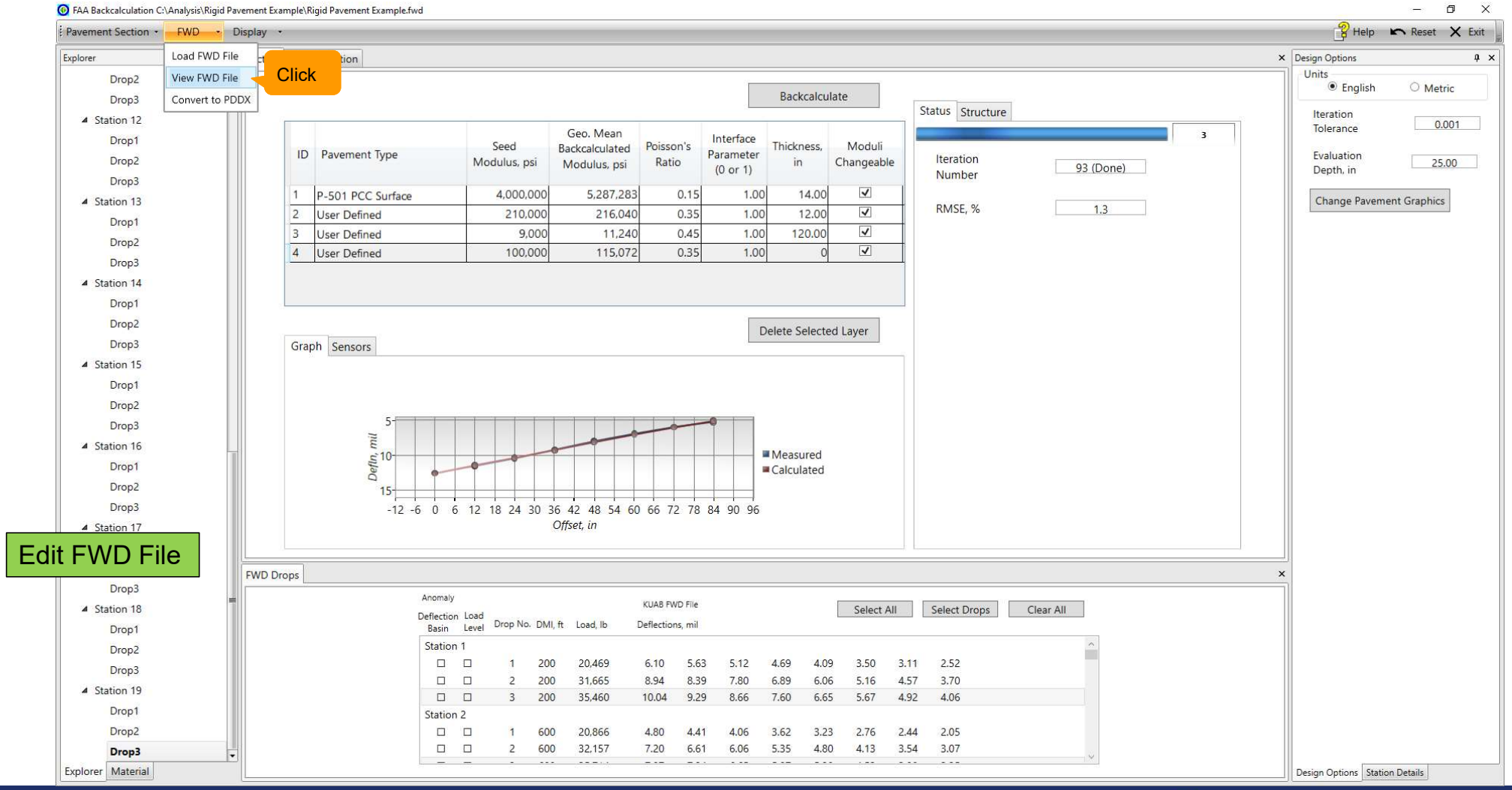

FAA Backcalculation C:\Analysis\Rigid Pavement Example\Rigid Pavement Example.fwd
Display
Help
Reset
Exit
Section
LTE Calculation
Backcalculate
Status
Structure
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
ID
Pavement Type
Seed
Geo. Mean Backcalculated
Poisson's
Interface Parameter
Thickness,
Moduli
1 P-501 PCC Surface
2 User Defined
3 User Defined
4 User Defined
Graph
Sensors
Defln, mil
FAA Backcalculation and LEAF Output
Close
FWD File Image
Backcalculation Results
LEAF Output
Structure Information
Geometric Mean Modulus, psi
Arithmetic Mean Modulus, psi
Poisson's
Interface
Thickness, in
Changeable?
1 5,287,283 5,567,394 0.15 1.00 14.00 Yes
2 216,040 256,860 0.35 1.00 12.00 Yes
3 11,240 11,562 0.45 1.00 120.00 Yes
4 115,072 126,227 0.35 1.00 12.00 Yes
In batch mode, FWD data displayed is for last point only.
Station = 7400
Plate Radius = 5.91 in
Plate Load = 35,746 lbf
ISM = 2,883 kip/in
Sensor 1 2 3 4 5 6 7 8
Offset, in 0.0 12.0 24.0 36.0 48.0 60.0 72.0 84.0
Meas Defl, mil 12.4 11.4 10.3 9.1 7.8 6.8 5.8 5.0
Calc Defl, mil 12.5 11.3 10.2 9.1 7.9 6.9 5.8 4.9
RMS Error (%) = 1.3
Number of Iterations = 93
Point Comment = Comment:
93 (Done)
1.3
FWD Drops
Anomaly
Deflection Basin
Load Level
Station 1
1 200 20,469 6.10 5.63 5.12 4.69 4.09 3.50 3.11 2.52
2 200 31,665 8.94 8.39 7.80 6.89 6.06 5.16 4.57 3.70
3 200 35,460 10.04 9.29 8.66 7.60 6.65 5.67 4.92 4.06
Station 2
1 600 20,866 4.80 4.41 4.06 3.62 3.23 2.76 2.44 2.05
2 600 32,157 7.20 6.61 6.06 5.35 4.80 4.13 3.54 3.07
Drop3
Explorer
Material
Design Options
Station Details


FAA Backcalculation C:\Analysis\Rigid Pavement Example\Rigid Pavement Example.fwd
Pavement Section
FWD
Display
Help
Reset
Exit
Show Output
Batch Graph
Select Load and Run LEAF
Click
Explorer
Drop2
Drop3
Station 12
Drop1
Drop2
Drop3
Station 13
Drop1
Drop2
Drop3
Station 14
Drop1
Drop2
Drop3
Station 15
Drop1
Drop2
Drop3
Station 16
Drop1
Drop2
Drop3
Station 17
Drop3
Station 18
Drop1
Drop2
Drop3
Station 19
Drop1
Drop2
Drop3
Explorer
Material
Backcalculate
ID
Pavement Type
Seed Modulus, psi
Geo. Mean Backcalculated Modulus, psi
Poisson's Ratio
Interface Parameter (0 or 1)
Thickness, in
Moduli Changeable
1 P-501 PCC Surface 4,000,000 5,287,283 0.15 1.00 14.00
2 User Defined 210,000 216,040 0.35 1.00 12.00
3 User Defined 9,000 11,240 0.45 1.00 120.00
4 User Defined 100,000 115,072 0.35 1.00 0
Delete Selected Layer
Graph
Sensors
Defln, mil
Offset, in
Measured
Calculated
Status
Structure
Iteration Number
93 (Done)
RMSE, %
1.3
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
Design Options
Station Details
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
KUAB FWD File
Deflections, mil
Select All
Select Drops
Clear All
Station 1
1 200 20,469 6.10 5.63 5.12 4.69 4.09 3.50 3.11 2.52
2 200 31,665 8.94 8.39 7.80 6.89 6.06 5.16 4.57 3.70
3 200 35,460 10.04 9.29 8.66 7.60 6.65 5.67 4.92 4.06
Station 2
1 600 20,866 4.80 4.41 4.06 3.62 3.23 2.76 2.44 2.05
2 600 32,157 7.20 6.61 6.06 5.35 4.80 4.13 3.54 3.07
Click

Batch Graph

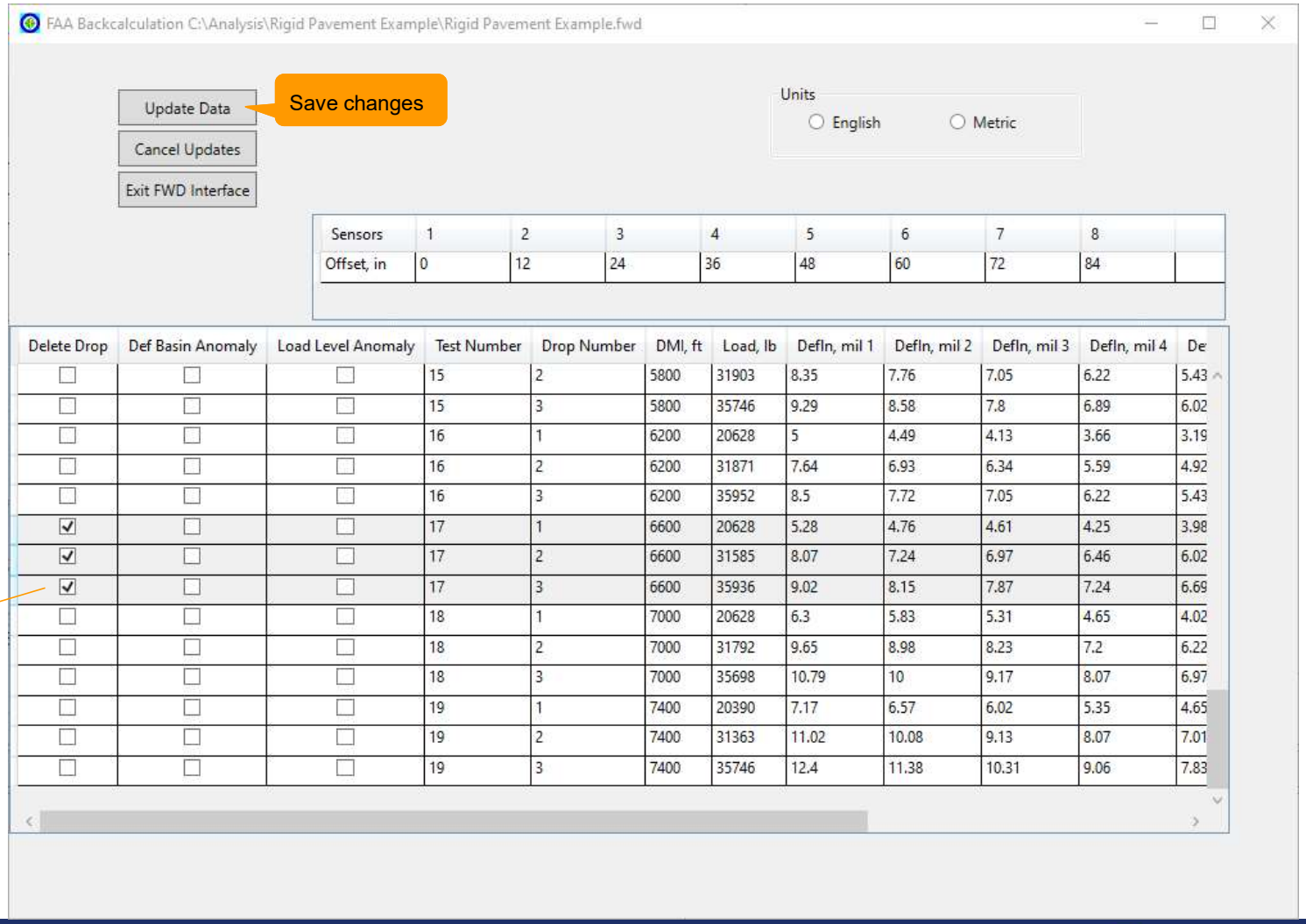
Batch Graph


FAA Backcalculation - Batch Graph
Deflections
Modulus
Layer
P-501 PCC Surface
User Defined1
User Defined2
User Defined3
Modulus
Outlier
Backcalculated Modulus, psi
100,000,000
10,000,000
1,000,000
100,000
10,000
1,000
700
1,400
2,100
2,800
3,500
4,200
4,900
5,600
6,300
7,000
7,700
DMI, ft
P-501 PCC Surface
User Defined1
User Defined2
User Defined3



Edit FWD File

Edit FWD File

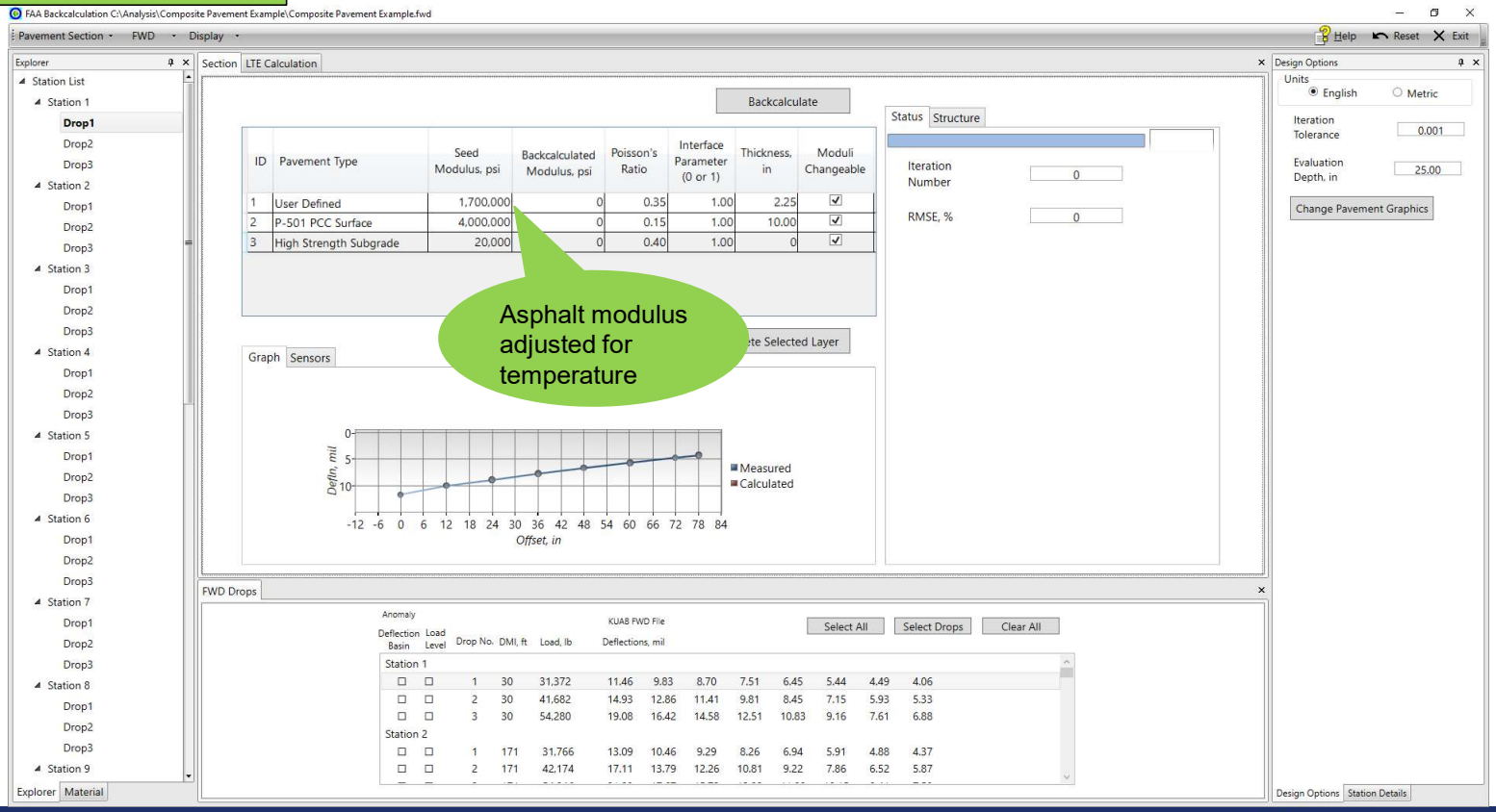


Click to delete outlier drops

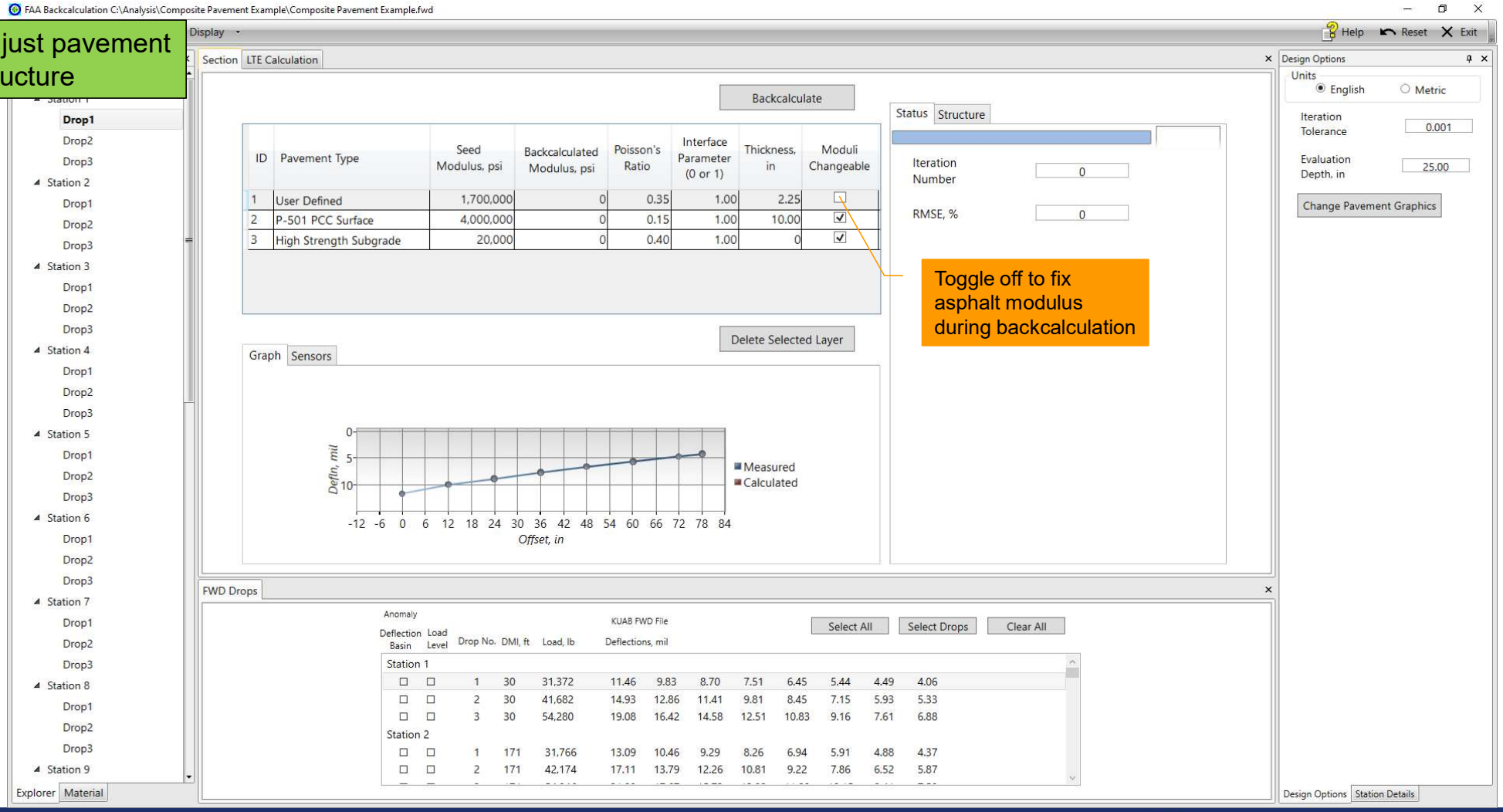
Composite Pavement Example

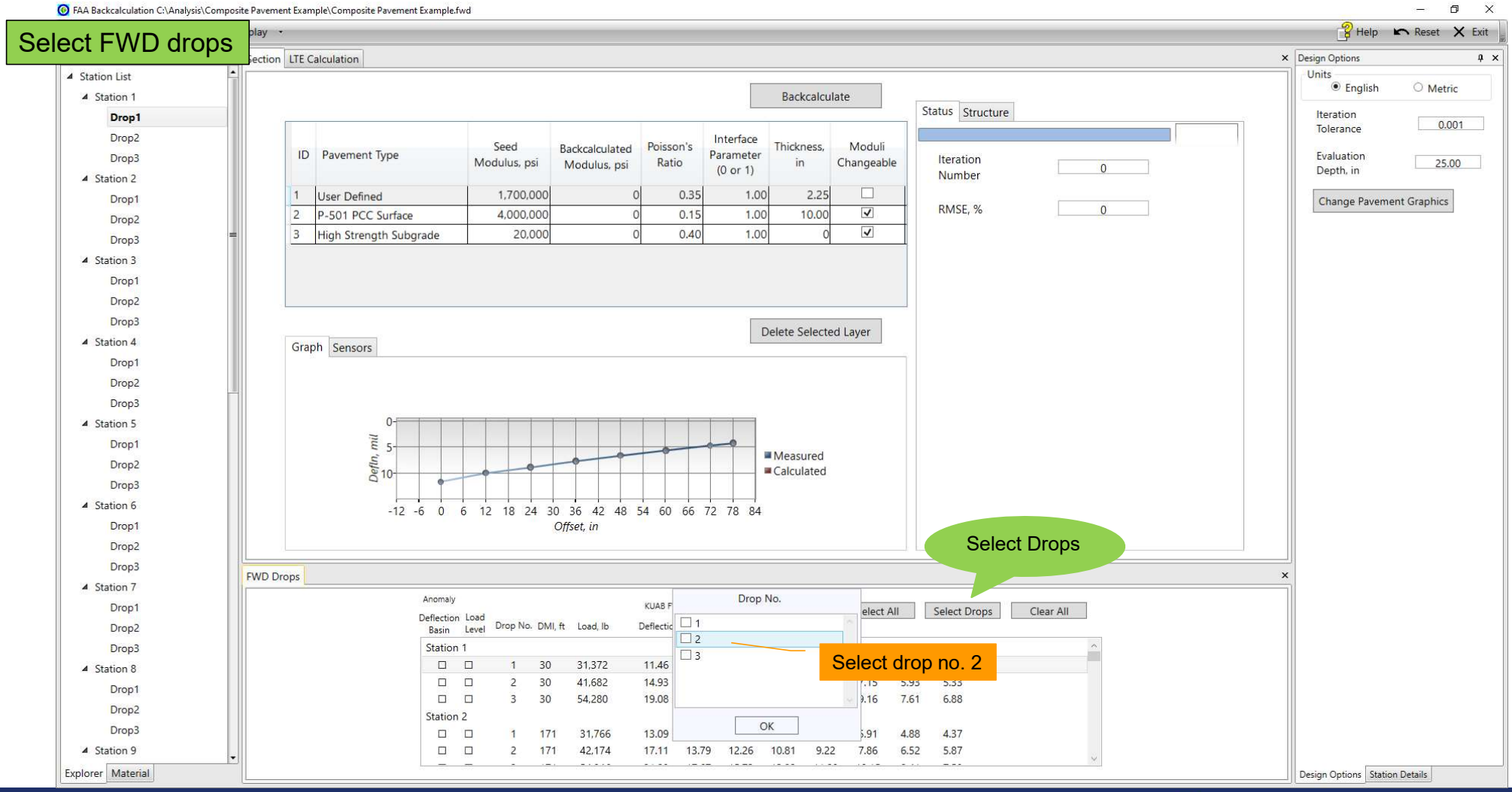
- Load pavement section
- Load FWD file
- Assign layers and thickness
- Assign seed modulus

Material Type	Thickness (in.)
P-401/P-403 hot mix asphalt	2.25
P-501 Portland cement concrete	10
Subgrade with a CBR of 15	

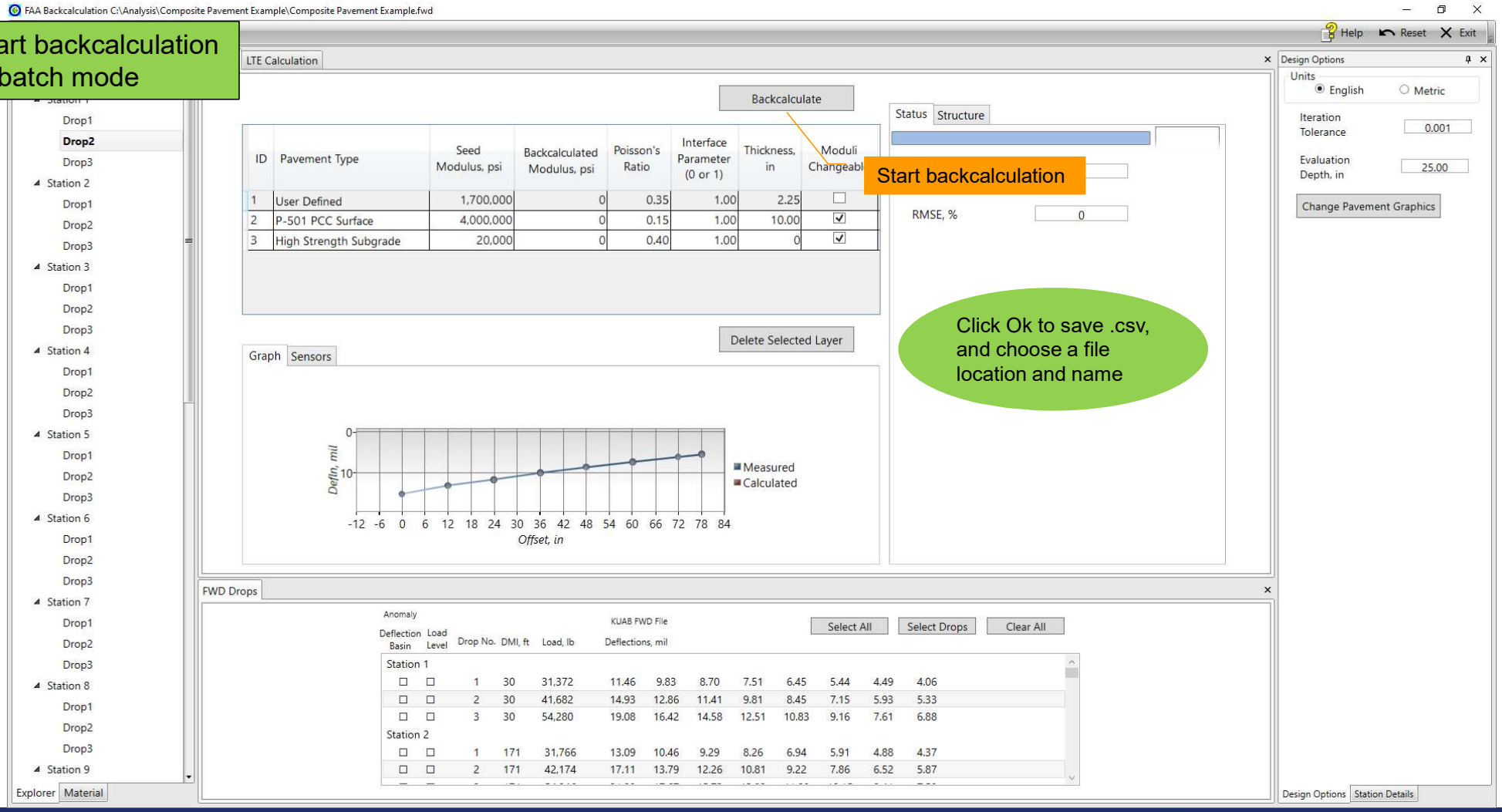


Adjust pavement structure




Select FWD drops
FAA Backcalculation C:\Analysis\Composite Pavement Example\Composite Pavement Example.fwd
Help
Reset
Exit
Station List
Station 1
Drop1
Drop2
Drop3
Station 2
Station 3
Station 4
Station 5
Station 6
Station 7
Station 8
Station 9
Explorer
Material
LTE Calculation
Backcalculate
ID
Pavement Type
Seed Modulus, psi
Backcalculated Modulus, psi
Poisson's Ratio
Interface Parameter (0 or 1)
Thickness, in
Moduli Changeable
1 User Defined 1,700,000 0 0.35 1.00 2.25
2 P-501 PCC Surface 4,000,000 0 0.15 1.00 10.00
3 High Strength Subgrade 20,000 0 0.40 1.00 0
Delete Selected Layer
Graph
Sensors
Defln, mil
Offset, in
Measured
Calculated
Status
Structure
Iteration Number
RMSE, %
Design Options
Units
English
Metric
Iteration Tolerance
0.001
Evaluation Depth, in
25.00
Change Pavement Graphics
Select Drops
FWD Drops
Anomaly
Deflection Basin
Load Level
Drop No.
DMI, ft
Load, lb
Station 1
Station 2
Drop No.
1
2
3
OK
Select All
Select Drops
Clear All
Select drop no. 2
Design Options
Station Details

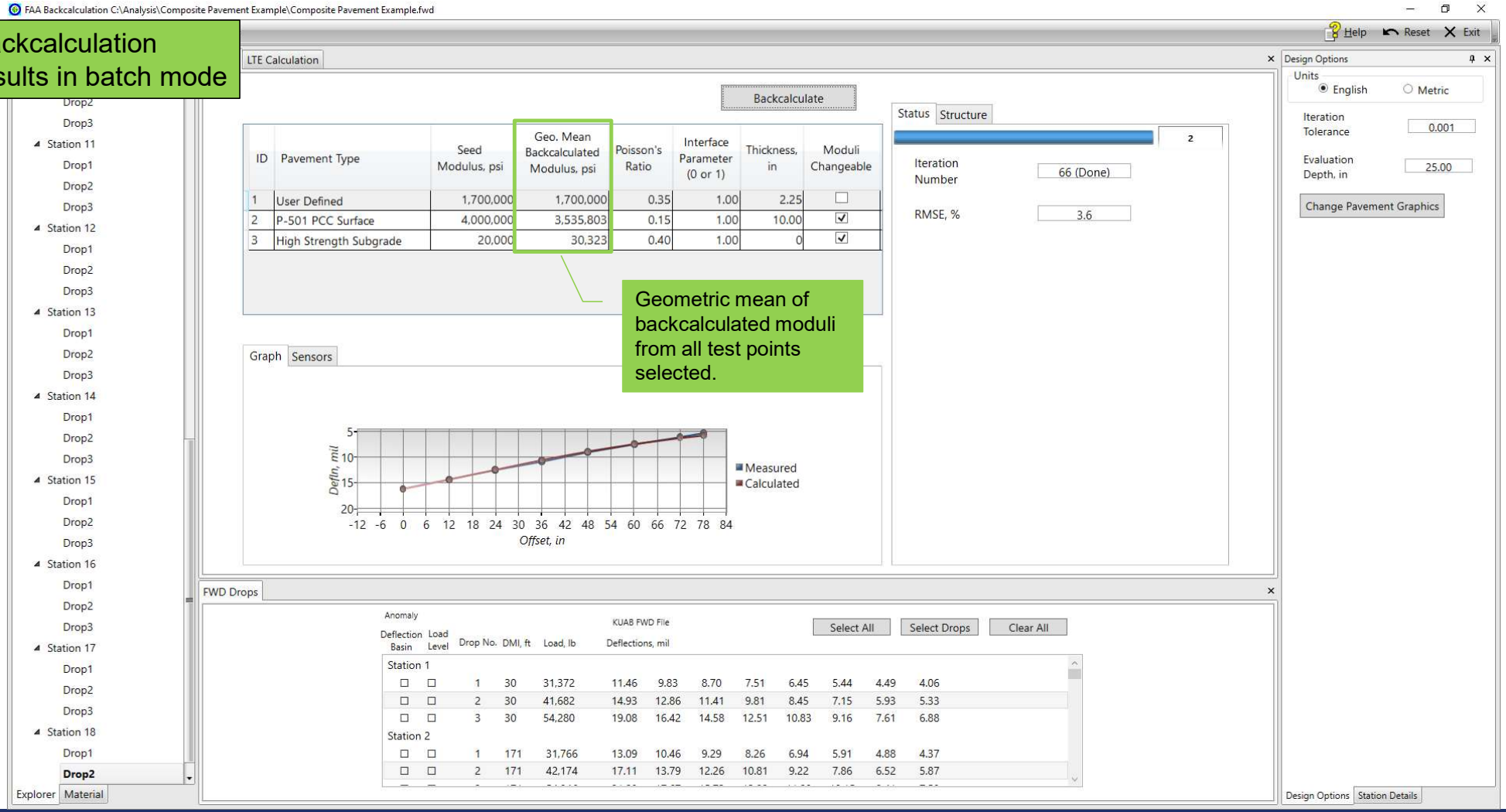
Start backcalculation in batch mode



ID	Pavement Type	Seed Modulus, psi	Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeabl
1	User Defined	1,700,000	0	0.35	1.00	2.25	☐
2	P-501 PCC Surface	4,000,000	0	0.15	1.00	10.00	☑
3	High Strength Subgrade	20,000	0	0.40	1.00	0	☑

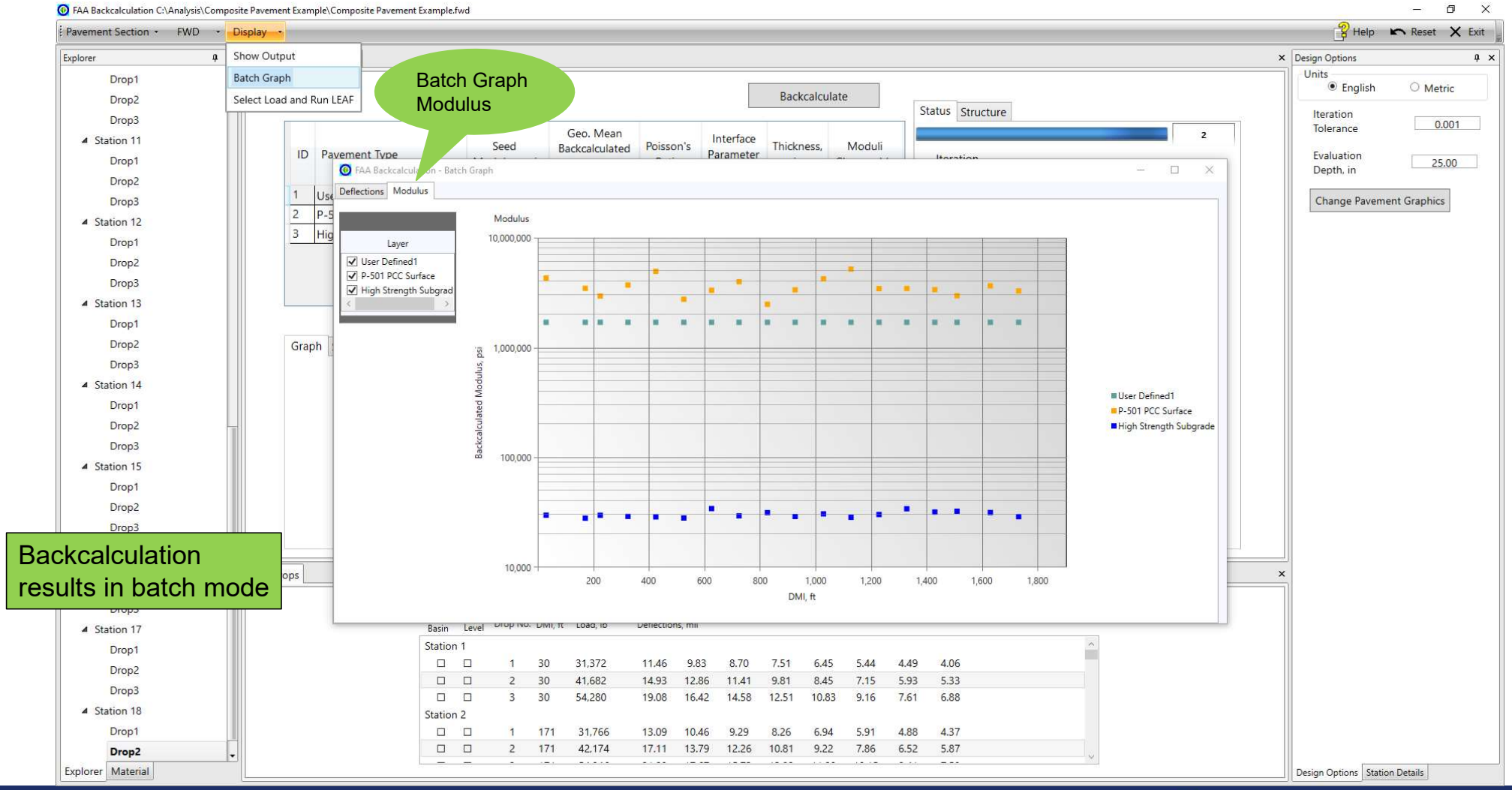
Anomaly Deflection Basin	Load Level	Drop No.	DMI, ft	Load, lb	KUAB FWD File Deflections, mil							
Station 1												
☐	☐	1	30	31,372	11.46	9.83	8.70	7.51	6.45	5.44	4.49	4.06
☐	☐	2	30	41,682	14.93	12.86	11.41	9.81	8.45	7.15	5.93	5.33
☐	☐	3	30	54,280	19.08	16.42	14.58	12.51	10.83	9.16	7.61	6.88
Station 2												
☐	☐	1	171	31,766	13.09	10.46	9.29	8.26	6.94	5.91	4.88	4.37
☐	☐	2	171	42,174	17.11	13.79	12.26	10.81	9.22	7.86	6.52	5.87

Backcalculation results in batch mode



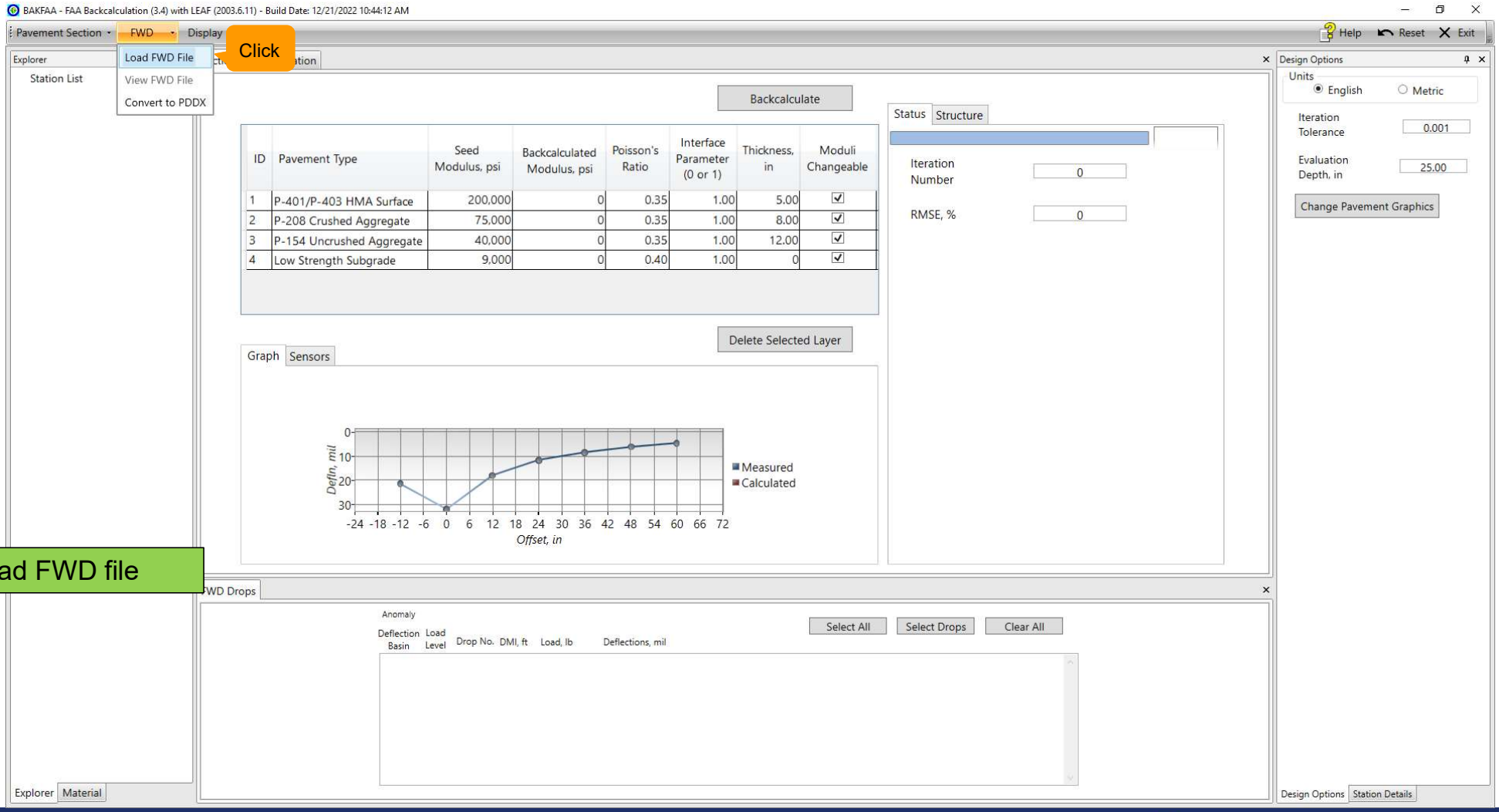
ID	Pavement Type	Seed Modulus, psi	Geo. Mean Backcalculated Modulus, psi	Poisson's Ratio	Interface Parameter (0 or 1)	Thickness, in	Moduli Changeable
1	User Defined	1,700,000	1,700,000	0.35	1.00	2.25	☐
2	P-501 PCC Surface	4,000,000	3,535,803	0.15	1.00	10.00	☑
3	High Strength Subgrade	20,000	30,323	0.40	1.00	0	☑

Anomaly Deflection Basin	Anomaly Load Level	Drop No.	DMI, ft	Load, lb	KUAB FWD File Deflections, mil							
Station 1												
☐	☐	1	30	31,372	11.46	9.83	8.70	7.51	6.45	5.44	4.49	4.06
☐	☐	2	30	41,682	14.93	12.86	11.41	9.81	8.45	7.15	5.93	5.33
☐	☐	3	30	54,280	19.08	16.42	14.58	12.51	10.83	9.16	7.61	6.88
Station 2												
☐	☐	1	171	31,766	13.09	10.46	9.29	8.26	6.94	5.91	4.88	4.37
☐	☐	2	171	42,174	17.11	13.79	12.26	10.81	9.22	7.86	6.52	5.87

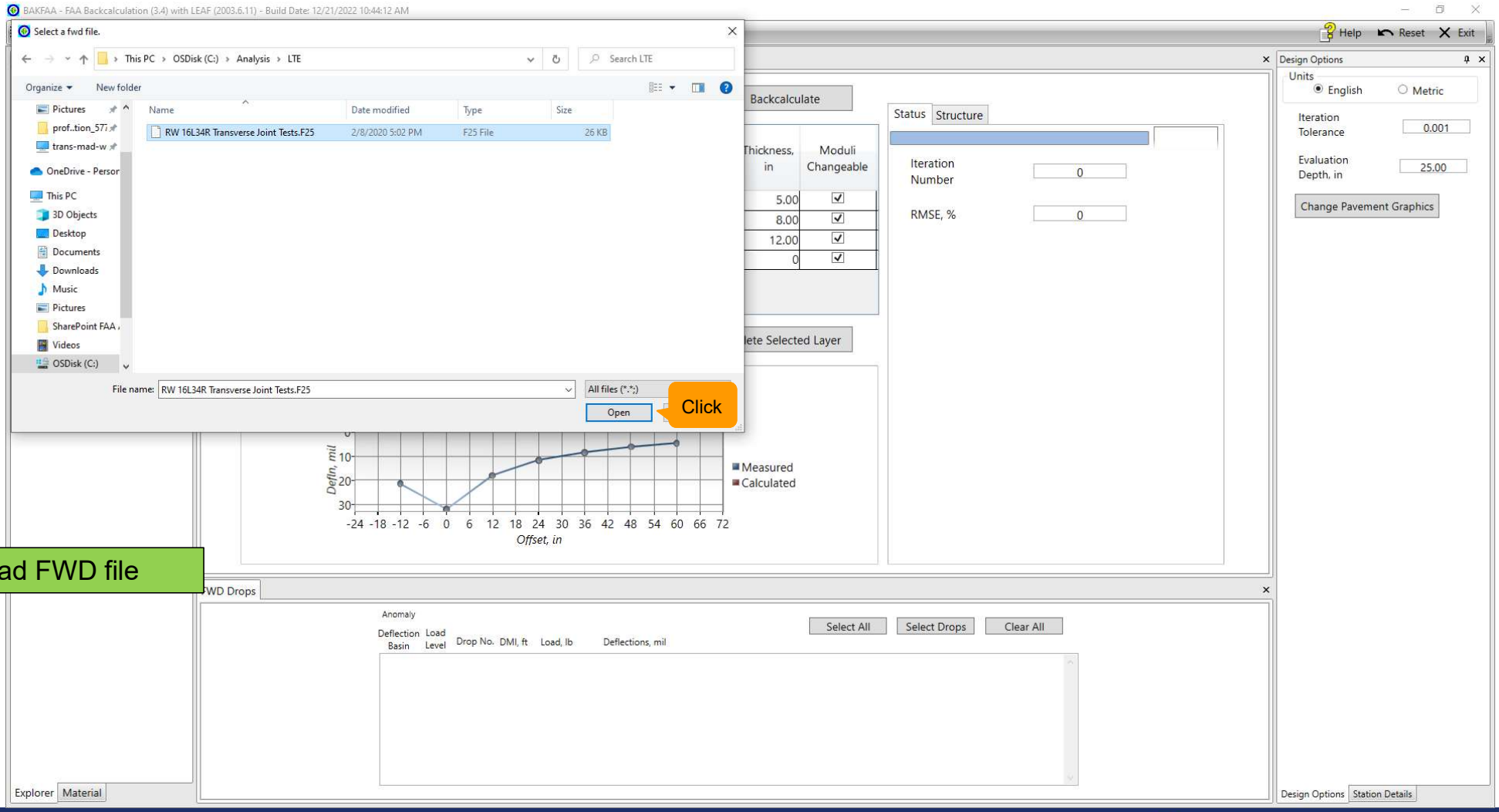


Backcalculation results in batch mode

Load Transfer Efficiency (LTE) Example

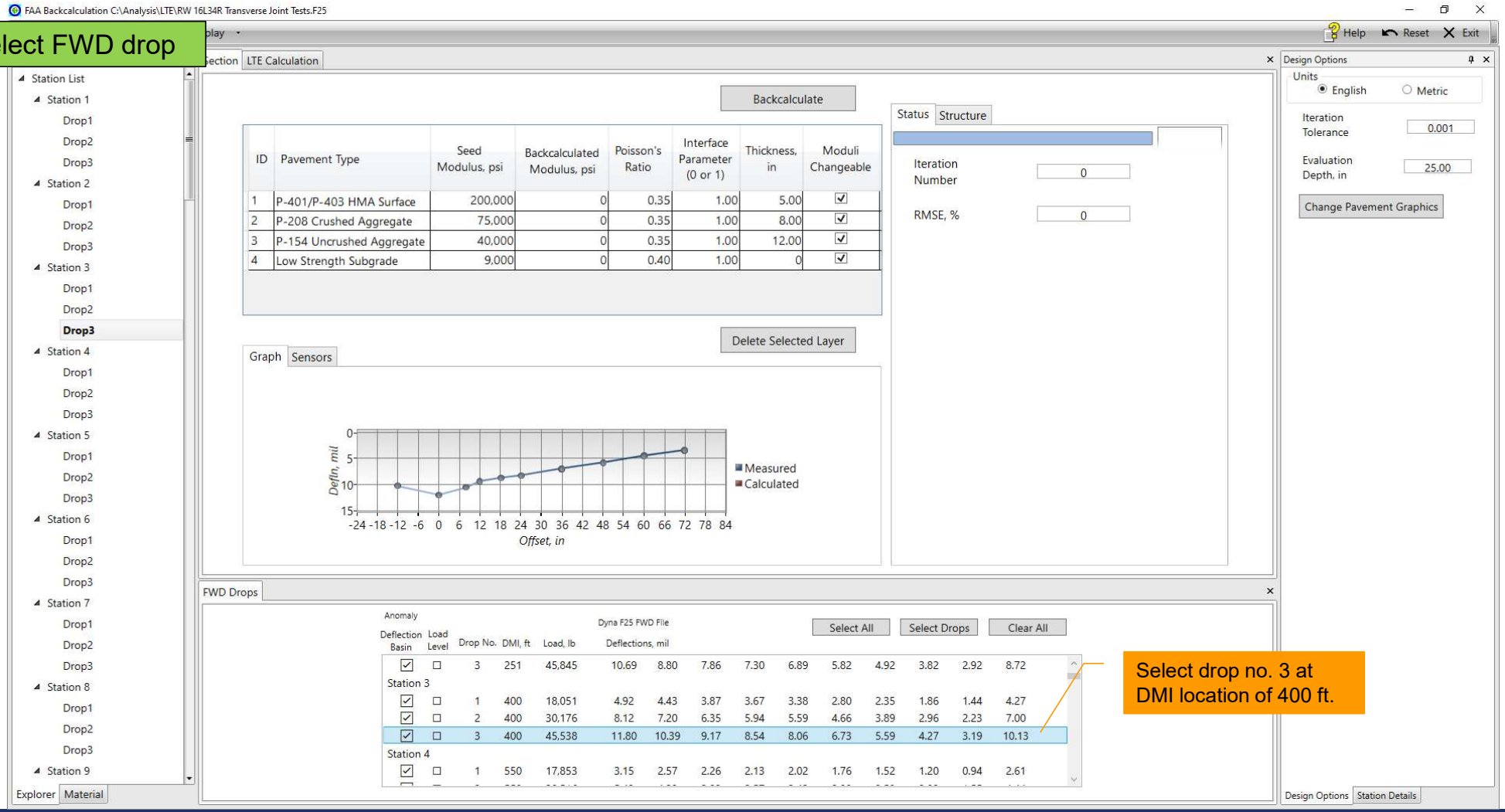


Load FWD file



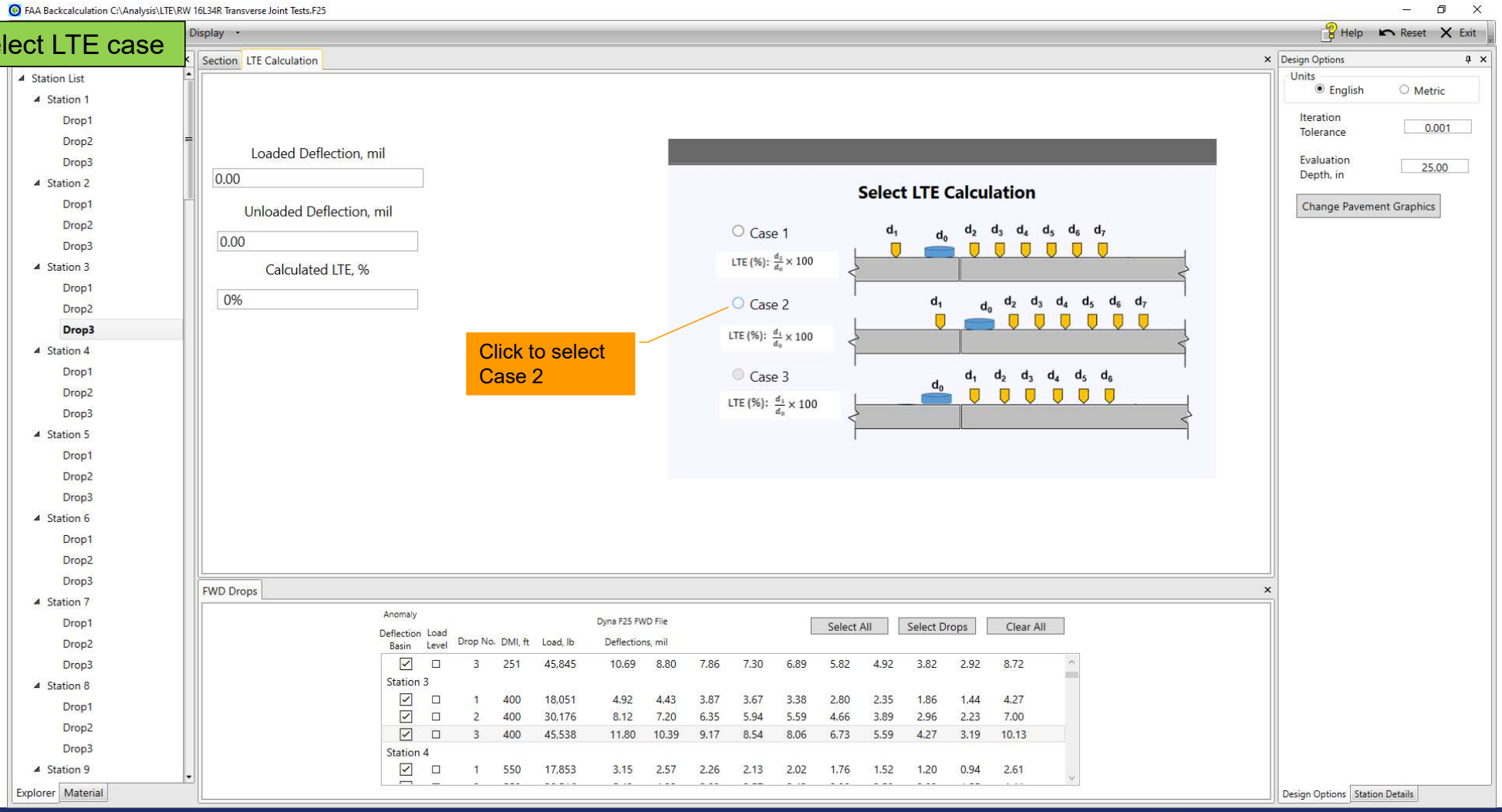
Load FWD file

Select FWD drop

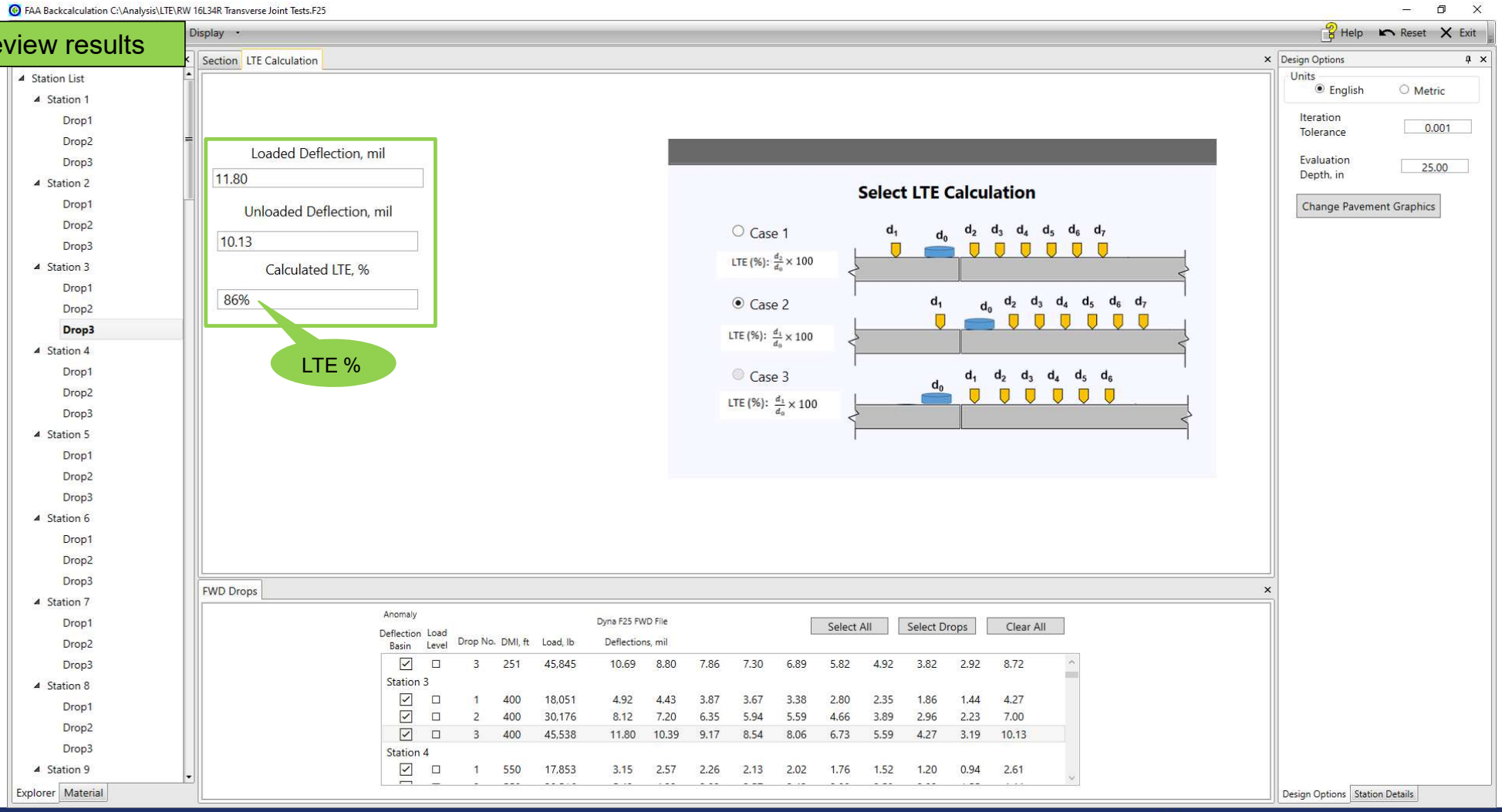


Select drop no. 3 at DMI location of 400 ft.

Select LTE case



Review results



Contact Us

- Richard Ji, PhD: Richard.Ji@faa.gov
- Matthew T. Brynick, PE: Matthew.T.Brynick@faa.gov